Unit – 1 Behavioral Medicine Introduction

Behavioral medicine has had a long history, but recent developments may have blurred the boundaries of behavioral medicine or caused a "drift" in the meaning of the term.

Prior to the 1960s, the prevalent view of health and disease was reductionist, focusing on organs and organ systems. By the 1970s, a more systematic and empirical evaluation of the interaction between behavior and illness generated a movement to integrate behavioral techniques and conferred credibility on the mind-body relationship. This movement evolved into a discipline now referred to as "behavioral medicine."

In 1977, behavioral medicine was defined as "the interdisciplinary (relating to more one batch of knowledge)field concerned with the development and integration (coordination f process in the nervous system)of behavioral and biomedical science knowledge and techniques relevant to health and illness and the application of this knowledge and these techniques to prevention, diagnosis, treatment, and rehabilitation."

Definition of Behavioral Medicine:

A broader, and more current, definition comes from the charter of the International Society of Behavioral Medicine, which states that behavioral medicine is "**the interdisciplinary field concerned with the development and integration of socio cultural, psychosocial, behavioral, and biomedical knowledge relevant to health and illness and the application of this knowledge to disease prevention, health promotion, etiology, diagnosis, treatment, and rehabilitation."**

(http://www.sbm.org/about)

Health

Health is the level of functional or metabolic efficiency of a living being. In humans, it is the general condition of a person's mind and body, usually meaning to be free from illness, injury or pain. Health is not simply the absence of disease, but is instead a state of functioning and well-being that encompasses both physical and mental aspects of an individual.

WHO, (1948) definition of Health

"Health" is complete state of physical, mental and social well-being; it is not simply the absence of illness or disease.

Illness

Illness is a conditional marked by the pronounced deviation from the normal healthy state. When ill an individual is unable to carry out normal day to day activities.(Dorland, 1965).Illness is a subjective phenomena. It depends upon an individual's evaluation of his/her own internal state ability to functions.

The subjectively of illness does not make it whimsical or unreal, but it does not complicate the phenomena by making it difficult to define objectively. And because people usually wait until they feel sick to go to the doctor.

Definition-"A person's bodily state is what typically brings him/her to seek help from health professionals.

Definition of Boundary:

Boundaries are mutually understood, unspoken physical and emotional limits of the professional relationship between the patient and the physician or student, or supervisor and student.

When these limits change, what is allowed in the relationship becomes ambiguous, and unethical conduct and other unprofessional behavior may occur. The essence of a patient-physician relationship is trust. This

relationship must have clear boundaries. Both physician and patient may misinterpret behaviors based on their own experiences.

Although the patient-physician relationship is emphasized throughout this presentation, clear boundaries also are necessary among the following groups: staff, trainees, and patient families. The medical student can be in the position of power and prestige, which can make the relationship uneven. The student must always be in control of the emotional elements of the relationship or seek guidance. It should be emphasized that every time a student sees a patient, the dynamics of a patient-physician relationship develop. When a student interacts with a supervisor in any manner other than mentor-trainee, a boundary violation may occur.

Boundary Violations:
A boundary violation may occur any time the professional relationship becomes anything other than patient and doctor/student, or doctor and student.

If a student uses his or her professional role related to a patient for a tangible or intangible benefit or gain, or the relationship becomes anything other than professional, he or she is committing a boundary violation.

Boundary Problems:
High Risk Situations

- Over-familiarity with patients
- Business relationships with patients
- Gift giving and accepting
- Treatment of family and friends
- Social contact with patients
- Inappropriate staff and trainee relationships

The health and well-being of patients depends upon a collaborative effort between patients and physicians. Students, residents, and physicians may be more susceptible to violating these boundaries due to their inexperience in the professional role.

Many patients find over-familiarity offensive. Business relationships and/ or gift giving and receiving may lead to a decrease of a physician's objectivity. Treating family and close friends often results in inadequate or overly involved patient care being provided and objectivity is sacrificed.

Social contacts with patients may blur the professional boundary. Boundary problems between office staff, hospital staff, and trainees lead to undesirable consequences.

Over-Familiarity

- Once crossed over – hard to return to original relationship
- May lead to further reciprocal intrusions and obligations
- May lead to coercion
- May cloud clinical judgment
- May lead to serious violations (sexual)

There are many problems with over familiarity, although at the outset it seems harmless. Over-familiarity goes both ways and may include physician or student self disclosure or inquiry about things outside the

clinical context. Once a line is crossed, it is difficult to reestablish the previous relationship. For example, how difficult is it to refuse a prescription to your in-laws once you have already been calling them in?

A patient whose boundaries are violated may become demanding and belligerent. Resentment or a sense of the patient being special may cloud the physician's judgment. In either case, patient care suffers.

Business Relationships

- Misuse of information from patients
- Personal gain
- Unnecessary treatments or referrals

Although not necessarily a medical student issue while in training, this could become an issue in the future. Under no circumstances may a physician place his or her own financial interests above the welfare of his or her patients. Taking advantage of information gained by means of the patient-physician relationship for personal gain is unethical. For example, you are treating a patient for hypertension. He offers you a partnership in a business venture during a clinic visit. The ethical response would be to respectfully decline the offer.

Business relationships should be kept separate from patient-physician relationships, although this may be difficult in small communities. Physicians should institute treatment or referrals based on clinical information, rather than for personal gain.

Gift Giving and Accepting

Patient to Physician/Student

- Small gift versus larger or more expensive gift
- Response to the gift should be individualized

Physician/Student to Patient

- Rarely would be appropriate
- Patient's sense of obligation may affect ability to communicate freely with physician/student

Patients frequently give physicians gifts. Small gifts given in gratitude may sometimes be welcomed and not viewed as a boundary violation. However, gifts accompanied by compliments and other evidence of seductive behavior may represent an attempt by the patient to consciously or unconsciously control the patient-physician interaction. Larger or more expensive gifts are clear and serious boundary transgressions, unless they are given as a charitable donation to a nonprofit institution, rather than as a personal gift to the physician. As implied by the saying, "There is no such thing as a free lunch," expectations arise from gifts. Physicians should make clear to patients that giving a gift will not secure preferential treatment.

The same can apply to the physician who gives patients gifts, with the patient feeling burdened by a sense of obligation that can never be openly discussed with the physician.

Treatment of Family and Friends

AMA Code of Medical Ethics

- Physicians should not treat themselves or immediate family members.

TMA Board of Councilors

- It is ethical to treat family/friends with appropriate medical records.

Texas Medical Board (TMB) Rule

- Inappropriate prescription of dangerous drugs or controlled substances to oneself, family members, or others without treatment relationship and documentation is a violation of the Medical Practice Act.

The AMA Code of Medical Ethics (E-8.19) states that physicians should not treat themselves or members of their immediate families. Professional objectivity may be compromised, and the physician's personal feelings may unduly influence his or her professional medical judgment.

The TMA Board of Councilors opinion states that it is ethical to treat family and friends, but that a medical record should be maintained. It notes that in urgent or episodic situations, generating a medical record may not be practical or possible. The opinion notes that legal requirements exceed the ethics opinion.

Texas Medical Board Rule 198(8)(1)(M) states that inappropriate prescription of dangerous drugs or controlled substances to oneself, family members, or others in which there is a close personal relationship is a violation of the Medical Practice Act. The rule states that prescribing or administering dangerous drugs or controlled substances without taking an adequate history, performing a proper physical examination, and creating and maintaining adequate records and prescribing controlled substances in the absence of immediate need (no more than 72 hours) constitute inappropriate prescribing.

Therefore, it may be appropriate to undertake self-treatment or treatment of immediate family members in emergency situations. In addition, while physicians should not serve as primary or regular care providers for immediate family members, there are situations in which routine care is acceptable in short-term, minor problems. There are significant risks for physicians who write prescriptions for controlled substances for themselves or immediate family members. Physicians should always document any treatment provided to avoid legal sanctions by TMB.

Social Contact With Patients

- Social contact with patients may blur professional boundaries
- Urban/Rural considerations
- Appropriate attire in clinic

It is certainly reasonable (and even expected) that in small communities, physicians and patients will encounter each other in the course of normal daily life activities. However, social contact with patients may become risky when the physician begins to perceive that relationship as "special" and beyond the range of providing adequate medical care (e.g., becoming intimate friends). Making special exceptions for certain patients (e.g., offering reduced fees for services, overlooking inappropriate behaviors) may also be an early indicator of problematic boundary violations.

Staff and Trainee Relationships

- Clinical staff
- Hospital staff
- Trainees

Consider these scenarios:
1. A male physician starts dating a female nurse in his clinic. He ends the relationship. She is very upset, resigns, then files unjustified complaints against him with TMB and the Centers for Medicare & Medicaid Services.
2. A female physician starts dating a male hospital administrator at a hospital where she has privileges. The administrator makes some very unpopular staff cuts, angering the medical staff members, who demand his resignation. The administrator begs the physician to side with him in opposition to the medical staff's attempts to make him resign.
3. A male faculty member hosts a holiday party, inviting medical students and house staff he knows. After an evening of drinking, the last person at the party is a female house staff member. Something allegedly

happens between the two of them, and the next morning the house staff member files a sexual harassment complaint against the faculty member.
Each of these scenarios is based on real events and demonstrates that boundary problems can lead to very difficult situations for physicians.

Here are some ideas on how you can help yourself when you have identified that a boundary violation has occurred or is likely to occur:

- Use your professional support network
- Be familiar with ethical and legal statutes
- Listen to your emotional response to a situation:
- Take care of your own physical, mental, and emotional well-being.

Health Psychology:
Health psychology is a field within Psychology that focuses on the behavioral, cognitive, psychosocial, and physiological factors that influence individual responses to health and illness.

Its **aims** are:
- **the promotion of good health,**
- **the prevention of poor health,**
- **the maintenance of quality of life and well-being within the context of illness.**

Psychological influences on health and illness:
Psychological determinants in health have been a focus since the beginning of the 20th century and results explain that people who eat regular meals, maintain a healthy weight, do not smoke, drink little alcohol, receive adequate sleep and exercise regularly are in better health and live longer.

Life can be stressful. Everyone faces a different mix of demands and adjustments in life, and any one of us may break down if the going gets tough enough. **Under conditions of over-whelming stress, even a previously stable person may develop temporary (transient) psychological problems and lose the capacity to gain pleasure from life (Berenbaum & Connelly, 1993).**

Psychologists were also discovering associations between psychological and physiological processes. These include the influence of anxiety on the cardiovascular and immune systems and finding that the functioning of the immune system could be modified by training..

The American Psychological Associations division 38 is health psychology and they **focus on understanding the relationship between health and illness**. This division concentrates on examining the psychological determinants that influence health, and they contribute information to the health care policy system. **The three areas that relate to health psychology are:**

Research: Health psychologists conduct studies on a variety of health-related concerns. For instance, researchers may concentrate on investigating effective preventative measures, explore health promotion techniques, study the causes of health problems, investigate how to motivate people to seek treatment, and how to assist people cope with illness or pain.

Public Policy Work: Health psychologists may work in private or government settings and have a role in making public policy on health related issues. Their business might point to advising executive groups on health care improvement, address disparities in health care, or lobby government agencies.

Clinical Work: In medical and clinical surroundings, health psychologists regularly administer behavioral evaluations, participate in clinical interviews and conduct personality tests. They often participate in managing interventions with individuals or groups which relate to training people about anxiety reduction methods, offer addiction cessation advice and teach people how to refrain from unhealthy ways of life.

Behavioral influences on health and illness:

A wide range of behaviours can influence health. Here we present a selective overview of some key behavioural factors that can influence health. First, there is a focus on the topic of stress and its effects. There is then a consideration of personality factors that can influence health and health behaviour. Finally, there is an outline of lifestyle factors that can have health consequences, and this section concludes by examining the main theories that have been developed to explain the variation in health-related behaviours.

Stress and health

The term ‘stress' is usually used to describe situations in which individuals are faced with demands that exceed their immediate ability to cope. Stressful situations are typically those that are novel, unpredictable, and uncontrollable as well as those involving change or loss. These situations can give rise to adverse psychological and physiological changes which, in turn, may result in disease.

A major problem for health psychologists is to understand the way in which stress is associated with the development of illness. Two broad possibilities have been proposed; these have been referred to as indirect and direct effects. Thus, stress may have indirect effects on health by increasing levels of risk behaviour (e.g. smoking, alcohol consumption), or may have direct effects on specific physiological mechanisms (e.g. increases in blood pressure) as well as affecting the individual's resistance to disease through suppression of the immune system, or by exacerbating or triggering a disease process in an already vulnerable individual.

A range of behavioural and emotional responses are shown by individuals as they attempt to cope with stressful situations and these are accompanied by autonomic, neuroendocrine, and immunological changes. **During stressful episodes, releasing factors from the brain cause the pituitary to release ACTH which gives rise to the release of corticosteroids from the cortex of the adrenal glands. In addition to producing a number of well-known changes associated with the mobilization of both short- and longer-term physical resources (e.g. release of adrenaline (epinephrine)or noradrenaline (norepinephrine), release of glucose, activation of endorphins/encephalins, etc.), these steroids can also have effects on the immune system**. Thus, fairly acute stressors, such as examinations, and more chronic stressors, such as caring for a dependent elderly relative, can lead to deleterious immunological changes with adverse effects on health. For example, chronic stress has been found to slow down the process of wound healing and increase susceptibility to infectious disease.

Lifestyle and health

The effects on health of behaviours such as smoking and high alcohol use are well documented. There is overwhelming evidence that smokers not only are much more likely to die from lung cancer and other cancers but also have much higher rates of cardiovascular disease and chronic respiratory disorders, particularly emphysema and chronic bronchitis. Moreover, the disease risk is dose related in that higher levels of smoking are more strongly associated with all these diseases. With sustained high levels of alcohol use a different but equally unpleasant spectrum of health problems can be seen. Drinking is a major cause of accidents particularly motoring accidents and can cause liver damage as well as having detrimental effects on brain functioning.

For health psychologists, the key questions about health-risk behaviours concern their origin, their maintenance, and their prevention or treatment. There are diverse determinants of these behaviours since they may start as ways of coping with stress, in response to peer pressure, for pleasure, and for a number of other reasons. Similarly, they will be maintained by a variety of psychological, social, and biological factors.

There are many other risky behaviours that cannot be discussed in detail in an overview; these include drug abuse, poor diet, and accidents, and the health effects of all these are also well documented. Although health psychology has an important role to play in describing, explaining, and intervening in all risk behaviours, it is very important not to think of these problems exclusively in individual, behavioural terms since they often reflect adverse social circumstances or particular cultural contexts.

The same caveats about the influence of social and cultural factors must also be applied to the understanding of health-protective or health-enhancing behaviours. Prospective cohort studies have confirmed that various daily behaviours (e.g. patterns of eating, sleeping, and exercise) can have significant long-term effects on health. For example, there is now a growing body of evidence to indicate that regular exercise has a beneficial effect on physical and psychological health. Exercise can reduce the incidence of physical health problems in elderly people and facilitate recovery from heart attack. However, there can be significant problems in ensuring that exercise and other health-promoting activities are adhered to. Interventions need to be planned carefully, because it has been shown that it is usually very difficult to make and maintain changes in health-related behaviour. Information provision is rarely sufficient to promote behaviour change since it is also necessary to elicit and modify beliefs (see below) as well as influencing social networks in order to ensure success.

Neurotransmitters and The Stress Response:
There are many neurotransmitters involved in the stress response. Some of the most important neurotransmitters are released from clusters of extrinsic neurons. These systems have disproportionate power to regulate human behavior, emotional functioning, and cognition. This is because these systems originate in the brainstem and have connections in virtually all brain areas. The brainstem regulates and mediates hundreds of crucial functions -- including the complexities of the stress response.

The Reticular Activating System (RAS)
The RAS originates in the brainstem and is a network of ascending, arousal-related neural systems. The RAS plays a major role in arousal, anxiety, and the modulation of limbic and cortical processing. These brainstem and midbrain monoamine systems, working together, provide the flexible and diverse functions necessary to modulate the variety of functions involved in anxiety regulation.

Locus Coeruleus
The locus coeruleus (LC) is a critical brain stem nuclei involved in initiating, maintaining, and mobilizing the total body response to threat. The LC plays a major role in determining the value of incoming sensory information, increasing in activity if the information is novel or potentially threatening. Acute stress results in an increase in LC activity. The LC plays a critical role in regulating arousal, vigilance, affect, irritability, locomotion, attention, the response to stress, sleep, and the startle response.

Hippocampus
The hippocampus is critical to the process of learning. It takes short-term memory and converts it into long-term memory. It plays a major role in memory, including what we call episodic, declarative, and

spatial learning and memory. The hippocampus also plays a key role in various activities of the autonomic nervous and neuroendocrine systems.

Stress hormones and stress-related neurotransmitter systems have the hippocampus as a target. Various hormones (e.g., cortisol) appear to alter hippocampus synapse formation, thereby causing actual changes in gross structure and size. Repeated stress inhibits the development of neurons and atrophy of the hippocampus can occur. These neurobiological changes are related to some of the problems with memory and learning found in stress-related neuropsychiatric syndromes, including post-traumatic stress disorder (PTSD).

Amygdala
In the recent past, the amygdala has emerged as the key brain region in the processing, interpreting, and integration of emotional functioning. The amygdala is where fear learned from past experience is permanently stored. In the same fashion that the LC plays the central role in orchestrating arousal, the amydgala plays the central role of processing afferent and efferent connections related to emotional functioning.

The amygdala receives input directly from sensory systems throughout the brain. The amygdala processes and determines the emotional value of simple sensory input, complex multisensory perceptions, and complex cognitive abstractions. The amygdala orchestrates the response to this emotional information by sending projections to brain areas involved in motor (behavioral), autonomic nervous system, and neuroendocrine areas of the CNS.

Cortex
The quality and intensity of any emotion, including anxiety, is dependent upon subjective interpretation or cognitive appraisal of the given situation. How an individual cortically interprets the limbic-mediated activity (i.e., their internal state) associated with arousal plays a major role in the subjective sense of anxiety.

Kluver-Bucy syndrome, the result of damage to or surgical ablation of temporal lobes, results in loss of fear for current and previously threatening cues. The general lack of inhibition demonstrated by this syndrome suggests a loss of the capacity to recall cortically stored information related to previous threat, or to efficiently store threat-related cues from new experience.

Neuroendocrine Responses to Stress:
Neuroendocrine activation belongs to the main characteristics of the stress response in general. Nowadays it is clear that exposure to stress conditions evokes a broad spectrum of changes resulting in stimulation or inhibition of many factors in the brain, endocrine glands and other organs. The number of factors under consideration is still increasing. In the original stress concept of Hans Selye, stress was considered to be a nonspecific response of the body to any demand (Selye 1936).

According to the data of others (Kopin 1995; Pacak et al. 1995) as well as our own research (e.g. VigaS et al. 1984), neuroendocrine response during stress is not uniform but depends on the stress stimulus involved and on many other conditions. As to the stress stimulus, both its nature and intensity determine resulting changes. There are several other modifying factors, such as species differences, the time of the day or environmental temperature (Jeiovh et al. 1995a). Neuroendocrine response is significantly determined by the previous stress history and by actual state of the body - the age, health and disease, motivation and other emotional preconditions. In general, a stress signal is recognized by neural centers and via neurotransmitters and neuropeptides transmitted to neuroendocrine centers located in the hypothalamus. Changes in hypothalamic releasing and inhibiting hormones induce pituitary response with

consequent changes in peripheral stress hormone secretion. All this results in cardiovascular, metabolic and immune changes aimed to overcome increasing demands of the body.

Further, hypothalamo-pituitary-adrenocortical (HPA) axis and sympathetic adrenomedullary system are considered to be peripheral limbs of the stress system (Stratakis and Chrousos 1995). Though hormones of the hypothalamo-pituitarygonadal axis are not considered to be typical stress hormones, the reproductive function is very directly linked to the stress system. It is well known that the reproductive axis is inhibited at several levels by the components of the stress system and this interaction is thought to be bidirectional (Rabin et al. 1988). On the other hand, the hormones prolactin and oxytocin, which both exert specific roles in the course of reproduction, are rapidly activated in response to many stress stimuli. Plasma testosterone levels are for a long time known to be markedly reduced during chronic and intensive stress situations both in men and animals (Carstensen et al. 1973, Nakashima et al. 1975, Dessypris et al. 1976, RepCekova and Mikulaj 1977). However, several physical stress stimuli, such as exercise on a bicycle ergometer are inducing an elevation of plasma testosterone (Brisson et al. 1977, Jeiovii-RepCekovB et al. 1982, Jeiova et al. 1985).

Stress and Immune Status:

Research on psychosocial mediation of the immune system suggests it is responsive to changing moods and behaviors. Research in human and animal models is consistent in suggesting that immune system activity is regulated by the CNS (Moynihan & Ader 1996). Further, evidence indicates that stress affects activity in the immune system. Research has been complicated by the fact that responses among different components of the immune system are variable (e.g. Zakowski 1995). Some cells, like natural killer cells, appear to respond differently depending on the chronicity of the stressor (Delahanty et al 1996). Some immune cells also appear to respond to stress differently at different times of day and to follow clear circadian rhythms (Wang et al 1998). Direct effects of stress include acute activation and chronic suppression of natural killer cells, increased latent viral activity, decreased lymphocyte proliferation, and cytokine production (Andersen et al 1998, Pariante et al 1997, Schedlowski et al 1995, Kiecolt-Glaser et al 1994). These influences are thought to be caused by neural and hormonal regulation, principally through arousal of the SNS, the HPA axis, and opioid peptide systems (e.g. Webster et al 1997).

Nerve growth factor and cytokine influences are also possible and suggest a broad interface between the nervous and immune systems (e.g. Jenkins & Baum 1995, Aloe et al 1997). The importance of the SNS in observed stress-immune relationships is suggested by studies reporting correlations between SNS indices and immune system change or by studies of the effects of adrenergic blockade (Bachen et al 1995, Manuck et al 1991, Zakowski et al 1992a,b). Exercise and exertion also affect the immune system, often in ways that resemble the impact of stress or emotional arousal (e.g. Perna et al 1997, Nieman 1997). The immune system communicates directly with the CNS and appears to have a regulatory function in much the same way that the endocrine system supplements and extends neural activation (Maier & Watkins 1998). In addition, some mental health problems are associated with abnormal immune system function (Weisse 1992, Herbert & Cohen 1993). Some of these effects are mediated by cytokines that are activated by neural stimulation (Muller & Ackenheil 1998). Pro-inflammatory cytokines in turn activate the HPA axis and induce fever and illness symptoms, but these effects may also be influenced by glucocorticoids and other stress-related agents (Goujon et al 1997).

We do not yet know whether observed patterns of change in the immune system reflect alterations that have implications for vulnerability to disease (e.g. Cohen & Rabin 1998). However, there is considerable evidence of changes within the immune system associated with social support, negative affect, stress, and other behavioral or psychological factors, and evidence that these factors affect infectious disease, progression of cancer and HIV disease, and other health outcomes (e.g. Leserman et al 1997, Antoni 1997, Andersen et al 1994, Cohen et al 1991).

Stress, HIV, and AIDS

HIV disease and AIDS are unusually important as examples of behavior influenced outcomes. They reflect both direct effects on bodily functioning and on behaviors that may affect disease risk and prognosis. They also affect caregivers in powerful ways that can harm their health as well (e.g. Catalan et al 1996, Folkman et al 1997, Folkman 1997, Park&Folkman 1997). The transmission of the virus that causes HIV disease is typically accomplished through behaviors (sexual activity, IV drug use), and because HIV attacks the immune system directly, behaviors or states that affect the immune system should affect HIV. Mental health problems appear to be associated with an increased risk for HIV infection (Hoff et al 1997, O.Leary&Jemmott 1996), presumably because of distress-related increases in risky behavior. High-risk behaviors such as unprotected sexual activity or IV drug use can be minimized, but stress can trigger impulsive behavior, reduce countervailing constraints against high-risk behaviors, or temporarily inhibit or interfere with intentions to avoid them (e.g. Fishbein et al 1998, O.Keefe et al 1990). These impulsive behaviors may occur because of stress-related drug and alcohol use (e.g. Testa & Collins 1997, Robins et al 1997). The implications of increased drug or alcohol abuse include reduced motivation, a false sense of invulnerability, and lack of attention to details or sanctions leading to increased risk and infection (Dingle & Oei 1997, Chandra et al 1996). As a result, people who are intoxicated by drugs or alcohol may not use good judgement, may be more likely to end up in compromising situations, or may simply not care about risks to their health at that moment. Emotional arousal, stress, and behaviors such as smoking or drug use may affect resistance to the disease or disease progression. Evidence of this possibility is scant and is largely limited to early outcomes related to testing (Evans et al 1997, Antoni et al 1994). However, stress and related states or behaviors are associated with changes in immune system activity that could affect the body.s ability to resist or combat the virus. A study of 104 HIV+ individuals enrolled in the Oslo HIV Cohort Study suggested that negative affect was related to somatic symptoms associated with progression, but there was no evidence of mediation of these effects by the immune system (Vassend et al 1997). Distress associated with concealment of sexual identity was positively related to cancer morbidity and incidence of pneumonia and other infectious illnesses (Cole et al 1996). Stress management interventions that enhance or buffer immune system activity have proven useful in treatment of HIV+ individuals (Littrell 1996, Goodkin et al 1997, Pomeroy et al 1997, Schneiderman et al 1992). Aerobic exercise interventions that also reduced distress have had similar effects (LaPerriere et al 1994).

Negative Affectivity:

Negative emotions should carry with them a health warning similar to found on packets of cigarettes. Certainly, evidence is accumulating that negative emotions can be associated with poor health (Kielcolt-Glaser et al.,2002a). Poor diet, conversely, appears to contribute to pathophysiology of disease. In addition, smoking, alcohol consumption, drug use, and other health impairing behaviors also have direct effects on disease processes and indirect effects on bodily systems and mood or behavior. Depression, as we have already emphasized, is associated with measurable and undesirable changes in immune functioning (Robles et al.,2005;Zorillo et al.,2001). Depression also seems to be associated with increased mortality from all causes in medical inpatients (Herrmann- Lingen et al.,2001). In women, depression appears to heighten the risk for osteoporosis (Michelson et al.,1996), and in men, one prospective study showed that depression at baseline predicted decline in muscle strength over a 3-year period (Rantanen et al.,2000).

Although this effect has not been as well studied, anxiety also seems to be associated, in both men and women, with the development of coronary heart disease. Finally, it is well documented that chronic anger and hostility can be risk for factors for coronary heart disease and death (Miller et al.,1996).

Behavioral Patterns:

Type A and Type B personality theory is a theory that describes two common, contrasting personality types — the high-strung Type A and the easy-going Type B — as patterns of behavior that could either

raise or lower, respectively, one's chances of developing coronary heart disease. It was originally published in the 1950s.

Type A personality behavior was first described as a potential risk factor for heart disease in the 1950s by cardiologists Meyer Friedman and Ray Rosenman. After an eight and a half year long study of healthy men between the ages of 35 and 59, Friedman and Rosenman estimated that Type A behavior doubles the risk of coronary heart disease in otherwise healthy individuals. The individuals enrolled in this study were followed well beyond the original time frame of the study. Subsequent analysis indicated that although Type A personality is associated with the incidence of coronary heart disease, it does not seem to be a risk factor for mortality. This research had a significant effect on the development of the health psychology field, in which psychologists look at how an individual's mental state affects his or her physical health

Type A
The theory describes a Type A individual as ambitious, rigidly organized, highly status conscious, can be sensitive, care for other people, are truthful, impatient, always try to help others, take on more than they can handle, want other people to get to the point, proactive, and obsessed with time management. People with Type A personalities are often high-achieving "workaholics" who multi-task, push themselves with deadlines, and hate both delays and ambivalence.

In his 1996 book, *Type A Behavior: Its Diagnosis and Treatment*, Friedman suggests that Type A behavior is expressed in three major symptoms: free-floating hostility, which can be triggered by even minor incidents; time urgency and impatience, which causes irritation and exasperation usually described as being "short-fused"; and a competitive drive, which causes stress and an achievement-driven mentality. The first of these symptoms is believed to be covert and therefore less observable, while the other two are more overt.

Type B
The theory describes Type B individuals as a contrast to those with Type A personalities. People with Type B personality by definition generally live at a lower stress level and typically work steadily, enjoying achievement but not becoming stressed when they are not achieved. When faced with competition, they do not mind losing and either enjoy the game or back down. They may be creative and enjoy exploring ideas and concepts. They are often reflective, thinking about the outer and inner worlds.

Finally, there is emerging evidence that general patterns of positive or negative emotional responses, associated with personality, can influence various aspects of health. Individuals who are high in negative affect (i.e. experience more negative emotions, particularly anxiety) do not seem to be more prone to disease, but they are more likely to notice bodily changes and symptoms and consequently seek medical help more frequently (Wiebe and Smith for a more detailed account of negative affect and the links between personality and health).

Coping:
Coping is a process that we as individuals employ every day. We engage in coping when we feel under stress or want to manage a taxing situation. The process of coping involves two components, appraisal and coping (Lazarus, 1966). Appraisal is the act of perceiving a stressor and analysing one's own ability to deal with the stressor. Appraisal can be made in three different conditions: when we have experienced a stressor, when we anticipate a stressor and when we experience a chance for mastery or gain (Lazarus, 1966). Once we appraise a stressful situation we must decide how we will respond or 'cope' with the stressor, either choosing to master it, reduce it or tolerate it. The coping style we engage in is ultimately determined by whether we believe we have the resources to resolve the stressor (Lazarus, 1966).

Coping Styles:
There appear to be three main coping styles that people employ when attempting to resolve or remove a stressor :

- Problem-focused coping
- Emotion-focused coping
- Avoidant coping

Problem-focused coping involves altering or managing the problem that is causing the stress and is highly action focused. Individuals engaging in problem-focused coping focus their attention on gathering the required resources (i.e. skills, tools and knowledge) necessary to deal with the stressor. This involves a number of strategies such as gathering information, resolving conflict, planning and making decisions (Lazarus & Folkman, 1984).

Emotion-focused coping can take a range of forms such as seeking social support, acceptance and venting of emotions etc (Carver et al., 1989). Although emotion-focused coping styles are quite varied they all seek to lessen the negative emotions associated with the stressor, thus emotion-focused coping is action-orientated (Admiraal, Korthagen, & Wubbels, 2000; Folkman & Lazarus, 1980).

The third main coping style is **avoidant coping**. Avoidant coping can be described as cognitive and behavioural efforts directed towards minimising, denying or ignoring dealing with a stressful situation (Holahan, Holahan, Moos, Brennan, & Schutte, 2005). Although some researchers group avoidant coping with emotion-focused coping the styles are conceptually distinct. Avoidant coping is focused on ignoring a stressor and is therefore passive, whereas emotion-focused coping is active (Admiraal et al., 2000, Holahan et al., 2005).

Health Behavior Models:
The four most commonly used models of individual health behavior change, relying primarily on the criteria of Glantz and colleagues. Their review of articles published between 1992-1994 in health education, medicine, and behavioral science that use any theoretical framework (only 45% used a theory)1 revealed that the most used models were the Health Belief Model, Theory of Reasoned Action/Planned Behavior, Social Cognitive Theory and The Transtheoretical Model.

The Health Belief Model
The Health Belief Model (HBM) has the longest history of all the theories reviewed. It was originally conceived by social psychologists in the public health arena as a way of predicting who would utilize screening tests and/or vaccinations. According to the HBM, the likelihood that someone will take action to prevent illness depends upon the individual's perception that:

- they are personally vulnerable to the condition;
- the consequences of the condition would be serious;
- the precautionary behavior effectively prevents the condition; and
- the benefits of reducing the threat of the condition exceed the costs of taking action.

These four factors, which are influenced by mediating variables, indirectly influence the probability of performing protective health behaviors by influencing the perceived threat of the illness and expectations about outcome. The HBM has been used for intervening with health screening, illness, sick role, and precautionary behaviors. The model has undergone some modifications since its original formulation. The model's four key components are conceptualized as: 1) perceived susceptibility, 2) perceived severity, 3) perceived effectiveness, and 4) perceived cost.

Perceived susceptibility refers to the probability that an individual assigns to personal vulnerability in developing the condition. The concept of perceived susceptibility has been found to be predictive of a number of health-protective behaviors. From an HBM perspective, the likelihood individuals will engage in precautionary behaviors to prevent cancer (e.g., quit smoking, eat a diet low in fat and high in fiber, exercise, get a mammogram or prostate exam) depends on how much they believe they are vulnerable to or at risk for cancer. In general, it has been found that people tend to underestimate their own susceptibility to disease.

Perceived severity refers to how serious the individual believes the consequences of developing the condition are. An individual is more likely to take action to prevent cancer if s/he believes that possible negative physical, psychological, and/or social effects resulting from developing the disease pose serious consequences (e.g., altered social relationships, reduced independence, pain, suffering, disability, or even death). Models of Health Belief frequently refer to perceived health threats. The combination of perceived susceptibility and perceived severity constitute a threat.

Perceived effectiveness refers to the benefits of engaging in the protective behavior. Motivation to take action to change a behavior requires the belief that the
Precautionary behavior effectively prevents the condition. For example, individuals who are not convinced that there is a causal relationship between smoking and cancer are unlikely to quit smoking because they believe that quitting will not protect against the disease.

Perceived cost refers to the barriers or losses that interfere with health behavior change. The combination of perceived effectiveness and perceived costs constitute the notion of outcome expectation. Belief alone is not enough to motivate an individual to act. Taking action involves cognitively weighing the personal costs associated with the behavior against the benefits expected as a result of engaging in the behavior. Benefits have to outweigh the costs involved.

The Theory of Reasoned Action/Planned Behavior

The Theory of Reasoned Action (TRA) is a widely used behavioral prediction theory which represents a social-psychological approach to understanding and predicting the determinants of health-behavior. Over the years, TRA has been applied to many diverse health-related behaviors including: weight loss, smoking, alcohol abuse, HIV risk behaviors, and mammography screening. The theory of reasoned action states that the intention to perform a particular behavior is strongly related to the actual performance of that behavior. Two basic assumptions that underlie the TRA are: 1) behavior is under volitional control, and 2) people are rational beings. From the perspective of TRA, we behave in a certain way because we choose to do so and we use a rational decision-making processing choosing and planning our actions. The TRA was designed to predict behavior from intention, and proposes quasi-mathematical relationships between beliefs, attitudes, intentions, and behavior. A modified version of TRA includes the addition of perceived control over the behavior and is referred to as the Theory of Planned Behavior (TPB) describes the main constructs used in TRA and the Theory of Planned Behavior. According to the TRA, behavior is influenced by the intention to perform the behavior. Intention is influenced by three major variables: subjective norms, attitudes, and self-efficacy. Subjective norms involve an individual's perception of what significant others believe about his or her ability to perform the behavior.

Bandura's Social Cognitive Theory (SCT), also referred to as Social Learning Theory, is a behavioral prediction theory that represents a clinical approach to health behavior change. This theory has been widely applied to health behavior with respect to prevention, health promotion, and modification of unhealthy lifestyles for many different risk behaviors.

Bandura conceptualized influences on behavior that involved the concept of person in terms of basic human capacities that are cognitive in nature. Key concepts associated with the person include: personal

characteristics, emotional arousal/coping, behavioral capacity, self-efficacy, expectation, expectancies, self-regulation, observational/experiential learning, and reinforcement.

Personal characteristics have been operationalized as multiple, interacting determinants such as demographics (e.g., gender, race/ethnicity, education), personality, cognitive factors (e.g., thoughts, attitudes, beliefs, knowledge), motivation, and skills.

Emotional arousal/coping can interfere with learning and thus influence behavior. This refers to an individual's ability to respond to emotional stimuli with various techniques, strategies, and activities that help one to deal with arousing situations (e.g., fear, anxiety).

Behavioral capacity refers to the individual's possession of both the knowledge and skills necessary to perform a behavior.

Self-efficacy refers to an individual's confidence in his or her ability to perform a behavior in various situations. Self-efficacy has been recognized as an important mediating variable between knowledge, attitudes, skills, and behavior.

Expectations are beliefs associated with the outcome of a behavior. *Expectancies* are the value an individual attributes to the anticipated outcome of performing a behavior.
Self-regulation refers to the individual's ability to manage or control behavior. Individuals use goal setting, self-monitoring, and self-reinforcement to regulate performance of a behavior.

Observational/experiential learning refers to the acquisition of a behavior through observation and experience. Learning can occur either through observation of another's performance of a behavior (modeling), or through personal experience, i.e., trial and error.

Reinforcement refers to the consequences that affect the probability a behavior will be tried again. Individuals are motivated to perform behaviors through rewards and incentives.

Bandura proposed that the actual performance of a particular behavior is highly related to an individual's belief in his/her ability to perform that behavior in specific situations. An individual with low self-efficacy is likely to have lower expectations of successfully performing the behavior and be more affected by situational temptations that are counterproductive to promoting and maintaining behavior change. In contrast, an individual who has high self-efficacy not only expects to succeed but is actually more likely to do so. For example, the likelihood that an individual will successfully perform a behavior like exercise is strongly dependent upon how confident that individual is that s/he can actually do activities, such as walking, jogging, swimming, or doing aerobics on a regular basis.

Trans theoretical Model (TTM) is a model of intentional behavior change that has produced a large volume of research and service across a wide range of problem behaviors and populations. This model describes the relationships among: stages of change; processes of change; decisional balance, or the pros and cons of change; situational confidence, or self-efficacy in the behavior change; and situational temptations to relapse. TTM research has found remarkable similarities across different kinds of behavior changes. We have found repeatedly that the stages of change have predictable relationships with the pros and cons of behavior change, confidence in behavior change, temptation to relapse, and the processes of change.

Stages of Change
Individuals do not change their behavior all at once; they change it incrementally or stepwise in stages of change. The stages most commonly used across research areas include: Precontemplation, Contemplation,

Preparation, Action, and Maintenance. Individuals do not typically move linearly from stage to stage, but often progress and then recycle back to a previous stage before moving forward again. This change process is conceptualized most meaningfully as a spiral, which illustrates that even when individuals do recycle to a stage they've been in before, they may still have learned from their previous experiences.

Precontemplation describes individuals who for many reasons do not intend to change within the next six months. Some of these individuals may want to change at some future time, but just not within the next six months. Others may not want to change at all and, in fact, may be very committed to their problem behavior(e.g., a lifelong smoker or someone who regularly cultivates a deep tan).

Contemplation describes individuals who are thinking about changing their problem behavior within the next six months. They are more open to feedback and information about the problem behavior than their counterparts in Precontemplation.

Individuals in the Preparation stage are committed to changing their problem behavior soon, usually within the next 30 days. These people have often tried to change in the past and/or have been practicing change efforts in small steps to help them get ready for their actual change attempt.

The Action stage includes individuals who have changed their problem behavior within the past six months. The change is still quite new and their risk for relapse is high, requiring their constant attention and vigilance.

Maintenance stage individuals have changed their problem behavior for at least six months. Their change has become more of a habit, and their risk for relapse is lower, but relapse prevention still requires some attention, although somewhat less than for individuals in Action.

Professional Scope and Application of Psychological Principles:

In addition to the scope of the profession as prescribed in the regulations, the following acts fall within the scope and application of psychological principles:

- assessing, diagnosing, and intervening in clients dealing with life challenges, particularly those with developmental and forms of psychological distress and/or psychopathology; identifying psychopathology in psychiatric disorders, and psychological conditions; identifying, and diagnosing psychiatric disorders and psychological conditions; applying evidenced-based psychological interventions to people with psychological, and psychiatric conditions; referring clients to appropriate professionals for further assessment or intervention;
- advising on the development of policies, based on various aspects of psychological theory and research; designing, managing, and evaluating programmes dealing with psychological, and psychiatric problems;
- training and supervising other registered clinical psychologists in clinical psychology;
- conducting psychological practice, and research in accordance with the Ethical Rules of Conduct for Practitioners registered under the Health Professions Act, 1974; adhering to the scope of practice of clinical psychologists;
- designing, managing, conducting, reporting on, and supervising psychological research; and
- providing expert evidence and I or opinions.

Health Care in Hospitals:
It begins with a general consideration of the psychological consequences of admission to hospital and is followed by a brief outline of some of the psychological problems associated with the hospitalization of

younger children. Then there is a selective overview of particular hospital treatments such as intensive care which can produce specific emotional reactions due to the limitations and demands they impose on patients.

Psychological Effects of Hospitalization:

Hospital patients vary in many ways including their age, personality, and social circumstances, as well as in the severity of their health problems and the duration of their stay. Also, their experience in hospital will vary greatly and this inevitably means that there are difficulties in attempting a general discussion of the impact of hospital admission. The psychological reactions which are described may reflect not only a response to hospitalization but also to the illness itself.

Physical and Social Environment:

There are few studies which have directly investigated how the physical environment of the hospital affects the patient's condition. The hospital environment has been identified as one of a range of stressors which people experience in hospital. It has been found that patients with a window view of trees had a better postoperative recovery than patients recovering from the same type of surgery but whose hospital window looked out on to a brick wall.

Enforced life style changes have also been identified as a key hospital based stressor in the Hospital Stress Index (Koenig et al., 1995). Other hospital stressors which have been identified by an older measure, the Hospital Stress Rating Scale (Volicer & Bohannon, 1975) concerns about investigations and treatment.

Communication in Hospital:

In addition to the general stresses associated with hospital admission, some of the fear and anxiety which is found in many hospital patients may stem from the uncertainty and lack of information which they may have about the nature of their illness and its treatment and prognosis. There are a number of studies of hospital patients which show clearly that they are greatly dissatisfied with the communications aspects of hospital life (Ley, et al., 1972).

Ley, Bradshaw, Kincey and Atherton (1976) conducted a study of hospital patients on three wards of a general hospital. The results showed that 80 % of patients who had received the informational visit were satisfied with the communication received, which was approximately twice the level of satisfaction with communication found in the other two groups.

Psychological Aspects of Specific Hospital Treatments:

In the same way that physical illness imposes physical and social limitations on the individual which can give rise to psychological reactions, some treatments are also very restricting and have been found to cause emotional and behavioral changes. In particular a number of studies have been made of patients in specific treatments settings such as intensive care units (ICUs), coronary care units (CCUs) and haemodialysis. Although these treatments environment can give rise to specific stressors, they also offer challenging opportunities for psychological interventions.

Experimental work with healthy volunteers ahas shown that long periods of sensory deprivation or sensory overload will often give rise to a state characterized by increased wakefulness, disorientation, and visual hallucination (Goldberger, 1982).

In the context of coronary care, a number of studies have identified possible areas for psychological intervention in facilitating the recovery of patients in CCUs. Using a self regulatory framework, one study has shown that patients with myocardial infraction have clear beliefs about cause, timeline, consequences, and controllability, of their heart condition during the acute phase of their hospital stay in the CCU. (Petrie, et. al., 1996).

Stressful Medical Procedure in Hospital:

In addition to the generally stressful effects of hospital admission, there is a range of medical procedures which can give rise to considerable discomfort and anxiety. These includes specific investigative procedures such as barium X-rays (Allan & Armstrong, 1984) endoscopy (Johnson & Leventhal, 1973) which may not be uncomfortable and sometimes physically distressing but which also carry the threat of uncovering a serious medical condition (Weinman & Johnston, 1988). The way in which a patient reacts to a medical procedure can also have a significant influence on the outcome, particularly in recovery from surgery (Johnston, 1986).

References:

Admiraal, W. F., Korthagen, F. A. J., & Wubbels, T. (2000). Effects of student teachers' coping behaviour. *British Journal of Educational Psychology, 70*(1), 33-52.

Ajzen I, Fishbein M. *Understanding Attitudes and Predicting Social Behavior*. Englewood Cliffs, NJ: Prentice-Hall; 1980.

Bandura A. *Social Foundations of Thought and Action: A Social Cognitive Theory*. Englewood Cliffs, NJ: Prentice-Hall; 1986.

Baranowski T. Beliefs as motivational influences at stages in behavior change. *International Quarterly of Community Health Education*. 1992;13:3-29.

Becker MH. The health belief model and personal health behavior. *Health Education Monographs*. 1974;2:324-473.

Becker MH, Maiman LA. Sociobehavioral determinants of compliance with health and medical care recommendations. *Medical Care*. 1975;13:10-24.

Carstensen H., Amer I. Wide L., Amer B. (1973) Plasma testosterone, LH and FSH during the first 24 hours after surgical operations. J. Steroid Biochem. 4: 605-61 1.

Carver, C., Scheier, M., & Weintraub, J. (1989). Assessing coping strategies: A theoretically based approach. *Journal of Personality and Social Psychology, 56*(2), 267-283.

Committee on Bioethics (Collaborators 14). From the American Academy of Pediatrics: Policy Statements—Pediatrician- Family-Patient Relationships: Managing the Boundaries. *Pediatrics* 124(6): 1685-1688, 2009.

Eysenck, H.J. (1990). Type A Behavior and Coronary Heart Disease: The Third Stage. Journal of Social Behavior and Personality, 5, 25-44.

Fishbein M, Bandura A, Triandis HC, Kanfer FH, Becker MH, Middlestadt SE. Factors influencing behavior and behavior change. In: Baum A, Reveson T, and Singer J, eds. *Handbook of Health Psychology*. Hillsdale, NJ: Lawrence, Erlbaum, and Associates; in press.

Folkman, S., & Lazarus, R. S. (1980). An analsyis of coping in a middle-aged community sample. *Journal of Health and Social Behavior, 21*(3), 219-239.

Friedman, M.; Rosenman, R. (1959). "Association of specific overt behaviour pattern with blood and cardiovascular findings". *Journal of the American Medical Association* (169): 1286–1296.

Friedman, M. (1996). *Type A Behavior: Its Diagnosis and Treatment*. New York, Plenum Press (Kluwer Academic Press), pp. 31 ff.

Gabbard, GO and Nadelson, C. Professional Boundaries in the Physician-Patient Relationship. *JAMA* 273(18), 1445-1449,
1995.

Gaufberg, E, Baumer, N, et al. Professional Boundaries: The Perspective of the Third Year Medical Student in Negotiating Three Boundary Challenges. *Teach Learn Med* 20(4): 334-339, 2008.

Glantz K, Lewis FM, Rimer BK, eds. *Health Behavior and Health Education: Theory, Research, and Practice.* 2nd ed. San Francisco, CA: Jossey-Bass, Inc.; 1997.

Holahan, C. J., Holahan, C. K., Moos, R. H., Brennan, P. L., & Schutte, K. K. (2005).
Stress Generation, Avoidance Coping, and Depressive Symptoms: A 10-Year Model. *Journal of Consulting and Clinical Psychology, 73*(4), 658-666.

International Society of Behavioral Medicine. www.isbm.info. Accessed May 8, 2008.

Janz NK & Becker MH. The health belief model: a decade later. *Health Education Quarterly*. 1984;11:1-47

JeiovaD., JurankovaE., Kvetiianskg R., Kaciuba-Uscilko H., Nazar K., VigaS M. (1995a) Low ambient temperature and neuroendocrine response to hypoglycemia in men. Obes. Res. 3: 713s-719s.

Kopin I.J. (1995) Definitions of stress and sympathetic neuronal responses. Ann. N.Y. Acad. Sci. 771: 19-30.

Lazarus, R. S., & Folkman, S. (1984). *Stress, appraisal and coping*. New York: Springer.

Pacik K., Palkovits M., Kvetfiansky R., Zadid G., Kopin I.J., Goldstein D.S. (1995) Effects of various stressors on in vivo norepinephrine release in the hypothalamic paraventricular nucleus and on the pituitary-adrenocortical axis. Ann. N.Y. Acad. Sci. 77 1 : 1 15- 130.

Perry C, Baranowski T, Parcel G. How individual's, environments, and health interact: Social Learning Theory. In: Glanz K, Lewis FM, Rimer BK, eds. *Health Behavior and Health Education: Theory, Research, and Practice*. San Francisco: Jossey-Bass; 1990: 161-86.

Rabin D., Gold P.W., Margioris A,, Chrousos G.P. (1988) Stress and reproduction: interaction between the stress and reproductive axis. In: Mechanisms of physical and emotional stress (Eds. P.W. Gold, D.L. Loriaux and G.P. Chrousos). Plenum Press, New York, p. 377-387.

Ragland, D.; Brand, R. (1988). "Type A Behavior and Mortality from Coronary Heart Disease". *The New England Journal of Medicine* **318**: 65–69.

Regan, S, Ferris, TG, et al. Physician Attitudes Toward Personal Relationships with Patients. *Med Care* 48(6): 547-552, 2010.

Selye H. (1936) A syndrome produced by diverse noxious agents. Nature 32: 138.

Society of Behavioral Medicine. www.sbm.org/about/definition.asp. Accessed May 8, 2008. Sonstroem RJ. Psychological models. In: Dishman R, ed. *Exercise Adherence.* Champagne, IL: Human Kinetics Publishing; 1988: 125-54.

Stratakis C.A., Chrousos G.P. (1995) Neuroendocrinology and pathophysiology of the stress system. Ann. N.Y. Acad. Sci. 771: 1-18.

Strecher VJ, Rosenstock IM. The health belief model. In: Glantz K, Lewis FM, Rimer BK, eds. *Health Behavior and Health Education: Theory, Research, and Practice.* 2nd ed. San Francisco, CA: Jossey-Bass; 1997:41-59.

VigaS M., Kvetfiansky R., JurCoviCovi J., Jeiovi D., Tatjr P. (1984) Comparison of catecholamine and adenopituitary hormone responses to various stress stimuli in man. In: Stress: the role of catechola~ninesa nd other neurotransmitters (Eds. E. Usdin, R. Kvetfianskf and J. Axelrod). Gordon and Breach Science Publishers, New York, p. 865-882.

www.childtraumaacademy.com/amazing_brain/.../page02.html

Unit II CNS

What is Cognition & Cognitive function?

Cognition is the word we use to refer to mental activities such as seeing, attending, remembering and solving problems. The study of cognition is the study of the ***cognitive processes*** that receive, transmit and operate upon information. These processes operate at every waking moment and they are also part of our personality, our intelligence and the way we interact socially.

One major problem in the study of cognition is that cognitive processes are private, covert processes, unobservable to the naked eye. Yet their existence is not doubted and they are studied scientifically on the basis of indirect evidence just as invisible entities (for example the sub-atomic particles or the black holes in space) are the mental activities associated with thought, decision- making, language and higher mental process.

CNS

The central nervous system (CNS) is the processing center for the nervous system. It receives sends information to the peripheral nervous system. The two main organs of the CNS are the brain and spinal cord.

The brain monitors and regulates the body's actions and reactions. It continuously receives sensory information, and rapidly analyzes this data and then responds, controlling bodily actions and functions.

The spinal cord is a long, conical structure and the principal function of the spinal cord, is to distribute motor fibers to the effectors organs of the body and to collect somatosensory information to be passed on the brain.

This cognitive ability in the brain may be compromised in various disorders of the central nervous system. Some of them are discussed in the following presentation.

COGNITIVE ASPECTS

Dementia

Dementia is a syndrome due to disease of the brain, and there is disturbance of multiple higher cortical functions, including memory, thinking, orientation, comprehension, calculation, learning capacity, language, and judgement and consciousness is not clouded. Memory impairment and at least one of the following cognitive disturbances:

1. aphasia,
2. apraxia,
3. agnosia, or
4. disturbance in executive functioning like memory

Dementia of the Alzheimer's Type

Alzheimer's disease (AD) is a slowly progressive disease of the brain that is characterized by impairment of memory and eventually by disturbances in reasoning, planning, language, and perception.

Vascular dementia

The development of multiple cognitive deficits manifested by both-

1. Memory impairment (impaired ability to learn new information or to recall previously learned information).
2. Disturbance in executive functioning (i .e. planning, organizing, sequencing, abstracting)

Dementia in Pick's disease

Pick's disease is a degenerative disease of the brain that particularly affects the frontal and temporal lobes. As in other frontal lobe dementias, Pick's disease is characterized clinically by changes in personality early in the course, deterioration of social skills, emotional blunting, behavioural disinhibition, and prominent language abnormalities.

- Difficulties with memory,
- Apraxia and other features of dementia usually follow later in the course.
- Individuals may develop such severe problems in language, attention, or behaviour that it may be difficult to assess their degree of cognitive impairment.

Dementia in Creutzfeldt-Jakob disease

In Creutzfeldt-Jakob disease, there is usually a progressive spastic paralysis of the limbs, accompanied by extra-pyramidal signs with tremor, rigidity. Other variants may include ataxia, visual failure, or muscle fibrillation and atrophy of the upper motor neuron type.

Dementia in Huntington's disease

Huntington's disease (HD) is a neurodegenerative genetic disorder that affects muscle coordination and leads to cognitive decline and psychiatric problems. Huntington's disease is an inherited progressive degenerative disease of cognition, emotion, and movement. Difficulties with memory

retrieval, executive functioning, and judgment are common. Disorganized speech and psychotic features are sometimes present.

Dementia in Parkinson's disease

. Parkinson's disease is a slowly progressive neurological condition, characterized by

- tremor,
- rigidity,
- bradykinesia, and
- postural instability.
- Occur in approximately 200/0-60% of individuals with Parkinson's disease and is more likely to be present in older individuals or those with more severe or advanced disease.

Dementia in human immunodeficiency virus [HIV] disease

A disorder characterized by cognitive deficits meeting the clinical diagnostic criteria for dementia, HIV dementia typically presents with complaints of

- Forgetfulness, slowness, poor concentration, and
- Difficulties with problem-solving and reading.
- Apathy, reduced spontaneity, and social withdrawal are common.

What are the signs and symptoms of cerebral palsy?

A person with cerebral palsy will generally show symptoms during the first three years of life. A child/baby with cerebral palsy may have some of the signs and symptoms below:

- Achieves developmental milestones, such as crawling, walking, or speaking, later than his/her peers.
- Crawls in an unusual way.
- Has difficulty feeding and sucking.
- Lies down in awkward positions.
- Favors one side of the body over the other.
- Has overdeveloped or underdeveloped muscles (has floppy or stiff movements).
- Has bad coordination and balance (ataxia).
- Has involuntary, slow writing movements (athetosis).
- Muscles are stiff and contract abnormally (spastic paralysis).
- Has hearing problems.
- Has problems with eyesight.
- Has bladder control problems.
- Has bowel movement control problems.
- Has seizures.
- Range of movements are limited.

<u>WHAT IS A SPINAL CORD INJURY ?</u>

A **spinal cord injury** (SCI) is typically defined as damage or trauma to the spinal cord that in turn results in a loss or impaired function resulting in reduced mobility or feeling.

Typical common causes of damage to the spinal cord, are trauma (car/motorcycle accident, gunshot, falls, sports injuries, etc.), or disease (Ataxia, spinal cord tumour, spinal stenosis, etc.).

Traumatic Brain Injury

A TBI is caused by a bump, blow or jolt to the head or a penetrating head injury that disrupts the normal function of the brain. Not all blows or jolts to the head result in a TBI. The severity of a TBI may range from "mild," i.e., a brief change in mental status or consciousness to "severe," i.e., an extended period of unconsciousness or amnesia after the injury. The majority of TBIs that occur each year are concussions or other forms of mild TBI.

MILD COGNITIVE IMPAIRMENT

Mild cognitive impairment (MCI) (Gautier *et al.* 2006; Portet *et al.* 2006; Winblad *et al.* 2004) is a recently described syndrome seen in elderly patients, characterized, as the name suggests, brain impairment in cognitive ability . Clinical features in cognitive impairments seen in MCI include

- deficits in short-term memory,
- Calculating ability, abstracting ability, visuospatial ability, etc…

DELIRIUM).

Clinical features

Delirium, as reviewed by Lipowski (1983, 1987, 1989), is characterized by

- confusion,
- disorientation, memory loss,
- And often other symptoms such as hallucinations and delusions.
- The syndrome is typically of acute or sub-acute onset, and most patients develop the full syndrome within a day or two (Levkoff *et al.* 1992).

OTHER DYSFUNCTIONS

Encephalitis

Encephalitis is defined as inflammation of the brain.

Meningitis

Meningitis is inflammation of the protective membranes covering the brain and spinal cord, known collectively as the meninges. The inflammation may be caused by infection with viruses, bacteria, or other microorganisms, and less commonly by certain drugs.

Amnesia

Patients have trouble retaining long term memories. Difficulty creating recent term lost of memories is called anterograde amnesia and is caused by damage to the hippocampus part of the brain which is a major part of the memory process. Retrograde amnesia is also caused by damage to the hippocampus but the memories that were encoded or in the process of being encoded in long term memory are erased.

Aphasia

Aphasia is a breakdown in language skills.

Apraxia

Apraxia is a breakdown in the ability to translate a verbal command into its motor expression.

Agnosia

Agnosia is a breakdown in a person's ability to associate an object with its meaning, to recognize objects, people and him/herself.

SIGNS AND SYMPTOMS OF NERVOUS SYSTEM DISORDERS

Cognitive impairment

- *Executive functioning*
- *Information processing speed*
- *Attention*
- *Memory*
- *Language*
- *Visuospatial/visuoperceptual functioning*
- *General intellectual functioning*

Mood disorders

- Depression and
- anxiety.

Psychosis

Psychosis is one of the most disabling and distressing symptoms .

Sleep disorders

Specific sleep disorders include insomnia, hypersomnia, parasomnia, and rapid eye movement (REM) sleep behaviour disorder.

Fatigue

Fatigue is one of the most common, distressing and disabling non-motor symptoms.

Neuro-behavioural disorders

Neuro-behavioral problems such as hyper-sexuality, preoccupation with complex motor acts such as disassembling electrical equipment, hypomania and mania, aggression and heightened irritability, an urge to walk considerable distances without purpose, pathological gambling and shopping, and food cravings.

The following are the most common general signs and symptoms of a nervous system disorder. However, each individual may experience symptoms differently. Symptoms may include:

- Persistent or sudden onset of a headache
- A headache that changes or is different

- Loss of feeling or tingling
- Weakness or loss of muscle strength
- Sudden loss of sight or double vision
- Memory loss
- Impaired mental ability
- Lack of coordination
- Muscle rigidity
- Tremors and seizures
- Back pain which radiates to the feet, toes, or other parts of the body
- Muscle wasting and slurred speech

The symptoms of a nervous system disorder may resemble other medical conditions or problems.

PSYCHOTHERAPEUTIC INTERVENTION

Psychotherapy for primary depression and anxiety disorders is a widely accepted treatment option (National Institute for Health and Clinical Excellence, 2007). In particular, cognitive-behavioural therapy (CBT) for depression in older adults (Laidlaw et al., 2008) has been proven to be an effective evidence-based treatment.

Psychosocial adjustment

Psychosocial adjustment depends on the interaction of cognitive, behavioural, physical, personality and social factors.

EPILEPSY

Epilepsy is a brain disorder in which clusters of nerve cells, or neurons, in the brain sometimes signal abnormally. Epileptic activity within the brain has an effect on the behavior, mood, and cognitive functions of the epileptic patient. Additionally, his/her behavior and mood may be affected by the attitude of others to his/her disability. Furthermore, the patient's social and psychological adaptations may modify his/her epileptic experiences. As a consequence of the interaction between these aspects patients with epilepsy face many cultural and psychosocial problems (Betts et al., 1976). Problems related to areas such as overprotection, education, employment, marriage, child bearing, and psychiatric disturbances will vary according to the stage of epilepsy and the level of understanding in the society.

Psychological effects of being epileptic

To be epileptic is to be stressed and stress can influence the incidence of fits. Under stress many people develop anxiety symptoms which may be seen as a fight or flight reaction, some become depressed, and others show obsessive ritualistic behavior. There are those who develop temporary psychotic states or acute conversion (hysterical) reactions which effectively isolate them from the stressful situation (Betts et al., 1976). The literature published has identified general mood disturbance, depression, and pathological distress, which are independent of seizure control, as significant predictors of quality of life impairment among people with epilepsy (Gilliam et al., 2003). Anxiety related emotions, which are very common in the epilepsy
Population, are amongst the most pathogenic secondary consequences (Mitan, 1983).

The social effects of being epileptic

To be epileptic means being exposed to the fear of having attacks, being at a disadvantage in terms of work and personal relationships, being open to prejudice, this exists both in the lay public and in the

medical and nursing professionals. Fear of social exposure of fits and feelings of disgrace often lead to social isolation. Public ignorance and fear of the unknown is another important source for the epileptic patients' problems.

INTERVENTION

- **relaxation therapy-**
- **Cognitive behaviour therapy-** (CBT) to be effective in reducing depression, among people with epilepsy with a depressed affect. Cognitive behaviour therapy is widely used to treat anxiety and depression. It emphasizes the influence of thoughts and their content on emotional state and treatment focuses on changing thought patterns and behaviour. As anxiety and depression are commonly experienced by people with epilepsy,
- **Bio-feedback-** galvanic skin response biofeedback reported significant reduction in seizure frequency. EEG bio-feedback was found to improve the cognitive and motor functions in individuals with greatest seizure reduction.
- **Educational interventions-** Educational interventions were found to be beneficial in improving the knowledge and understanding of epilepsy, coping with epilepsy, compliance to medication and social competencies.
- **Family counseling**- It is considered as adjunctive treatment for epilepsy.

COMMON THERAPIES USED IN CEREBRAL PALSY TREATMENT

AQUA THERAPY

Aqua therapy (also known as Aquatic therapy), under the supervision of a trained and certified professional therapist, provides deep, intense exercise within a soothing and comforting environment. This form of therapy promotes physical functioning with the aid of water's restorative and detoxifying properties. Water buoyancy makes aerobic and anaerobic exercises safe and effective by allowing an individual to ambulate freely in a way that doesn't place undue stress on the musculoskeletal system from forces such as gravity and body weight. Aqua therapy takes place in both heated and non-heated environments, although warm water therapy provides a massage effect on muscles, joints and ligaments.

PHYSICAL THERAPY AND PHYSIOTHERAPY (REHABILITATION)

These forms of therapy are part of the branch of rehabilitative health devoted to restoring, maintaining, and promoting optimal movement, physical function, and health. Physical therapy and physiotherapy provide therapeutic intervention to treat activity limitations, participation restrictions, and environmental barriers experienced by those with physical impairment. They are considered two of the most important therapies for the treatment of cerebral palsy. Physical therapy or physiotherapy begins soon after diagnosis is made and is an integral part of early intervention and maintenance programs. They optimize physical functioning. They can incorporate functional training, manual therapy, assistive technologies, and electrotherapeutic modalities.

BEHAVIORAL THERAPY (PSYCHOTHERAPY)

Behavioral therapy, also known as cognitive behavioral therapy, is rooted in the belief that emotional challenges and unproductive behaviors are learned and can therefore be changed. In behavioral therapy, troubling situations are identified and thoughts, emotions and beliefs about those situations are explored, challenged and ultimately altered. Behavioral therapy empowers the individual to respond to challenging situations in a more effective and acceptable manner.

CONDUCTIVE EDUCATION

Conductive education is a comprehensive method of learning by which individuals with neurological and mobility impairment, like cerebral palsy, learn to specifically and consciously perform actions that children without such impairment learn through normal life experiences. Conductive education takes the position that motor disorders are learning disabilities, children are taught to see themselves as active participants in their own education; they are encouraged to be problem-solvers and develop a self-reliant "ortho-functional" personality that fosters participation, initiative, determination, motivation, independence, and self-sufficiency. Because of this, conductive education programs are not medically-based, but task oriented.

RECREATION THERAPY (THERAPEUTIC RECREATION)

Recreation therapy, also known as therapeutic recreation, focuses on designing ways in which an individual can fully participate in recreational activities of their choice. When children with impairment are presented with an obstacle to perform a life-enhancing activity, recreation therapists work to identify the interest level, capabilities, adaptive approaches, and in some cases modified processes required to complete the activity successfully. Inclusion in life-enhancing activities improves a child's physical, mental and social experiences.

SENSORY INTEGRATION THERAPY

Children with cerebral palsy or those born prematurely can display symptoms of sensory integrative dysfunction, an abnormal degree of sensory processing, whether decreased or increased. Sensory integrative dysfunction can make a child appear clumsy when he or she walks, or create difficulty with fine motor skills such as pencil control, which, in turn, can lead to delay in writing ability. Sensory dysfunction can also cause over-sensitivities or under-sensitivities that lead to mental and physical distraction or fatigue. Problems with sight, hearing and balance can be addressed through sensory integration therapy.

SOCIAL THERAPY

Children with impairment may experience any number of social challenges, such as social anxiety, relationship problems, dependent personality disorders or social skill difficulties. Anxiety, panic, shyness, or emotional pain can be replaced with emotional growth and new ways of perceiving, thinking and interacting. Social therapy can provide a child with new skills to use towards healthier relationships with others and within their surroundings. The effect of social therapy can be empowering. Often, pain is recognized and released. The child then develops new ways to respond to people, situations and moments in his or her life.

HIPPOTHERAPY

Hippotherapy is a form of physical, occupational and speech therapy that uses equine (horse) movement to develop and enhance neurological and physical functioning by channeling the movement of the horse. This further develops physical and cognitive abilities. Hippotherapy is not to be confused with therapeutic horseback riding, in which individuals are taught specific riding skills. Hippotherapy is built on the concept that the individual and variable gait, tempo, rhythm, repetition and cadence of a horse's movement can influence human neuromuscular development in humans. Horseback riding triggers a

series of complex physical and mental reactions; such as making physical adjustments to maintain proper alignment on the horse. Riders must also plan movements to maintain balance on the horse, and be able to interact with the animal.

SPEECH AND LANGUAGE THERAPY

Language, speech, breathing, chewing and swallowing difficulties can be present with various health conditions, including brain injury, cerebral palsy, cognitive impairment, hearing loss, and learning disabilities. Oral motor dysfunction, facial muscle impairment, is common in children with cerebral palsy. Oral motor functioning depends on an intricate process of sending and receiving messages to various facial, throat and neck muscles to coordinate breathing while talking, chewing, swallowing and digesting.

SPINAL CORD INJURY

The rehabilitative care may include:

- Assistive (or Adaptive) technology devices (assistive devices, such as wheelchairs, braces and special computer equipment, and technology) for easier home and work activities
- Augmentative communication assessment and training to determine communication needs and appropriate aids and techniques
- Aquatic therapy (rehabilitation exercises performed in a warm water therapeutic pool)
- Arts therapy (uses the creative process to help patients improve physical skills, thinking skills and emotional well-being)
- Cognitive therapy (helps with thinking and understanding skills)
- Functional electrical stimulation (uses low levels of electrical current to stimulate physical or bodily functions lost through paralysis)
- Locomotor training (LT) (a method of physical therapy where the patient is suspended in a harness over a treadmill, while specially-trained therapists move their legs to simulate walking)
- Medications for cognitive, communication and motor skill recovery
- Neurological diagnostic tests (such as CT scan, MRI and EMG)
- Neuropsychological evaluations (measure concentration, learning and other skills)
- Pain management (care that alleviates or reduces pain)
- Rehabilitation psychology (behavioral and psychological treatment for patients and families coping with chronic illness, chronic pain and disability)
- Occupational therapy (helps patients participate in activities of daily living)
- Physical therapy (helps patients improve strength, mobility and fitness)
- Optokinetic training (exercises and activities to increase visual field and visual perception)
- Real world simulations (such as community reentry and independent living apartment)
- Respiratory therapy (treatment of breathing disorders using respirators or aerosol medications)
- Recreational therapy (helps patients engage in recreational and leisure activities to enhance change to community participation and satisfaction)
- Vision evaluation and vision therapy (therapy for the eyes and brain)
- Counseling

STROKE×××

Forty percent of stroke patients are left with moderate functional impairments and 15% to 30% with severe disability. Effective rehabilitation interventions initiated early after stroke can enhance the

recovery process and minimize functional disability. Improved functional outcomes for patients also contribute to patient satisfaction and reduce potential costly long-term care expenditures.

In either situation, rehabilitation can address a number of issues including:

- Fall prevention and home safety training
- Managing medications
- Self-care skills and mobility
- Strengthening exercises
- Adaptive equipment
- Nutrition counseling
- Patient and family education
- Fatigue and endurance
- Swallowing problems
- Impaired memory, problem solving or behavioral issues
- Communication and cognitive problems
- Respiratory impairments
- Balance problems
- Equipment needs

HOW TO IDENTIFY COGNITIVE IMPAIREMENT

- **CT and MRI scans.**
- **Electroencephalogram (EEG).**

What is NEUROPSYHOLOGICAL ASSESSMENT?

In order to understand and quantify the effects of brain on intellectual, motor or emotional function, a clinician administers a set of tests hat is designed to measure these effects.
This procedure is called Neuropsychological assessment and often complements the Neurological examination which assesses the patient's central nervous system.

Some common neuropsychological tests

- **Mini-Mental State Exam (MMSE):**
- **Wechsler adult intelligence scale (WAIS)**
- **P.G.I Battery of Brain Dysfunction (PGI – BBD):**
- **The Bender Visual Motor Gestalt test**
- **Halstead-Reitan Neuropsychological Battery (HRNB**
- **Luria-Nebraska Neuropsychological Battery (LNNB):**
- **Raven's Progressive and Coloured Progressive Matrices:**
- **Recognition Memory Test (RMT) (NFER – Nelson**
- **Card-sorting (for example, the Wisconsin Card Sorting Test):**

TREATMENT

- **PHARMACOLOGICAL TREATMENT**
- **PSYCHOLOGICAL TREATMENT**

Psychological Intervention and rehabilitation

Cognitive rehabilitation (CR) aims to enable people with cognitive impairments to achieve their optimum level of well-being by helping to reduce the functional disability resulting from damage to the brain. Central to this process is the collaborative identification of personally-meaningful goals and the

development of interventions to address these goals. CR interventions draw upon a mixture of approaches aimed at restoration of function, implementation of compensatory strategies and environmental modification, and these can be integrated with approaches directed at dealing with the emotional responses to impairment and other psychosocial difficulties to provide an holistic rehabilitation framework. While initially developed primarily for people with non-progressive brain injury, this approach is equally applicable to people with progressive conditions such as Alzheimer's disease (AD).

What is cognitive rehabilitation?

Rehabilitation involves 'enabling people who are disabled by injury or disease to achieve their optimum physical, psychological, social and vocational well-being' (McLellan 1991, p. 785). Cognitive rehabilitation applies this concept specifically to people with cognitive impairments (Mateer 2005), taking into account the particular needs and challenges resulting from damage to the brain. Cognitive rehabilitation interventions aim 'to enable clients or patients, and their families, to live with, manage, by-pass, reduce or come to terms with deficits precipitated by injury to the brain' (Wilson 1997, p. 488). Where the neurological damage cannot be ameliorated, there is still scope to reduce the resulting disability (limitations on engaging in activity) and handicap (restrictions on social participation) (World Health Organisation 1980, 1998), and to address any excess or unnecessary disability arising from secondary responses to the situation, for example depression or loss of self-esteem and self-confidence.

Cognitive rehabilitation interventions may involve a mixture of approaches aimed at restoration of function, implementation of compensatory strategies and environmental modification, and these can be integrated with approaches directed at dealing with the emotional responses to impairment (Mateer 2005) and other psychosocial difficulties to provide an holistic neuropsychological rehabilitation framework.

Why is cognitive rehabilitation relevant for people with dementia?

While initially developed for people with non-progressive brain injury, cognitive rehabilitation is equally applicable to people with progressive conditions such as Alzheimer's disease and other dementias, at any stage or degree of severity, although the focus will differ as priorities change over the course of the disorder and may shade into a palliative care approach towards the end of life (Clare 2008; Wilson 2008). It is appropriate to think of dementia in terms of disability rather than simply disease. The manifestation and progression of dementia in any one individual are influenced by a number of factors alongside the progression of neurological impairment; these include personality, biographical experience, social relationships, communication and interaction, and environmental context (Kitwood 1997).

What kind of assessment is needed?

Implementing cognitive rehabilitation requires a careful assessment of the person's neuropsychological profile to establish current strengths and difficulties. This will be important in selecting specific methods and strategies. For example, awareness of the presence of perceptual difficulties would suggest caution in the use of visual cues. A general knowledge of the likely pattern of impaired and relatively preserved functions for the given stage of dementia is valuable; for example it can be anticipated that episodic memory will generally be severely affected in the early stages of Alzheimer's disease, while procedural memory will be relatively well preserved.

Once a goal has been agreed, the therapist can then select from available evidence-based strategies in order to devise a personalised intervention. The following brief overview outlines some of the most frequently-used strategies:

1. Facilitating remaining episodic memory functioning. Where the aim is to build on the remaining episodic memory ability to encourage learning of important new information, or re-learning of previously-known information, a number of guiding principles can be followed.
2. Supporting procedural memory. Where the aim is to improve or restore the ability to carry out selected activities of daily living, action-based learning can be used (Hutton, Sheppard, Rusted, and Ratner 1996). Prompting methods can be used to encourage and support performance of an activity.
3. Supporting semantic knowledge. Approaches used with people who have semantic dementia include repeated rehearsal combined with contextual information (Reilly, Martin, and Grossman 2005; Snowden and Neary 2002), and demonstration of object use (Bozeat, Patterson, and Hodges 2004).
4. Addressing risky or problematic behaviours. This is especially relevant for people with frontal dementia, and a range of behavioural and environmental modifications can be employed to address problematic behaviours while maintaining the person's quality of life (Lough and Hodges 2002).

The process of setting and addressing goals proceeds through a number of stages, which can be summarized as follows:

1. Determine whether the person is able to identify something that s/he would like to be different.
2. Identify the area to focus on – for example, memory difficulties, participation in activities, or family relationships,
3. Identify the specific issue to focus on – for example, remembering the names of people met when participating in an activity,
4. Establish the baseline level of performance.
5. Identify the goal expressed in clear behavioural terms.
6. Identify what level of performance will indicate that the goal is
    1. wholly or
    2. partially achieved.
7. Plan the intervention to address the goal, using appropriate methods and techniques.
8. Implement the intervention, with appropriate attention to emotional responses and contextual issues.
9. Monitor progress and adjust the intervention if necessary.
10. Evaluate the outcome of the intervention and decide on any further steps to be taken.

The intervention addresses six areas:

1. *Rehabilitation goals*. The aim is to identify and work on one or two personal rehabilitation goals. Goals are identified at the start of therapy by following the process outlined above, and an individual approach to addressing these goals is designed and implemented over the course of the sessions. Work on personal goals forms the central element of the program and continues throughout all sessions, supported by the other components of the program.
2. *Practical strategies*. This component involves a review of the participant's current use of memory aids and practical coping strategies, explores whether it might be possible to build on these to promote more efficient use, and supports the introduction of new aids or strategies where appropriate (Bourgeois 1992).
3. *Memory*. The therapist introduces techniques for learning new associations and information, provides practice in these, identifies the person's preferred strategy, and encourages the wider application of this strategy in everyday life.

4. *Attention and concentration*. Practice is provided in maintaining attention and concentration while processing information, drawing on rehabilitation methods devised for people with impairments of executive function (Levine et al. 2000).
5. *Stress management*. The person's current ways of coping with stress and anxiety are explored, suggestions are made to build on these strategies, and relevant practice is provided using simple stress management and relaxation techniques (Suhr, Anderson, and Tranel 1999).
6. *Family involvement (where possible)*. If the participant has a spouse or other family member or carer, and accepts his/her involvement, that person is invited to join the last 15 minutes of each session. This part of the session is devoted to reviewing the content of the session, agreeing the home practice to be undertaken in preparation for the next session, and discussing ways of facilitating progress with the personal rehabilitation goals. Family involvement helps to ensure the intervention has maximum impact in the everyday context.

COGNITIVE RETRAINING- This is commonly used technique for the patient with central nervous disorder.

Cognitive retraining aims to enhance the cognitive functions of a patient with cognitive deficits. Cognitive retraining tasks aim to improve various cognitive functions like attention and concentration, information processing, planning and organization, memory, response inhibition.etc. of the patient.

Before introducing cognitive retraining tasks, it is recommended to assess the patient to understand areas of cognitive deficits of the patient. Tasks are introduced to the patient based on the need of the patient. Some of the tasks which can be used in cognitive retraining are mentioned below:

Attention:

Letter cancellation tasks:

These tasks targets sustained, focused and divided attention. The task was to cancel any two given letters placed randomly in an array of letters. Time taken in seconds to complete, errors of omission and commission has to record for each day.

Sustained attention:

Letters of the English alphabet in lower case of any fixed font size randomly provide on an A4 sheet .a specific number of rows and number of letters should be maintained with single spacing between each row.

Focused attention:

A passage from a book can be selected with each row having an average of 15 words with appropriate font size, and rows .Two letters can be the target to cancel and time taken to cancel with errors committed are recorded.

Divided attention

A group of nine symbols can be placed randomly on an A4 sheet .There should be 30 symbols in each row with no space between symbols and a total of 50 such rows. The time taken to cancel any two symbols each day and errors committed should be recorded. This targeted the divided attention as verbal fluency task can also give simultaneously. The examiner can randomly select a letter and patient can be instructed to tell orally as many words as possible starting from the letter introduced by the therapist.

Verbal fluency task:

Therapist has to provide at least two alphabets to the patient and ask the patient to tell as many words as possible starting with the letter provided by the therapist for a period of 2 minutes. Average number of words produced by the patient in each session has to be recorded.

Information processing:

Grain sorting task (Amita Mishra, 1994): This task targets information processing. Patient has to sort the grains. The difficulty level of the task can be increased as the patient reach improvement. Grains use for this test should match the concepts of size, colour and shape.

Card sorting task:

Task includes two decks of playing cards with only number cards. The cards with faces are not included.The therapist has to keep a rule and patient is not informed about it. Patient is instructed to place the cards into groups and try getting the therapist has at the earliest. Patient can be provided with feedback by the therapist.

Planning and organization:
Koh's Block design test
Planning:
Planning task of a real life situation:
Memory:
Hashar & Zacks(1979) has adopted four tasks

1. **Elaborate encoding:**
   The aim of this task is to improve the depth of encoding. A list of words has to compile. Start with a baseline of 10 words. For each word presented the patient has to produce an association word. At the end of the list, the patient has to recall the words with the help of association words.
2. **Frequency encoding**
   Here the task was the auditory presentation of lists of common words.
   At level 1: a list of six words was presented only once. After the presentation the patient was asked to report target stimulus and how many times it was repeated.
   At level 2, 3, & 4 the number of words can also increase to a fixed number and target stimuli can also be increase.
3. **Spatial encoding**
   This task consisted of visual presentation of common objects (pen, watch, key, pencil etc) .The objects were kept in front of the patient and after 15 seconds of exposure the objects covered. Then the patient has to name the objects and their spatial location. The difficulty of the task was increased by increasing the number of objects presented. Number of objects and their locations correctly given was the score.
4. **Temporal encoding**
   This task consisted of oral presentation of words. The list was divided into three groups.ie.begining words, middle words and last words. The words were read out at the rate of 2 seconds per word. After the presentation ,the patient was told word from the same list and patient has to tell whether the word belong to first,second,or third category. The list of words has to change to avoid the effect of practice.The complexity of the task has to increase by increasing the size of the list.Number of correct responses were the score.

IMMEDIATE VISUAL MEMORY

Expose patient a design for 10 seconds and then ask the patient to reproduce it after taking the design back.

Response Inhibition:

Patient has to colour a given design. Colour pencils can be used. Response inhibition was hypothesized to the process mediating the task in addition to sustained attention. The patient has to regulate the pressure and the direction of strokes while colour ing to attain neatness in colour ing task. The time taken has to be recorded. Stroop task with movement:
This task targeted response inhibition. In this task patient has to perform a pattern of motor action. For eg. Clapping one's hand, lifting the arm and letting it down.
The therapist then suddenly gives a stop signal; patient is instructed to stop performing the particular task immediately and restart when he hears therapist gives start signal. The start-stop signal is given randomly. The patient has to perform this task at least five minutes a day during cognitive retraining.

Unit –III: Cardiovascular system: Psychosocial, personality, lifestyle, and health practice issues, psychobehavioural responses including coping with illness and functional loss in hypertension , MI, following CABG and other cardiovascular conditions, salient issues with regard to quality-of-life and well-being, empirically proven methods of psychological management of CVS diseases.

CARDIOVASCULAR SYSTEM:

The cardiovascular system comprises the blood, the blood vessels of the circulatory system, and the heart. The heart which is about the size of a clenched fist and weighs on average only about 11 ounces, consists of 3 layers of tissue: a thin outer layer,called the ***epicardium,*** a thin inner layer called the ***endocardium***, and a thicker middle layer, the heart muscle itself, or ***myocardium*** { derived from the Greek roots ***myo*** (muscle) & ***kardia*** (heart) }. The myocardium is separated into four chambers that work in coordinated fashion to bring blood into the heart and then to pump it throughout the body. Like all muscles in the body the myocardium requires a steady supply of oxygen and nutrients to remain healthy. And the harder the heart is forced to work to meet the denmands of other muscles in the body , the more nutrients and oxygen it needs.

The heart's blood supply comes from two branches of the ***aorta*** (the major artery of the heart) lying on the surface of the epicardium. These left and right *coronary arteries* branch into smaller and smaller blood vessels called ***arterioles*** until they become the capillaries that supply the myocardium with the blood it needs to function.

CARDIOVASCULAR DISEASE:

When the blood supply from the coronary arteries is impeded beyond a critical point, the risk of developing cardiovascular disease increases substantially. About 60 million Americans suffer from some kind of disorder of the heart and blood vessel system, collectively referred as **cardiovascular disease (CVD). CVD** appears in many guises, including stroke and **coronary heart disease(CHD),** a chronic illness in which the arteries that supply the blood become narrowed or clogged and cannot supply enough blood to the heart. The underlying physical causes of these diseases are: **Atheroscelorosis** and **Arteriosclerosis.**

The Causes: Atherosclerosis and Arteriosclerosis

Most cases of CVD result from **atherosclerosis,** a condition in which the linings of the arteries thicken with an accumulation of cholesterol and other fats. As these a***therosclerotic plaques*** develop, the arterial passageways become narrowed, impeding the flow of blood through the coronary arteries. Although

plaques tend to develop in most people in their thirties and forties, these plaques will not threaten their health- at least not until age 70 or older.

Inflammation in the circulating blood (systemic inflammation)can contribute to the development of atherosclerosis and help trigger heart attacks & strokes. Inflammation is the body's response to injury, and blood clotting is often part of that response. Although researchers are not certain what causes the low-grade inflammation that seems to put otherwise healthy people at increased risk for atherosclerosis, many believe that a chronic bacterial or viral infection might be the underlying cause (Desvarieux and others, 2003).One of the proteins that increases during the inflammatory response, C-reactive protein (CRP), is increasingly being used to assess a person's risk of CVD. The risk for heart attack in people with the highest CRP levels is twice that of people whose CRP levels are at the lowest levels (Genest, 2004; Mazer & Rabbani, 2004).

Closely related to atherosclerosis is **arteriosclerosis, or "hardening of the arteries".** In this condition, the coronary arteries lose their elasticity , making it difficult for them to expand and contract. This makes it difficult for them to handle the large volumes of blood needed during physical exertion. In addition, a blood clot is much more likely to form in, and block, a coronary artery that has lost its elasticity due to arteriosclerosis.

Risk factors for CHD include high blood pressure, diabetes, cigarette smoking, obesity, high serum cholesterol level, and low levels of physical activity (American Heart Association, 2004).Identifying patients with metabolic syndrome also predict heart attacks. Metabolic syndrome is diagnosed when a person has three or more of the following problems: obesity centered around the waist; high blood pressure; low levels of HDL, the so-called good cholesterol; difficulty metabolizing blood sugar, an indicator of risk for diabetes; and high levels of triglycerides, which are related to bad cholesterol. High cardiovascular reactivity may also be a component of this cluster (Waldstein & Burns, 2003). Heart disease has a family history component, being more common among the offspring of people who have heart disease. This component may include a genetically based predisposition to cardiovascular reactivity, which may emerge early in life (Boyce, Alkon, et al.,1995; Yamada et al.,2002) and which is exacerbated by lifestyle-related risk factors, including exposure to stress. However, taking all known risk factors together accounts for less than half of all newly diagnosed cases of CHD; accordingly, a number of risk factors remain to be identified, which could target people who are at risk for CHD early in the disease process.

THE DISEASES : ANGINA PECTORIS, MYOCARDIAL INFARCTION, AND STROKE:

Left unchecked, atherosclerosis and arteriosclerosis may advance for years before a person experiences any symptoms. Once the process gets, underway, however, the risk of developing one of three diseases increases with time.

The first begins with a gradual narrowing of the blood vessels. Any part of the body that depends on blood flow from an obstructed artery is subject to damage. For example, if the narrowing affects arteries in the legs, a person may experience leg pain while walking. When the arteries that supply the heart are narrowed with plaques, restricting blood flow to the heart – a condition called **ischemia** – the person may experience a sharp, crushing pain in the chest, called **angina pectoris**. Although most angina attacks usually pass within a few minutes without causing permanent damage, ischemia is a significant predictor of future coronary incidents.

Although angina attacks can occur anytime- including while a person is sleeping – they typically occur during moments of unusual exertion, because the body demands that the heart pump more oxygenated blood than it is accustomed to handling – for example, when a casual runner tries to complete a 26- mile marathon. Angina may also occur during strong emotional arousal or exposure to extreme cold or heat.

Mental stress during daily life, including feelings of tension, frustration, and sadness, increases the risk of ischemia (Rosenfeldt and others,2004).

The second, much more serious cardiac disorder occurs when a plaque ruptures within a blood vessel, releasing a sticky mass that can further reduce blood flow or even obstruct it completely. Within seconds of the complete obstruction of a coronary artery, a heart attack , or **myocardial infarction (MI),** occurs, and a portion of the myocardium begins to die **(an infarct is an area of dead tissue).** Unlike angina, which lasts only a brief time, MI involves a chronic deficiency in the blood supply and thus causes permanent damage to the heart.

The third possible manifestation of cardiovascular malfunction is cerebrovascular disease, or **stroke.** They are the third leading cause of death, after myocardial infarctions and cancer (Arias and others, 2003). The most common type of stroke – **ischemic stroke** – occurs when plaques or a clot obstruct an artery, blocking the flow of blood to an area of the brain. **Hemorrhagic stroke** occurs when a blood vessel bursts inside the brain, increasing pressure on the cerebrum and damaging it by pressing it against the skull. Hemorrhagic stroke is associated with high blood pressure, which stresses the artery walls until they break, or exposes a weak spot in an artery wall (***aneurysm***) that balloons out because of the pressure of the blood circulating inside.

The effects of stroke may include loss of speech or difficulty understanding speech, numbness, weakness or paralysis of a limb or in the face, headaches, blurred vision, and dizziness. Strokes usually damage neural tissue on one side of the brain, with a resulting loss of sensation on the opposite side of the body.

HYPERTENSION

Hypertension, or high blood pressure, occurs when the supply of blood through the vessels is excessive. It can occur when cardiac output is too high, which puts pressure on the arterial walls as blood flow increases. It also occurs in response to peripheral resistance – that is, the resistance to blood flow in the small arteries of the body.

Hypertension is assessed by the levels of systolic and diastolic blood pressure as measured by a sphygmomanometer. **Systolic blood pressure** is the greatest force developed during contraction of the heart's ventricles. **Diastolic pressure** is the pressure in the arteries when the heart is relaxed; it is related to resistance of the blood vessels to blood flow.

CABG – In **coronary artery bypass graft** surgery, an incision is made in the patient's breastbone, and a small piece of a vein is removed from elsewhere in the body (typically from a leg but sometimes from an arm or the chest) and grafted around the region of a blocked or narrowed artery.The bypass allows blood to circumvent the blockage and flow more freely to the undernourished section of myocardium. Bypass surgery is typically recommended when blockages are severe and when the patient has not responded to other forms of treatment.

PSYCHOSOCIAL FACTORS IN CARDIOVASCULAR DISEASE: The Type A Personality

In their efforts to identify psychological risk factors for cardiovascular disease, behavioural scientists initially identified a behavioural style termed the Type A behavior pattern. The Type A behavior pattern was formulated by two physicians (M.Friedman & Rosenman, 1974) as a behavioural and emotional

style marked by an aggressive, unceasing struggle to achieve more and more in less time, often in competition with other individuals or forces. The Type A behavior pattern was thought to be characterized by three components: easily aroused hostility, a sense of time urgency, and competitive achievement strivings. So-called Type Bs, with whom Type As are frequently compared, are less driven individuals who do not show these behavior patterns.

In the 1960s and 1970s, hundreds of studies supported the association between Type A behavior and risk of future CVD in both men and women. In an effort to explain this relationship, researchers have focused on physiological differences between Type A and Type B people.Among their findings: Type A people have more rapid blood clotting and higher cholesterol and triglyceride levels under stress than their Type B counterparts (Lovallo & Pishkin,1980).Type A people also often display greater autonomic arousal, elevated heart rate, and higher blood pressure in response to challenging events (Jorgensen and others, 1996).

A secondary cluster of traits combining depression with high levels of anxiety – both of which are independently linked to hypertension and CVD (Rutledge & Hogan, 2002) – has also been observed in Type A individuals (Frasure- Smith and others, 1995).Depression appears to be particularly lethal, as shown by a 12- year Centers for Disease Control study (Anda and others, 1993).Depressed people were more vulnerable to CVD, even after the researchers controlled for differences in age, gender, smoking and other CVD risk factors. Moreover, depressed people have four times the risk of recurrence of a heart attacks compared to MI survivors with no depressive symptoms (Frasure- Smithand others, 1999).

Believing that the Type A syndrome was too global , researchers began to analyze component behaviours, including competitiveness, hostility, time urgency and anger, to determine whether one or more of these components might more accurately predict CVD.

Competitiveness, Hostility, and Time Urgency

In a probe of the key variables in the Type A – CVD relationship, researchers used data from the 20- year Coronary Artery Risk Development in Young Adults (CARDIA) study of over 5000 women and men between the ages of 18 and 30 years to investigate the relationships among three components of Type A behavior (time urgency, hostility, and competitiveness) and risk of developing hypertension. Regardless of participants'age, sex, race, education, body mass index, alcohol consumption, and fitness levels, those characterized by the highest levels of two variables – time urgency and hostility (but not competitiveness) – had an 80 percent greater risk of developing hypertension (Lijing and others, 2003).

Lately, researchers have focused most closely on hostility as the possible "toxic core" of Type A behavior. Redford Williams and his colleagues administered a questionnaire called the Cook- Medley Hostility Scale (Ho Scale) to a large group of coronary patients. They found a striking correlation between patients' scores on the questionnaire and the severity of blockage in their coronary arteries. Hostile patients had significantly more severe coronary artery blockages than did less hostile patients (Williams, 1996).People with the highest scores on the Ho Scale are more than 1.5 times as likely to suffer an acute MI as are people with the lowest scores (Barefoot and others, 1995).

Although hostility is related to CVD mortality , its status as an independent risk factor for cardiovascular disease continues to be debated (Smith & Gallo, 2001). Hostility is also related to other behaviours that promote CVD, include obesity hypertension, alcohol and tobacco use, negative life events, and little social support (Siegler and others,1992). Children and adolescents who exhibit high hostility scores are also more likely to develop the metabolic syndrome than those who score low on measures of hostility (Raikkonen and others,2003). These relationships have caused researchers to narrow their investigations to a specific component of hostility – anger.

Anger

Strong negative emotions such as anger may be as dangerous to the heart as smoking, a high fat diet, or obesity. By one estimate, 20 percent of fatal MIs occur in response to an angry outburst (Ferroli, 1996). In the massive Atherosclerosis Risk in Communities study, 256 of the 13,000 middle-aged participants had heart attacks. Janice Williams and her colleagues (2000) found that people who scored highest on an anger scale were three times more likely to have a heart attack than those with the lowest scores. People who scored in the moderate range on the anger scale were about 35 percent more likely to have a heart attack. This elevated risk was true even after taking into account the presence of other risk factors such as smoking, diabetes, elevated cholesterol and obesity.

Arguments with family members were the most frequent cause of anger, followed by conflicts at work and legal problems (Hilbert, 1994).

Suppressed anger may be as hazardous to health as expressed anger. James Pennebaker (1992) has developed a general theory of inhibition that is based on the idea that to hold back one's thoughts or feelings requires work that, over time, results in levels of stress that can create or exacerbate illness. In support of this theory, cardiac patients who deny their anger or frustration are 4.5 times more likely to die within 5 years than are other cardiac patients (Bondi, 1997).Suppressed anger was an even stronger predictor of mortality than elevated cholesterol level or cigarette smoking.

Why Do Hostility and Anger Promote Cardiovascular Disease?

An angry, hostile personality predicts an increased risk of CVD, but how do these traits work their damage ? The key theoretical models differ in their relative emphasis on biological, psychological, and social factors.

Psychosocial Vulnerability

Some theorists maintain that hostile adults lead more stressful lives and low levels of social support, which, over time, exerts a toxic effect on cardiovascular health. In support of this *psychosocial vulnerablility hypothesis*, researchers have found that chronic family conflict, unemployment , social isolation, and job-related stress are all linked to increased risk of CVD (Kop and others, 2001).

The Work Environment

The work environment can be an important source of satisfaction or stress (Mills and others, 2004). Jobs associated with high productivity demands, excessive overtime work, and conflicting requirements accompanied by little personal control tend to be especially stressful. Data from the 20-year CARDIA study reveal that ***job strain,*** predicts the incidence of hypertension, even after adjusting for baseline blood pressure, education, body mass index and age (Markovitz and others, 2004).

Social Support

Coping with stressful events is especially difficult when an individual feels cut off from others. William Ruberman and his colleagues (1984) found that 3 years after surviving an acute MI, those with a combination of high stress and social isolation had four times the death rate of people with low stress and strong social support.

Living alone after suffering a heart attack is also associated with a higher risk of a recurrence of CHD. Redford Williams (1996) found that coronary disease patients who were unmarried and / or had no one with whom they could share their innermost concerns were three times more likely to die in the next 5 years than patients who had a confidante – a spouse or a close friend.

The Health Behaviour Explanation

Some researchers believe that hostility , anger, job strain , and isolation may also have an indirect effect on health. For example, people with poor support may not take care of themselves as those who have someone to remind them to exercise, eat in moderation, or take their medicine. Similarly, a person a cynical attitude may perceive health- enhancing behaviours, such as adhering to a healthy diet and active lifestyle, as unimportant and may ignore warnings about smoking and other health- compromising behaviours. Hostility and anger have indeed been linked with excessive alcohol and caffeine consumption, greater fat and caloric intake, elevated LDL cholesterol, lower physical activity, greater body mass, hypertension, sleep problems, and nonadherence to medical regimens (Miller and others, 1996).

Psychophysiological Reactivity Model

Stress, hostility, and anger may act slowly over a period of years to damage the arteries and heart. When we vent our anger, our pulse quickens, the heart pounds more forcefully, and blood clots more quickly. In addition, blood vessels constrict, blood pressure surges, and blood levels of free fatty acids increase.Our immunity also decreases as adrenaline, cortisol, and other stress hormones suppress the activity of disease-fighting lymphocytes.

To pinpoint the physiological bases of hostility, researchers have studied hostile men and women who were harassed while trying to perform a difficult mental task. The stress caused an unusually strong activation of the fight- or- flight response in these people. When challenged, they displayed significantly greater cardiovascular reactivity (CVR) in the form of larger increases in blood pressure and greater outpourings of epinephrine, cortisol and other stress hormones (Kop & Krantz, 1997).

The Biopsychosocial Model

Health psychologists have combined the insights of these findings to provide a biopsychosocial explanation for how hurriedness, hostility, and anger contribute to cardiovascular disease. This model suggests that in order for a chronic disease such as CVD to develop, a person must first have a physiological predisposition. This is determined by family history of CVD and previous health history (other diseases, poor diet, tobacco use, and so on). Whether or not CVD develops then depends on a variety of psychosocial factors in the person's life, including the level of stress from the work and home environments, and the availability of social support. Hostile individuals with a strong sense of time urgency tend to elicit aggressive behaviours from others ,producing interpersonal conflict and more hostility. This in turn leads to a reduction in social support, more negative affect, and artery-damaging cardiac reactivity. Thus, hostile attitudes create a self-fulfilling prophecy for the mistrusting, hostile person by producing a hostile environment.

Other Psychosocial Risk Factors and CHD

Vigilant coping – that is, chronically searching the environment for potential threats - has also been associated with risk factors for heart disease (Gump & Matthews, 1998). Anxiety has been implicated in sudden cardiac death, perhaps because anxiety appears to reduce vagal control of heart rate (L.L.Watkins, Grossman, Krishnan,& Sherwood, 1998).

Recently, researchers have explored whether social dominance contributes to risk for coronary heart disease. Social dominance reflects a pattern of attempting to dominate social interactions through verbal competition, a fast speaking rate, and the tendency to jump on other people's responses before they have had a chance to finish. Evidence suggest that social dominance may be related to all-cause mortality (Houston, Babyak, Chesney, Black, & Ragland,1997), and it may be especially related to mortality due to coronary heart disease.

Investigators have related vital exhaustion, a mental state characterized by extreme fatigue, a feeling of being dejected or defeated, and enhanced irritability to cardiovascular disease(Wirtz et al.,2003); vital exhaustion, in combination with other risk factors, predicts the likelihood of a heart attack (Bages, Appels, & Falger, 1999) and of a heart attack after initial recovery (Kop, Appels, et al.,1994).

Psychological Management of CVS Diseases

Managing Angina

Angina pectoris, or chest pain, occurs because the muscle tissue of the heart must continue its activity without a sufficient supply of oxygen or adequate removal of carbon dioxide and other waste products, and can often be triggered and exacerbated by stress, emotional extremes and physical exertion. Angina has been shown to be associated with depression, anxiety , neuroticism, type A behavior and raised levels of somatic awareness. Assessments should emphasise these points, using self-report measures when available. Recent episodes hould be reviewed in detail, with particular attention being devoted to the duration of the episode, the reaction of the patient and significant others, as well as their medication use. Activity levels are often implicated as key mediators of problems with physical deconditioning among patients with angina. Inactivity tends to lower the anginal threshold (Lewin, 1997).Many of the concepts which are part of the cognitive behavioural management of chronic pain can be used in the assessment, conceptualization and management of angina. The use of pain diaries is particularly relevant and help patients and therapists to identify the key influences on the experiences of angina. Problem-level formulations of avoidance of activity often specify the primary mediating role of predictions about worsening chest pain. Case level formulations should aim to link neuroticism, avoidance and anginal interference with activities. Exercise has been shown to result in significant benefits for patients with angina; however, many patients do not feel able to engage in exercise because of beliefs which promote avoidance related to 'taking it easy'. This is often made more difficult by the fact that misconceptions may be shared by health care staff (Lewin, 1999).

CBT is helpful in enabling patients to modify thoughts and beliefs which block access to exercise, and patients can use graded task assignment and activity scheduling to build exercise into their routine. Patients may avoid returning to work because of overcautiousness, which is usually manifest as thoughts about overexertion and links with death or an acute cardiac event. Research also suggest that interventions designed to promote relaxation and the management of stress are beneficial. Cognitive mechanisms relating to patient experience of stress can be useful as ways of structuring intervention. Whatever variables are targeted in CBT, angina and activity levels should be monitored as key outcome variables.

Cognitive – Behavioural Treatments for Hypertension

Cognitive–behavioural modification techniques are increasingly being used in the treatment of hypertension.

A variety of behavioural and cognitive-behavioural methods have been evaluated for their potential success in lowering blood pressure(M.S.Glasgow & Engel,1987).Methods that draw on relaxation include biofeedback, progressive muscle relaxation,hypnosis and meditation, all of which reduce blood pressure via the induction of a state of low arousal.Deep breathing and imagery are often added to accomplish this task.Evaluations of these treatments suggest modestly positive effects (Nakao, et al.,1997), although hypertensive patients may not practice them as much as they should(Lichstein & Rosenthal, 1986).Giving patients feedback about exactly how their compliance efforts are can improve blood pressure control (Zuger, 1999).

Other stress management techniques have also been employed in the treatment of hypertension.Such techniques train people to identify their particular stressors and to develop plans for dealing with them.

The programs include ***training in self-reinforcement, self-calming talk, goal setting anf time-management.***

The fact that anger has been tied to hypertension implies that teaching people how to manage their anger might be useful.Infact, studies suggest that training hypertensive patients how to manage confrontational scenes through such ***behavioural techniques as role playing*** can produce better skills for managing such situations and can lower blood pressure reactivity (Larkin & Zayfert,1996).

CARDIAC REHABILITATION

Cardiac Rehabilitation is defined as the active and progressive processby which individuals with heart disease attain their optimal physical, medical, psychological, social, emotional, vocational and economic status (Dracup,1985).

The goals of rehabilitation are to produce relief from sytmptoms, to reduce the severity of the disease, to help patients to feel confident in making the lifestyle changes which can reduce their risk of further problems and enhance their quality of life.It may also have a beneficial impact on health service utilization factors such as rehospitalisation rates (Black et al.,1998).

Assessment

Patients should be assessed with specific regard to the predominant components of their cardiac illness representation, its relationship to their lifestyle choices premorbidly, and how this might relate to which risk factors to target during rehabilitation.Beliefs about heart disease are central to understanding cardiac-related disability and quality of life (Lewin,1999).Much of the emphasis of cardiac rehabilitation involves enabling patients to appreciate why lifestyle changes might be advantageous.This is almost impossible if therapists do not first appreciate the extent to which patients' lives involved risk factors for cardiac problems.

Depression is often more prevalent at discharge from hospital and is now believed to be an independent risk factor for ischaemic heart disease(Hippisley-Cox,1998).It is conceivable therefore that the treatment of depressed mood may feature as a distinct component of cardiac rehabilitation for some patients.

Formulation

Problem-level formulations of problems associated with cardiac rehabilitation commonly focus upon risk behaviours and facilitative beliefs regarding their presence.Formulations should also contain detail on beliefs about motivation to modify them. **Case-level formulations** of psychosocial elements of cardiac rehabilitation should outline the critical nature of recent acute cardiac events (eg. *myocardial infarction*), the presence of other life events and psychosocial problems which may relate to patient strategies for the management of behaviours and feelings resulting from cardiac events.

Education

Patients may distort or misinterpret statements made by nursing or medical staff regarding their illness and staff advice about the process of rehabilitation. Patients often draw conclusions about the aetiology of heart disease and its treatment based on the statements of staff. These may be factually inaccurate and can become the targets for educational interventions. Inaccurate understanding is also seen in the relatives of patients referred for cardiac rehabilitation. Many of the beliefs of patients and family members relate to the beliefs which link lifestyle factors and the occurrence of future cardiac events. The provision of accurate information early in the course of a patient's cardiac illness trajectory (i.e.during admission or at the first out-patient appointment) can result in belief changes which might be more difficult to achieve months after initial hospital admission. In cases of patients referred many months after the initial event, it

may be necessary to combine educational components with behavioural experiments or thought change strategies.

Modifying Risk Behaviour

Patients often state that they are motivated to make changes to risk behaviours, often citing their shock which their 'near miss' has produced as the main reason for this. It is important that patients must consider the reasons why it might be advantageous to them to modify the risk behaviours and the obstacles which might exist to change. Therapists should not take statements about life-style changes at face value.Treatment goals should emphasise the reinforcement of small changes.Approaches which consisitently remind patients of the need for large changes can fail because of the demotivating effects inherent in this approach.Risk factor modification should aim to enhance a patient's self-efficacy – something which is often compromised by the shock of a sudden cardiac event.Diary monitoring of risk behaviours, self-efficacy and regular goal-setting will help to keep patients focused upon small changes.

CONCLUSION

Thus, there are many ways in which CBT can contribute to the psychological well-being of people with cardiac problems. Some patients have multiple cardiac problems and in such cases interventions consists of a range of CBT strategies.

Topic: PSYCHO PHYSIOLOGICAL BASIS OF RESPIRATORY AIRWAY DISEASE AND MANAGEMENT PLAN

What is the Respiratory system?

Respiratory system is the system that helps to breathe in and out, so oxygen (0_2) can be pumped through the body and carbon dioxide (CO_2) can be removed from the blood stream. Therefore, this system has the 3 main functions:

- To take O2
- To excrete CO2
- To regulate the composition of blood.

The systems include a number of organs:

1. **Nose & mouth:** This is where it all begins. This is where the **oxygen** first enters your body and also where **Carbon Dioxide** leaves.
2. **The Pharynx and Trachea:** This is where the air passes from the nose to the bronchi tubes and lungs.
3. **The Bronchi Tubes and Bronchiole:** These two tubes keep splitting up and form bronchiole.

- **The** inspiration of air is an active process brought about by the contraction of muscles. Inspiration causes the **lungs to expand** inside the thorax (chest wall).

- **Expiration** is a passive function brought about the relaxation of the lungs which reduces the volume of lungs within the thorax.
- The lungs fill most of the space within the thorax, which is called thoracic cavity; and are very elastic, depending on the thoracic wall support. Therefore if the air were to get into the space between the thoracic wall and the lungs, one or more lungs would collapse.
- **Area:** Major controlling area is **Medulla.** Its function depends upon the chemical composition of blood. Eg, if blood's O2 level is too high, the respiratory centre will be stimulated and respiration will be increased. If the CO2 level falls too low, the respiratory centre will slow down until CO2 level is back to normal.

Disorders of Respiratory System:

- **Reduced air flow** **:** Asthma, emphysema, bronchitis
- **Infections** **:** Pneumonia, tuberculosis, botulism

What is psychosomatic disorder?

Psychosomatic (Psycho physiological) disorder is based on 2 assumptions-

- There is a unity of mind and body
- Psychological factors must be taken into account when considering all diseases.

The **interdisciplinary** approach to all disorders which fits relevant biological, psychological and sociocultural data into a coherent picture- is also referred to as the psychosomatic approach.

A very important history

- In 19th century it was widely it was widely accepted that psychological factors can play a part in the etiology of physical illness
- In 20th century psychoanalyst Ferenzi, Adler suggested that Freud's theory of conversion of mental disorder into physical symptoms could be applied to physical illness.
- Hainroth coined the term psychosomatic.

Theory related to psychosomatic disorder:

- In the **first theory**, it was proposed that the specific types of emotional conflict and personality structure could cause specific physical pathology.
- In second theory it was proposed that non specific emotional conflict or stressors could contribute to pathology in constitutionally predisposed people.

Two prominent exponents of the first theory were Franz Alexander and Flanders Dunvar. Alexander proposed 7 diseases as psychosomatic-

1. Bronchial Asthma
2. Rheumatoid arthritis
3. Ulcerative colitis

4. Essential hypertension
5. Neurodermatitis
6. Thyrotoxicosis
7. Peptic ulcer

Bronchial Asthma

- It is a respiratory disorder characterized by difficulty in breathing due to reversible airway obstruction, airway inflammation, and increase in airway responsiveness to a variety of stimuli.
- It is a severe allergic reaction which can be caused by a variety of foreign substance, including dust dog cat dander, pollens and fungi and also by physical stimuli such as smoke.

Multifactorial etiology → Organ specific
→ Psychological

Double vulnerability → Organ- bronchial
→ Psychological (anxiety, irritability, depression, frequency
Illness induced)

Somatopsychic kickback → QOL
→ Psychological disorder

Favorable influence → Psychotherapy
→ Psychotropic drug

High risk psychological factor:

Sometimes, asthma can cause anxiety or a panic attack. Other times, anxiety or a panic attack can trigger or aggravated an asthma attack. Chronic stress, anxiety, or panic attacks have a direct impact on breathing. Patients complaints about "shortness of breath," "smothering," "choking," "can't get enough air," along with complaints of dizziness, lightheadedness, chest tightness or severe chest pains. Many adult asthma patients report that having had several weeks or months of absence of symptoms, they begin to feel that their asthma is well under control, and then, suddenly, **"out of the blue," it's back.** This leads to a renewed phase of worries and concerns. They get very anxious and feel they've lost control over it

because it can come from behind and grab them unawares. Often **stress** has something to do with these "out of the blue" asthma episodes.

How asthma is related to stress and pain?

Stress can be described by **Lazarus and Folkman** (1984) as consequences of a person's appraisal processes**:** the assessment of whether personal resources are sufficient to meet the demands of the environment. Stress, then is determined by person-environment fit.

Primary appraisal is the attempt to determine the meaning of new event or change. This appraisal may lead to perceiving the environment as positive, neutral, negative. Potentially negative events are further appraised for their harm, threat or challenge.

- Harm is the damage done already(leads to loss of self esteem)
- Threat is assessment of possible future damager(high BP, anxiety)
- Challenge is the potential to overcome or even profit from the event(favorable emotional reaction)

Secondary appraisal is the assessment of coping ability and resources, whether they will be sufficient to meet the harm, threat and challenge. The subjective experience of stress is a balance between primary and secondary appraisal.

Potential responses to stress include:

- **Physiological**
- **Cognitive**
- **Emotional**
- **Behavioural**

PHYSIOLOGICAL MECHANISMS

The activation of Autonomic Nervous System:

Two interrelated systems

- Sympatho Adreno-medullary System **(SAM)**
- Hypothalamic- pituitary adrenocortical Axis **(HPA)**

Hans Selye's (1956, 1976) general adaptation syndrome:

- Alarm reaction
- Stage of resistance
- Stage of exhaustion

Hypothalamus and limbic structure and endocrine response to stress and impact on immune system

- The core brain parts of the brain stimulated during emotional arousal such as fear, anger are hypothalamus and limbic structures.
- ACTH is secreted into the blood stream by certain cells of pituitary gland. The rate of ACTH is secreted by another hormone like chemical substance called corticotrophin releasing factor (CRF) that is made by certain cells in the brain structure known as hypothalamus.
- Stressors are able to activate the nerve cells of the hypothalamus so that more CRF is sent to pituitary gland, thus increasing the secretion of ACTH into blood.
- ACTH stimulates the outer layer of adrenal glands so that cortisol are secreted into the blood stream.
- Cortisol promotes the formation of glucose- a fuel needed for nerve and muscles by breaking fats and proteins which are required for the manufacturing of new cells.
- Long term effect of presence of stressors will be detrimental for immune system.

Fig :

COGNITIVE MECHANISMS

It includes changes in perception, attention, memory decision making. Attentional processes are particularly vulnerable and failure to notice important but peripheral stimuli under stressful condition is a common cause of human performance. Corticosteroids play an important role in memory and other cognitive functioning during stress.

EMOTIONAL MECHANISMS

It includes feeling of distress, anxiety, fear depression .

BEHAVIOURAL MECHANISMS

Either passive behaviour (learned helplessness) or active response (flight or fight) with associated physiological adjustment. Other effects like smoking, alcohol consumption, unhealthy food choice, sleep disturbances also important.

Specific Vs non specific factors

Researchers generated the following **specific personality factors** or unconscious conflict to be associated with psychosomatic disease.

Disease	***Causative factors***
Bronchial Asthma	Separation anxiety
Peptic ulcer	Ungratified dependency need
Coronary heart disease	Type A personality
Hypertension	Guilt regarding hostile feelings

Non specific theory suggests chronic stress, intervened by anxiety predisposes certain persons to psychosomatic disorder- the vulnerable organ of the body becomes the target (heart, stomach, skin). This vulnerability or diathesis is probably genetic in nature but may be acquired too.

The major factors contribute to this disorder:

Biological	**Psychosocial**
Genetic	Personality factors
Autonomic reactivity	Defences
Somatic weekness	Kinds of stress
Inadequate cortico visceral control mechanism	Interpersonal relationship
	Effects of learning

The psychopathology (the mind- body unity)

Constitutional predisposition (genetic/other biological factor like autonomic reactivity, week organ)

↓

Early family relationships are disturbed

↓

Lack of emotional control, maladaptive learning (conditioning of ANS)

↓

Prolonged stress (inadequate coping)

↓

Leading to psycho physiological disorder

Some Studies and facts

- Asthma is a chronic childhood disease in united states- and affects more boys than girls (Dey and others, 2004).
- WHO (2000) - worldwide there are more than 180,000 asthma deaths each year and highest prevalence of asthma is in western countries.
- The disease is particularly prevalent among African Americans, Hispanic Americans and individual with lower S-E-S (Lehrer and others 2002).

Algorhythm of psychosomatic approach to Asthmatic patient

- Medication
- Psychological approach –

  -Self monitoring

  -Cognitive restructuring

  -Healthy life style

- Relaxation techniques
  - Biofeedback
  - Jacobson Progressive Muscle Relaxation

The psychotherapeutic approach

- **Documentation:** Detail history of the illness with the course and frequency of attack.
- **Psycho education:** identification of triggering substances. Basic asthma facts and requirement of medications. Avoiding allergenic properties like smoke, animal danders, dust etc.

- **Self monitoring and cognitive strategies** in order to **<u>isolate the psychological factors</u>** from asthma trigger.

Depression	Anxiety
➡ Behavioural strategies like graded task assignments, monitoring activities, graded task assignments ➡ Cognitive strategies like maintaining thought diary in order to get rid of negative automatic thought, dealing with cognitive distortion ➡ Maintaining sleep diary	➡ Behavioural and cognitive strategies like activity schedule mainly for time management, identifying negative automatic thought and generating alternatives. ➡ Maintaining sleep diary

- **Working on quality of life:** Its components are social belonging; psychological status and overall how much the patient's daily activities and functioning are compromised due to disease. In order to deal with these compromised ability social support intervention like joining new community group, informal social group, self help group can be implemented.

- **Problem solving ability:** working on problem solving ability and developing new problem solving strategies can be implemented. The major focus and the target would be to the coping strategies during stress, whether it is problem focused or it is emotion focused. Anticipating the consequences of the emotion focus coping strategies by providing adequate information and psycho educating the patient about the purpose is an important step for therapeutic approach.

- **<u>Relaxation training:</u>**

Rationale: to protect body organs from unnecessary wear, and in particular, the organs involved in stress-related disease (Selye, 1956, 1974).it helps in relieving stress in conditions in asthma (Henry et al 1993).

- Providing the information about **biofeedback** and how it is effective for the client.
- Determining the number of session especially it continues until the patient is able to achieve the success without the monitor or electrodes.
- Home work on other relaxation training or mental exercise can be given to patient

<u>GASTROINTESTINAL DISORDER : PSYCHOSOCIAL ISSUES & MANAGEMENT</u>
<u>Unit V</u>

<u>INTRODUCTION:</u>

The psychiatric and psychological connections to gastrointestinal (GI) diseases have a rich history that continues to evolve as knowledge accumulates about both medical and psychiatric contributions to

disease. Peptic ulcer disease and ulcerative colitis emerged as two of the earliest proposed examples of psychosomatic illnesses. Psychological factors such as personality characteristics were felt to be of primary importance in the development of both peptic ulcer and ulcerative colitis.

Recent understanding of the role of Helicobacter pylori in peptic ulcer disease now supports a primary medical mechanism for this gastrointestinal disease. However, this does not eliminate the need to examine biopsychosocial factors in the disease management of peptic ulcer. Gastrointestinal disorders underscore the importance of a bio psychosocial model of illness management.

The term psychosomatic indicates simply a mind-body relationship. More accurately, psychosomatic syndromes or disorders are physical disorders in which emotional processes play a role in some individuals.

Understanding psychiatric contributions and their importance to gastrointestinal disorders is growing. This is particularly true for the importance of psychological factors in the functional gastrointestinal disorders. Psychiatric illnesses frequently produce somatic symptoms including gastrointestinal symptoms. Patients frequently consult medical physicians for the gastrointestinal symptoms produced by psychiatric disorders. Medications used for gastrointestinal disorders can induce psychiatric symptoms, and medications used for psychiatric disorder can induce gastrointestinal symptoms. Understanding the interaction between gastrointestinal disorders and psychiatric disorders is important for primary care physicians, gastroenterologists, general psychiatrists, and consultation-liaison psychiatrists.

HISTORY:

In the mid–nineteenth century, William Beaumont conducted longitudinal observational studies of the stomach of a man who had accidently shot himself in the abdomen. This accident resulted in the opportunity to directly observe the gastric mucosa while modifying external factors such as sight, taste, smell, and emotional arousal. Beaumont noted that emotional factors directly influence the appearance and function of the stomach.

Ivan Pavlov used a direct observational model in studying the principles of behavior. Pavlov studied gastrointestinal function, including digestion, in the dog by developing a surgical technique that allowed observation and sampling of the stomach. He found that the sight of food elicited the salivation response. Other cues could also be linked to the salivation response by pairing them with the sight of food. For example, if a tone accompanied the sight of food or feeding, the tone itself would eventually elicit a salivation response when presented without food. This finding led to the development of the concept of a conditioned stimulus and conditioned response. Pavlov developed a model for behavioral therapy using the results of his studies on the link between the gastrointestinal tract and the brain.

In the twentieth century, George Engel and others continued the study of the gastrointestinal tract and emotions. Engel was able to study the role of developmental factors on gastrointestinal function in a girl with a gastric fistula whom he observed from infancy to adulthood. **Engel noted that developmental factors, interpersonal events, and emotional state all affect gastrointestinal function.** Changes in gastrointestinal secretion, motility, and color were linked to specific emotions of anxiety, depression, and anger. Disruption in interpersonal function adversely affected gastrointestinal function.

Today, the study of the link between gastrointestinal function and emotional state involves analysis of the interaction between the cortex, hypothalamic axis, neurotransmitter function, and the gastrointestinal tract. Many gastrointestinal hormones and transmitters are also found in the brain and appear to mediate some of the complex interactions between the brain and gut. Emotional factors continue to appear to influence the function of the gastrointestinal tract and hence the gastrointestinal symptoms humans experience.

DEFINITION & CLASSIFICATION:

It is important to understand the distinction between the terms gastrointestinal disease and functional gastrointestinal disorders. **Gastrointestinal disease** indicates a medical condition that can be *documented by changes in pathophysiology, f*or example, peptic ulcer disease. Peptic ulcer disease can be diagnosed by direct examination of the anatomical features by endoscopy or radiography. Additionally, blood and breath tests confirm the presence of H. pylori, a bacterial agent that frequently causes peptic ulcer. The term **functional gastrointestinal disorders** *indicate clinically significant distress producing a symptom related to the gastrointestinal tract (*e.g., heartburn, dyspepsia, and diarrhea) that does not demonstrate a pathophysiological mechanism. *The term functional implies a disturbance of function without a disturbance of structure.*

Functional Gastrointestinal Disorders-

Functional gastrointestinal disorders are common syndromes associated with **significant subjective distress, abnormalities of bowel function, without evidence of structural abnormalities**. Functional gastrointestinal disorders frequently have high rates of psychiatric comorbidity. Psychological factors can contribute significantly to the level of subjective gastrointestinal distress.

Successful treatment of co morbid psychiatric conditions appears to reduce the severity of many functional gastrointestinal symptoms. Psychotropic drugs are commonly used by primary care physicians and subspecialists who treat functional bowel syndromes.

Classification of Functional Gastrointestinal Disturbances-

The 10th revision of International Statistical Classification of Diseases and Related Health Problems (ICD-10) classification system shares and differs in some ways from the diagnoses and categories of the fourth edition of Diagnostic and Statistical Manual of Mental Disorders (DSM-IV). ICD-10 recognizes a category of somatoform disorders with somatization disorders and hypochondriacal disorders similar to their DSM-IV counterparts. However, ICD-10 also includes a diagnosis called *somatoform autonomic dysfunction.* Two subtypes are noted in this category relevant to gastrointestinal disorders: **upper gastrointestinal tract** and **lower gastrointestinal tract.**

Functional Gastrointestinal Disorders and Somatothymia-

Somatothymia has been proposed by ***Alan Stoudemire*** as a way to understand the mechanism of unexplained somatic symptoms in some patients. Somatothymia is the tendency to communicate psychological distress (i.e., affective distress) to others through somatic symptoms. Patients who tend to somatise may be using their own culturally determined communication strategies to acknowledge psychological distress.

Patients from lower socioeconomic classes and third-world countries may be more likely to have somatothymia.

Psychological Factors Affecting Medical Conditions

Psychiatric disorders can influence gastrointestinal diseases and functional gastrointestinal disorders in a variety of ways. DSM-IV defines psychological factors affecting physical conditions to address the spectrum of ways that psychiatric disorders and psychological mechanisms influence medical conditions. Psychological factors affecting medical conditions include the following subtypes: (1) mental disorder affecting a medical condition, (2) psychological symptoms affecting a medical condition, (3) personality traits or coping style affecting a medical condition, (4) maladaptive health behaviors affecting a medical condition, and (5) stress-related physiologic response affecting a medical condition.

The corresponding classification category in ICD-10 is F54 psychological and behavioral factors associated with disorders or diseases classified elsewhere. All of the psychological factors affecting medical condition categories can be found with gastrointestinal diseases and disorders.

Drossman and his colleagues examined the prevalence and correlates of functional gastrointestinal disorders in a community survey of U.S. households. **Sixty-nine percent of this sample reported at least one of 20 functional gastrointestinal symptoms in the 3 months prior to the survey.**

Risk Factors

Women are more likely to report globus, functional dysphagia, irritable bowel syndrome, functional constipation, functional abdominal pain, functional biliary pain, and dyschezia. Men are more likely to report aerophagia and functional bloating. Functional gastrointestinal syndromes increase the rate of absenteeism at work or school. Occupational impairment appears to be specifically associated with chronic abdominal pain, functional biliary pain, functional dyspepsia, and irritable bowel syndrome.

Psychiatric disorders and the behaviors associated with them can increase the risk of gastrointestinal disease. A good example of this is alcohol abuse or dependence. Chronic alcohol consumption is associated with a variety of gastrointestinal disorders including esophageal reflux, gastritis, pancreatitis, and several of the gastrointestinal cancers.

Nicotine and caffeine produce direct physiological changes in gastrointestinal function. Both compounds can increase gastric acid secretion and exacerbate gastritis, esophagitis, or peptic ulcer disease.

PSYCHIATRIC COMORBIDITY IN FUNCTIONAL GASTROINTESTINAL DISORDER:

A large part of the literature reports the link between stress, anxiety, and physiological responsivity of the gastrointestinal system. Anxiety can produce disturbances in gastrointestinal function through a central control mechanism or via catecholamines. Parasympathetic autonomic responses also influence gastrointestinal function. Parasympathetic impulses originate in the periventricular and lateral hypothalamus and travel to the dorsal motor nucleus of the vagus, the main parasympathetic output pathway.

Acute stress can induce physiological responses in several gastrointestinal target organs. In the esophagus, acute stress increases resting tone of the upper esophageal sphincter and increases contraction amplitude in the distal esophagus. In the stomach acute stress induces decreased antral motor activity, potentially producing functional nausea and vomiting. Reduced migrating motor function can occur in the small intestine while in the. These effects in the small and large intestine may be responsible for bowel symptoms associated with irritable bowel syndrome.

Irritable Bowel Syndrome

Irritable bowel syndrome is the prototypical functional gastrointestinal disorder characterized by abdominal pain and diarrhea or constipation. The International Congress of Gastroenterology has developed a standardized set of criteria for irritable bowel syndrome.

1. Abdominal pain relieved by defecation or associated with change in frequency or consistency of stool.

2. Disturbed defecation involving two or more of the following:

altered stool frequency

altered stool form (hard or loose and watery)

altered stool passage (straining or urgency, feeling of incomplete evacuation)

passage of mucus

Irritable bowel syndrome can often be categorized into diarrhea-predominant, constipation-predominant, and mixed subtypes. Medical treatment often targets the predominant symptom. Some studies suggest that irritable bowel syndrome accounts for up to 50 percent of all outpatient evaluations done by gastroenterologists. Comorbid psychiatric disorders appear to increase the likelihood of health-care-seeking behavior for people with symptoms of irritable bowel syndrome.

Some patients with irritable bowel syndrome may demonstrate physiological abnormalities including abnormal intestinal myoelectrical activity, gastrointestinal hormonal abnormalities, or allergic responses to some foods. Most clinicians agree that both physiological and psychological factors contribute to the clinical picture of irritable bowel syndrome in most.

Psychiatric disorders complicate the diagnosis and management of many patients with irritable bowel syndrome. However, recent studies of general population samples found evidence of increased psychiatric comorbidity in community subjects with unexplained gastrointestinal symptoms.

In the National Institute of Mental Health (NIMH) Epidemiologic Catchment Area (ECA) Study, over 18,000 adults in the **United States** received a direct interview for psychiatric illness. Six gastrointestinal symptoms from the somatization disorder section were reviewed with the participants (abdominal pain, diarrhea, gas and bloating, constipation, and nausea and vomiting). Subjects who experienced multiple gastrointestinal symptoms were compared with those without gastrointestinal symptoms.

One gastrointestinal symptom increased the rate of lifetime major depressive disorder (7.5 versus 2.9 percent), **panic disorder** (2.5 versus 0.7 percent), and **agoraphobia** (10.0 versus 3.6 percent). Community subjects with two gastrointestinal symptoms had even higher rates of major depressive disorder (13.4 percent), panic disorder (5.2 percent), and agoraphobia (17.8 percent).

Other studies of psychiatric co morbidity in irritable bowel syndrome estimate comorbidity rates of 42 to 64 percent of all irritable bowel syndrome patients.

Four models have been proposed to explain the relation between irritable bowel syndrome and high rates of psychiatric co morbidity.

1. ***Somatization disorder hypothesis***, which proposes that some people display anxious and depressed mood with multiple nonspecific somatic symptoms. This model classifies irritable bowel syndrome as one of a group of diagnoses that can be made from a primary somatization disorder or other somatoform disorder. Other similar functional medical disorders that could occur in these patients include fibromyalgia. Although somatization disorder and other somatoform disorders occur with irritable bowel syndrome, they do not occur in all patients.

2. A second model to explain the role of psychiatric co morbidity in irritable bowel syndrome is the ***Somatopsychic hypothesis***. This hypothesis states that psychological symptoms are the result of chronic gastrointestinal distress and unsatisfactory interaction with health care providers who do not accurately diagnose and treat irritable bowel syndrome.

3. A third explanatory model, the ***psychogenic hypothesis.*** states that specific psychiatric disorders cause irritable bowel syndrome in a significant proportion of patients. Panic disorder in particular is proposed to cause secondary irritable bowel syndrome.

4. The final model is the ***self-selection model.*** This model proposes that psychiatric co morbidity increases the rate of treatment seeking by patients who have irritable bowel syndrome. Under this model, irritable bowel syndrome without psychiatric comorbidity would be accompanied by the lowest rate of health care use, while irritable bowel syndrome with psychiatric comorbidity would be associated with the highest rate. There is some support for this model in the medical clinic setting; psychiatric co morbidity does seem to be more prevalent in clinic samples than in community samples. However, population studies (including the NIMH ECA study) suggest that this model cannot fully explain the relation. A significant number of community subjects with irritable bowel syndrome who do not seek medical attention suffer from significant psychiatric co morbidity. Possibly several of the proposed models contribute to the overlap between irritable bowel syndrome and psychiatric illness.

One method of sorting out the relation between irritable bowel syndrome and psychiatric illness is the **family study method**. If psychiatric disorders simply follow the distress of irritable bowel syndrome, one would not expect a higher rate of psychiatric illness in the relatives of patients with the syndrome. Twenty patients with irritable bowel syndrome and 20 patients undergoing laparoscopic cholecystectomy were compared. Family rates of depressive disorders and anxiety disorder were higher in family members of the irritable bowel syndrome probands than in control probands. This study supports the psychogenic hypothesis for irritable bowel syndrome, although the association could be due to a treatment selection bias since probands in this study were identified in a tertiary-care center.

Physiological abnormalities also appear to contribute to the symptom profile of irritable bowel syndrome. Autonomic nervous system abnormalities in irritable bowel syndrome may vary by the predominant symptom noted by individual patients

COMORBIDITY

In an interesting study using an existing database **(from the Epidemiologic Catchment Area Study; Robins & Regier, 1991) Walker, Katon, Jemelka, and Roy-Byrne (1992**) examined the data from 18,571 cases. They classified cases as probably IBS when the individual had two of the following symptoms that were otherwise medically unexplained: abdominal pain, diarrhea, and constipation. They identified 412 cases of probable
IBS (with no other pain problem) and compared them with cases with no GI symptoms. Individuals with probable IBS showed higher rates of major depression (13.4%), panic disorder (5.2%), and agoraphobia (17.8%). Clearly, there is a fair degree of diagnosable psychiatric morbidity among IBS patients.\

Peptic Ulcer Disease

Peptic ulcer refers to mucosal ulceration involving the distal stomach or proximal duodenum. Symptoms of peptic ulcer disease include a gnawing or burning epigastric pain that occurs 1 to 3 hours after meals and is relieved by food or antacids. Accompanying symptoms can include nausea, vomiting, dyspepsia, or signs of gastrointestinal bleeding such as hematemesis or melena. Lesions are generally small, one centimeter or less in diameter.

Early theories identified excess gastric acid secretion as the most important etiologic factor, but the importance of infection with H. pylori is becoming more acknowledged. H. pylori is associated with 95 to 99 percent of duodenal ulcers and 70 to 90 percent of gastric ulcers. Antibiotic therapy that targets H. pylori results in much higher healing and cure rates than antacid and histamine inhibitor therapy. Standard regimens for the treatment of H. pylori infection often include combinations of two or three antibiotic

agents. Commonly used antibiotic agents with efficacy against H. pylori include amoxacillin (Amoxil), metronidazole (Flagyl), tetracycline (Achromycin), and clarithromycin (Biaxin).

Crohn's Disease

Crohn's disease is an inflammatory bowel disease affecting primarily the small intestine and colon. Common symptoms in Crohn's disease include diarrhea, abdominal pain, and weight loss. The yearly incidence of Crohn's disease is about half that of ulcerative colitis, about 5 per 100,000 population. The course is chronic, often with periods of remission followed by periods of acute symptoms. Treatment consists of the use of antibiotic agents such as metronidazole, sulfasalazine (Azulfidine) or mesalamine (Asacol). Immunosuppressive drugs also are commonly used to control flare-ups. Prednisone is the most frequently used corticosteroid, and azathioprine (Imuran) a commonly used immunosuppressive agent.

Because Crohn's disease is a chronic illness, most studies of psychiatric comorbidity focus on psychiatric disorders occurring after the onset of the disorder. A study of psychiatric symptoms in Crohn's disease prior to the onset of physical symptoms found high rates (23 percent) of preexisting panic disorder compared with control subjects and subjects with ulcerative colitis. No statistically significant preexisting psychiatric co morbidity in ulcerative colitis occurred in this study.

DIAGNOSIS AND CLINICAL FEATURES

Gastrointestinal Symptoms in Psychiatric Disorders

Gastrointestinal symptoms occur commonly in a variety of psychiatric disorders. This association is so well known that many of the diagnostic criteria for some mental disorders include the presence of specific gastrointestinal symptoms and behaviors.

Somatization disorder and the associated somatoform disorders are commonly present with gastrointestinal complaints. In DSM-IV abdominal and rectal pain are two possible symptoms for meeting the criteria for medically unexplained pain in four separate sites. In addition, somatization disorder requires two additional gastrointestinal symptoms. Examples given to meet these criteria include nausea, bloating, vomiting other than during pregnancy, diarrhea, or intolerance to several varieties of food.

Changes in appetite are a hallmark feature of major depressive disorders. **Nausea frequently accompanies a panic attack**, as does abdominal distress. **Diarrhea or fear of uncontrollable diarrhea in a social situation can be present in social phobia**. Eating disorders usually include a disturbance of appetite, and self-induced vomiting is the most common cause of purging behavior in bulimia. A variety of substances produce gastrointestinal disturbances during the intoxication and withdrawal phases. Physicians need to keep these psychiatric disorders in mind when considering the differential diagnosis of patients presenting with gastrointestinal symptoms.

The Role of Early Abuse in IBS

A possible explanation for the relatively high levels of psychological distress and psychiatric comorbidity found in IBS patients may lie in a body of work that began with an article published in 1990 by Douglas Drossman and his colleagues (Drossman, Leserman, et al., 1990) that documented a higher level of early (preadolescent) sexual and physical abuse among female patients with functional GI disorders than among comparable female patients with organic GI diseases. In every instance there was a higher prevalence of some form of early sexual abuse among those with functional disorders (53%) than among those with organic GI disorders (37%).

Only the early history of physical abuse was statistically different between the two samples. Half of the group with functional GI disorders had IBS (38 of 75). Of the IBS patients, 19 of 38 (50%) reported early

sexual or physical abuse. It seems clear that a large proportion of IBS patients will have a history of early sexual or physical abuse as well as a history of meeting the criteria for some Axis I psychiatric disorders (the prevalence of Axis II disorders among IBS patients has not been well studied as best we can determine.

TREATMENT

Psychotropic Treatment

Psychotropic drug use is common in the treatment of a variety of gastrointestinal disorders and is complicated by disturbances in gastric motility, absorption, and metabolism related to the underlying gastrointestinal disorder.

Many gastrointestinal effects of psychotropic drugs can be used therapeutically in functional gastrointestinal disorders. For example, *tricyclic antidepressant agent* can be used to reduce gastric motility in patients with irritable bowel syndrome with diarrhea. However, psychotropic gastrointestinal side effects can exacerbate a gastrointestinal disorder (e.g., prescribing a tricyclic drug to treat a depressed patient with gastro esophageal reflux).

Psychopharmacological agents provide one step in a stepped-care approach to the management of irritable bowel syndrome. For milder cases of irritable bowel syndrome, education, reassurance, and avoidance of foods that increase symptoms forms the base of treatment. Moderate cases of irritable bowel frequently respond to non psychopharmacological drug treatment targeted at the patient's primary symptom.

Tricyclic antidepressants can significantly reduce irritable bowel symptoms for many more-severe cases of irritable bowel syndrome and usually are reserved for more severe cases.

Retrospective studies of outpatients with irritable bowel syndrome found that 89 percent of patients had improvement in bowel symptoms with use of tricyclics or anxiolytics. Sixty-one percent of patients reported complete remission of symptoms. The most frequently used tricyclic drugs in this series were amitriptyline (about 50 mg a day) and doxepin (Adapin) (about 25 to 50 mg daily).

Antidepressant treatment can also reduce symptoms in other functional disorders.

Psychotherapy

Psychotherapy can be a key component in the stepped-care approach to the treatment of irritable bowel syndrome and other functional gastrointestinal disorders. Multiple different models of psychotherapy have been applied to clinical samples of patients with irritable bowel syndrome, including short-term, dynamically oriented individual psychotherapy, supportive psychotherapy, hypnotherapy, relaxation techniques, and cognitive therapy. There appears to be a significant placebo effect for psychotherapy in this population for both pharmacological and psychological treatment approaches. At least one study found the response to psychological treatment alone for irritable bowel syndrome to be comparable to that of pharmacological treatment.

Patients with significant Axis I psychopathology appear to be less responsive to psychological intervention than those without. This may be related to the increased severity of irritable bowel syndrome symptoms associated with psychiatric comorbidity.

Most studies of psychotherapeutic approaches to the treatment of irritable bowel syndrome have been case series, not controlled trials. Some well-controlled studies are beginning to be published that use various models of psychotherapy. A cognitive therapy model for the treatment of irritable bowel

syndrome has been developed that considers irritable bowel syndrome to be autonomic nervous system response to stress, with three components: cognitions, behaviors, and physiological response.

There are at least three (and possibly six) distinctly different psychological treatments for IBS that have been evaluated: brief psychodynamic psychotherapy, hypnotherapy, and various combinations of cognitive and behavioral treatments (CBT). Two of the components frequently included in CBT, relaxation training and cognitive therapy alone, have also been evaluated as individual treatments.

1. Brief psychodynamic psychotherapy

Guthrie, Creed, Davison, and Tomenson (1991) compared their version of brief psychodynamic psychotherapy plus home relaxation with conventional medical care. The treated group showed greater reductions than the controls in anxiety and depression. For women (about 75% of the total sample), the treated patients' global ratings of GI symptoms showed more improvement than the controls; there were similar results on physician ratings. Thus, both RCTs of psychodynamic psychotherapy show clear significant advantages over routine medical care

2. ***Hypnotherapy***

In 1984 Whorwell, Prior, and Faragher reported on the successful treatment of relatively refractory cases of IBS using hypnotherapy. Treatment included an initial hypnotic induction using arm levitation and then further sessions (between 7 and 12) with attention to general relaxation and gaining control of intestinal motility with some attention to ego strengthening. It was revealed that hypnotherapy was superior to a supportive psychotherapy control in reduction of pain, bowel habit disturbance, and bloating and that it led to an increase in sense of well-being. Hypnosis, commonly done in England and at UNC where hypnotic suggestion is used to relax the bowel and reduce symptoms.

3. CBT:

Cognitions form the key element in this therapy. Therapy focuses on identifying life stressors and the thoughts associated with these stressors. **Subjects are taught to identify threatening stimuli** and **modify their appraisals** and interpretations of such stimuli. Subjects **record automatic thoughts in their daily lives**, and the therapist focuses on identifying central themes. Therapeutic work centers around three processes: **rational self-analysis, decentering, and experimental disconfirmation.**

Using this model, a study of 20 patients compared gastrointestinal symptom response of 10 subjects receiving cognitive therapy with that of 10 subjects who simply monitored their gastrointestinal symptoms; 80 percent of the cognitive therapy group showed clinically significant improvement compared with only 10 percent of the control. These small, early studies suggest that specific psychotherapy approaches may become a key intervention in irritable bowel syndrome and other functional gastrointestinal syndromes.

Bennett and Wilkinson (1985) reported on the first CBT trial comparing a combination of progressive muscle relaxation, education, progressive muscle relaxation was superior to a drug treatment on global ratings. and modification of self-talk with a combination of drugs. The treatments were equally effective except that the CBT condition led to significantly greater reduction in state anxiety.
Blanchard and colleagues started off evaluating a cognitive behavioral treatment package consisting of education, relaxation training (progressive muscle relaxation), thermal biofeedback, and elements of cognitive therapy in comparison with symptom monitoring (Blanchard & Schwarz, 1987; Neff &

Blanchard, 1987). They next evaluated the CBT combination in comparison with an attention placebo condition (pseudomeditation and EEG biofeedback for alpha suppression) and with a symptom-monitoring condition (Blanchard et al., 1992)

Stress Management/Relaxation Training, can be a component of the other psychological treatments, where imaging and relaxation methods are used to reduce autonomic (blood pressure, pulse) activity and muscle tension

CONCLUSION

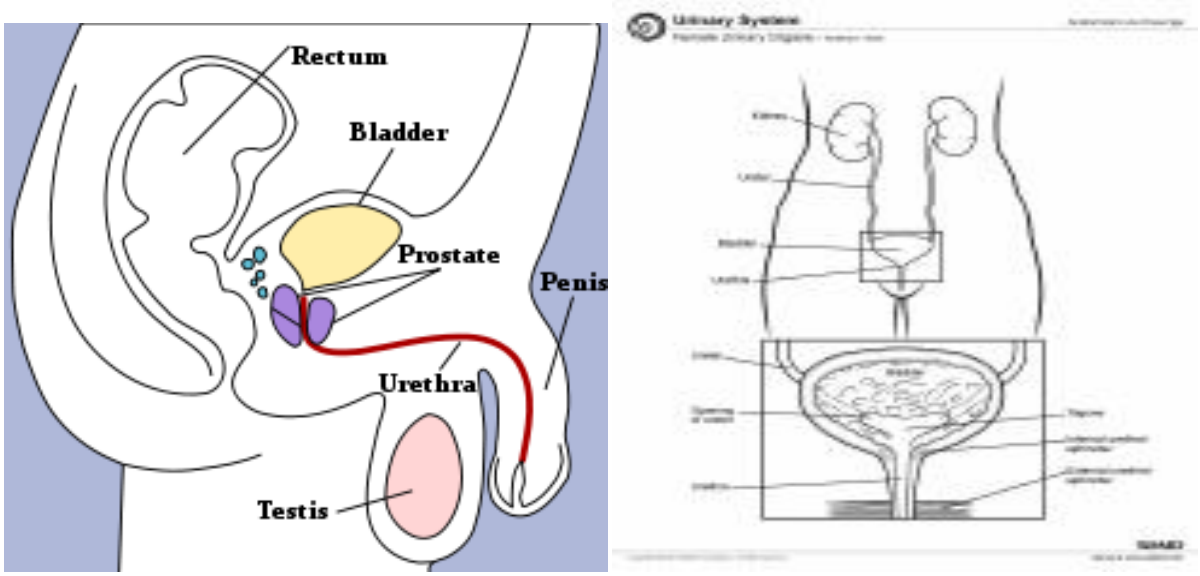
There are a variety of treatments that can be used for treating patients with IBS. A general approach includes an effective physician-patient relationship, proper education, and dietary or lifestyle modifications necessary for any treatment plan. In addition, the options for treatment are based on the nature of the symptoms as well as their severity and frequency.

Genitourinary/renal/reproductive system (unit-6):

"Psychological issues in male/female sexual dysfunctions, micturition/voiding problems including primary/ secondary enuresis, end-stage renal diseases, dialysis treatment, primary and secondary infertility, empirically validated psychological and behavioral interventions in these conditions."

Genitourinary system:

In anatomy, the **genitourinary system** or **urogenital system** is the organ system of the reproductive organs and the system. These are grouped together because of their proximity to each other, their common embryological origin and the use of common pathways, like the male urethra. Also, because of their proximity, the systems are sometimes imaged together.



Some components of the male and female genitourinary system.

Functions of the Genitourinary System

The urinary system maintains the composition and properties of the body fluid that establishes the internal environment of the body cells. The end product of the urinary system is urine, which is voided from the body during micturition (urination).

It consists of two kidneys, two ureters, the urinary bladder, and the urethra. Tubules in the kidneys are intertwined with vascular networks of the circulatory system to enable the production of urine. After the urine is formed, it is moved through the ureters to the urinary bladder for storage. Micturition, or voiding of urine from the urinary bladder, occurs through the urethra.

The urinary system, along with the respiratory, digestive, and integumentary systems, excretes substances from the body. For this reason, these systems are occasionally referred to as excretory systems. In the process of cellular metabolism, nutrients taken in by the digestive system and oxygen from inhaled air are used to synthesize a variety of substances while providing energy needed for body maintenance. Metabolic processes, however, produce cellular wastes that must be eliminated if homeostatis is to be maintained. Just as the essential nutrients are transported to the cells by the blood, the cellular wastes are removed through the circulatory system to the appropriate excretory system. Carbon dioxide is eliminated through the respiratory system; excessive water, salts, nitrogenous wastes, and even excessive metabolic heat are removed through the integumentary system; and various digestive wastes are eliminated through the digestive system.

The urinary system is the principal system responsible for water and electrolyte balance. Electrolytes are compounds that separate into ions when dissolved in water. Electrolyte balance is achieved when the number of electrolytes entering the body equals the number leaving. Hydrogen ions, for example, are maintained in precise concentration so that an acid-base, or pH, balance exists in the body.

A second major function of the urinary system is the excretion of toxic nitrogenous compounds—specifically, urea and creatinine. Other functions of the urinary system include the elimination of toxic wastes that may result from bacterial action and the removal of various drugs that have been taken into the body. All of these functions are accomplished through the formation of urine by the kidneys.

Blood to be processed by a kidney enters through the large renal artery. After the filtration process (see the kidneys), it exits through the renal vein. The importance of filtration of the blood is demonstrated by the fact that during normal resting conditions the kidneys receive approximately 20% to 25% of the entire cardiac output. Every minute, the kidneys process approximately 1,200 ml of blood.

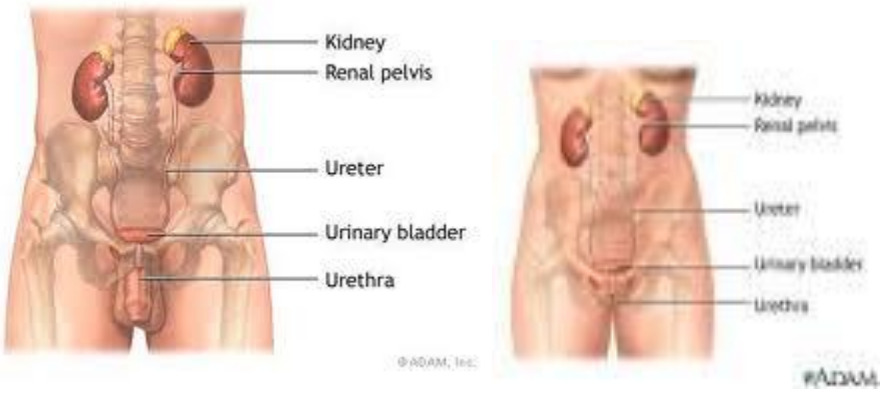
Renal system or urinary system:

The kidneys are essentially regulatory organs which maintain the volume and composition of body fluid by filtration of the blood and selective reabsorption or secretion of filtered solutes.

The kidneys are retroperitoneal organs (i.e. located behind the peritoneum) situated on the posterior wall of the abdomen on each side of the vertebral column, at about the level of the twelfth rib. The left kidney is slightly higher in the abdomen than the right, due to the presence of the liver pushing the right kidney down.

The kidneys take their blood supply directly from the aorta via the renal arteries; blood is returned to the inferior vena cava via the renal veins. Urine (the filtered product containing waste materials and water) excreted from the kidneys passes down the fibromuscular **ureters** and collects in the **bladder**. The

bladder muscle (the **detrusor muscle**) is capable of distending to accept urine without increasing the pressure inside; this means that large volumes can be collected (700-1000ml) without high-pressure damage to the renal system occurring.
When urine is passed, the **urethral sphincter** at the base of the bladder relaxes, the detrusor contracts, and urine is voided via the **urethra**.



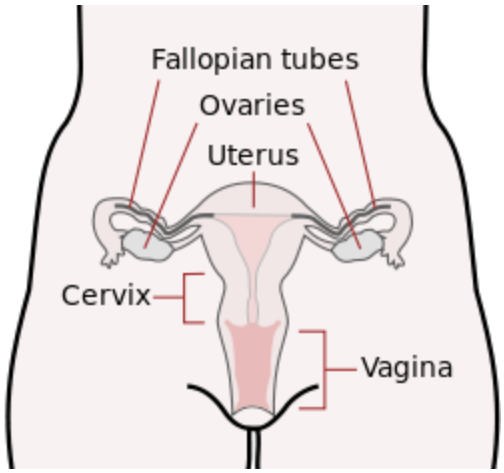
Male and female renal system.

Reproductive system:

The **human female reproductive system** (or **female genital system**) contains two main parts: the uterus, which hosts the developing fetus, produces vaginal and uterine secretions, and passes the male's sperm through to the fallopian tubes; and the ovaries, which produce the female's egg cells. These parts are internal; the vagina meets the external organs at the vulva, which includes the labia, clitoris and urethra. The vagina is attached to the uterus through the cervix, while the uterus is attached to the ovaries via the Fallopian tubes. At certain intervals, the ovaries release an ovum, which passes through the Fallopian tube into the uterus.

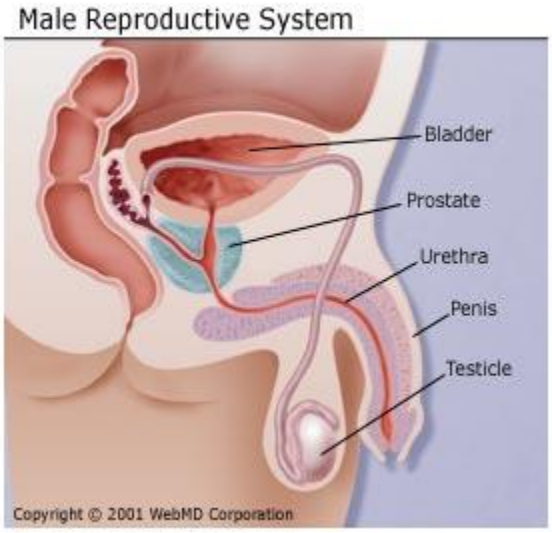
If, in this transit, it meets with sperm, the sperm penetrate and merge with the egg, fertilizing it. The egg releases certain molecules that that are essential to guiding the sperm and these allow the surface of the egg to attach to the sperm's surface then the egg can absorb the sperm and fertilization begins. The fertilization usually occurs in the oviducts, but can happen in the uterus itself. The zygote then implants itself in the wall of the uterus, where it begins the processes of embryogenesis and morphogenesis. When developed enough to survive outside the womb, the cervixdilates and contractions of the uterus propel the fetus through the birth canal, which is the vagina.

The ova are larger than sperm and have formed by the time a female is born. Approximately every month, a process of oogenesis matures one ovumto be sent down the Fallopian tube attached to its ovary in anticipation of fertilization. If not fertilized, this egg is flushed out of the system through menstruation.



The **human male reproductive system** consists of a number of sex organs that form a part of the human reproductive process. In this type of reproductive system, these sex organs are located outside the body, around the pelvic region.

The main male sex organs are the penis and the testes which produce semen and sperm, which, as part of sexual intercourse, fertilise an ovum in the female's body; the fertilised ovum (zygote) develops into a fetus, which is later born as a child.



What Causes Sexual Problems?

There are many factors which may result in a person experiencing a sexual dysfunction. These may result from emotional or physical causes.

Sexual dysfunction may arise from emotional factors, including interpersonal or psychological problems. Interpersonal problems may arise from marital or relationship problems, performance anxiety, or from a lack of trust and open communication between partners, and psychological problems may be the result of depression, sexual fears or guilt, past sexual trauma, sexual disorders, among others.

Sexual dysfunction is especially common among people who have anxiety disorders. Ordinary anxiousness can obviously cause erectile dysfunction in men without psychiatric problems, but clinically

diagnosable disorders such as panic disorder commonly cause avoidance of intercourse and premature ejaculation. Pain during intercourse is often a co morbidity of anxiety disorders among women.

Sexual activity may also be impacted by physical factors. These would include use of drugs, such as alcohol, nicotine, narcotics, stimulants, antihypertensive, antihistamines, and some psychotherapeutic drugs. For women, almost any physiological change that affects the reproductive system—premenstrual syndrome, pregnancy, postpartum, menopause—can have an adverse effect on libido. Injuries to the back may also impact sexual activity, as would problems with an enlarged prostate gland, problems with blood supply, nerve damage (as in spinal cord injuries). Disease, such as diabetic neuropathy, multiple sclerosis, tumors, and, rarely, tertiary syphilis may also impact on the activity, as would failure of various organ systems (such as the heart and lungs), endocrine disorders (thyroid, pituitary, or adrenal gland problems), hormonal deficiencies (low testosterone, estrogen, or androgens), and some birth defects.

Thus, sexual dysfunction can be a result of a physical or psychological problem.

- **Physical causes:** Many physical and/or medical conditions can cause sexual problems. These conditions include diabetes, heart disease, neurological diseases, hormonal imbalances, menopause, chronic diseases such as kidney disease or liver failure, and alcoholism and drug abuse. In addition, the side effects of certain medications, including some antidepressant drugs, can affect sexual desire and function.
- **Psychological causes:** These include work-related stress and anxiety, concern about sexual performance, marital or relationship problems, depression, feelings of guilt, and the effects of a past sexual trauma.
  - Female sex is very psychological. There may be so many psychological causes. Female sexual desire naturally fluctuates over the years. Highs and lows commonly coincide with the beginning or end of a relationship or with major life changes such as pregnancy, menopause or illness. If you are concerned by a low or decreased sex drive there are life style changes and sex techniques that may put you in the mood more often.
  - According to some studies more than 40% of females complain of low sexual desire at some point. Still researchers acknowledge that it is difficult to measure what is normal and what is not. If you want to have sex less often than your partner does neither one of you is necessarily outside the norm for people at your stage in life although your differences may cause you distress.
  - Similarly if your sex drives is weaker that it once was your relationship maybe stronger than ever. There is no magic number to define low sex drive. It varies from woman to woman. Causes for low sex drive are based on a complex interaction of many components affecting intimacy including physical well being, emotional well being, experiences, beliefs, life styles and current relationships. Alcohol and drugs, surgery, fatigue, menopause, pregnancy and breast feeding and then **psychological causes** such as anxiety, depression, physical appearance, low self esteem etc. are also causes of low sex drive.
    There can be many reasons of this condition in men also, such as hormonal imbalances, stress, drugs, sleep deprivation etc.
  - **Hormone Imbalances (Testosterone):**
    Testosterone levels are the most common root of libido issues in men. As the principle male sex hormone, testosterone maintains and drives your libido. When testosterone levels get too low, a man can lose much of his sex drive. Luckily, through testosterone hormone therapy, men can establish a more normal level, helping to reinstate their sex drive.
  - **Sleep:**
    The amount of sleep a man gets may also affect his libido. Sleep is necessary to help the body recharge and stay energized. A regular loss of sleep can result in a man being too tired to have sex. Getting at least eight hours of uninterrupted sleep a night can quickly boost a lowered libido.

- **Stress:**
  Whether it be job, money, or relationship related, stress can have a big impact on your libido. When your body becomes stressed it releases two main hormones called adrenaline and cortisol. These hormones are fine in normal doses, but if produced by the body too frequently it can lead to chronic stress. Eventually these hormones begin to interfere with testosterone, causing a lowered libido.
- **Alcohol:**
  You may be tempted to believe that alcohol increases your libido. However, chronic alcohol use can actually lower your libido. When you drink alcohol it depresses many important parts of your body, including your hypothalamus and pituitary gland. These parts of your body regulate the function of your gonads. Once they become depressed by alcohol they can cause a noticeable drop in libido. Alcohol may also affect the blood vessels handicapping the body's ability to achieve a firm erection.

- **Disease:**
  Certain disease may also cause a noticeable drop in libido. These diseases include diabetes, cardiovascular disease, Parkinson's disease and anemia. Diseases that specifically attack the cardiovascular system are especially prominent causes of libido problems in men as they can affect the ability to achieve an erection.
- **Medication:**
  Certain medications may also cause a drop in libido for men. Antidepressants and muscle relaxants are common causes of lowered libido in men. Illegal drugs can also lower a man's libido, especially marijuana, cocaine and heroin.
  Please contact a good urologist for complete checkup and appropriate management and/or referral to endocrinologist or psychologist.

Who Is Affected by Sexual Problems?
Both men and women are affected by sexual problems. Sexual problems occur in adults of all ages. Among those commonly affected are those in seniors, which may be related to a decline in health associated with aging.

How Are Sexual Problems Treated?
The ideal approach to treating sexual problems involves a team effort between the patient, doctors, and trained therapists. Most types of sexual problems can be corrected by treating the underlying physical or psychological problems. Other treatment strategies focus on the following:

- **Providing education.** Education about human anatomy, sexual function, and the normal changes associated with aging, as well as sexual behaviors and responses may help a woman overcome her anxieties about sexual function and performance.
- **Enhancing stimulation.** This may include the use of erotic materials (videos or books), masturbation, and changes to sexual routines.
- **Providing distraction techniques.** Erotic or non-erotic fantasies, exercises with intercourse, music, videos, or television can be used to increase relaxation and eliminate anxiety.
- **Encouraging non-coital behaviors.** Non-coital behaviors (physically stimulating activity that does not include intercourse), such as sensual massage, can be used to promote comfort and increase communication between partners.
- **Minimizing pain.** Using sexual positions that allow the woman to control the depth of penetration may help relieve some pain. The use of vaginal lubricants can help reduce pain caused by friction, and a warm bath before intercourse can help increase relaxation.

Treatment depends on the cause of the sexual dysfunction. Medical causes that are reversible or treatable are usually managed medically or surgically. Physical therapy and mechanical aides may prove helpful for some people experiencing sexual dysfunction due to physical illnesses or disabilities.

Sildenafil (Viagra) may be helpful for men who have difficulty attaining an erection. The medication increases blood flow to the penis. It must be taken 1 to 4 hours before intercourse. Men who take nitrates for coronary heart disease should not take sildenafil.

Mechanical aids and penile implants are an option for men who cannot attain an erection and find sildenafil is not helpful.

Women with vaginal dryness may be helped with lubricating gels, hormone creams, and -- in cases of premenopausal or menopausal women -- with hormone replacement therapy. In some cases, women with androgen deficiency can be helped by taking testosterone. Kegel exercises may also increase blood flow to the vulvar/vaginal tissues, as well as strengthen the muscles involved in orgasm.

Vulvodynia can be treated with numbing cream, biofeedback, or low doses of certain antidepressants that also treat nerve pain. Surgery has not been successful.

Behavioral treatments involve many different techniques to treat problems associated with orgasm and sexual arousal disorders. Self-stimulation and the Masters and Johnson treatment strategies are among the many behavioral therapies used.

Simple, open, accurate, and supportive education about sex and sexual behaviors or responses may be all that is required in many cases. Some couples may benefit from joint counseling to address interpersonal issues and communication styles. Psychotherapy may be required to address anxieties, fears, inhibitions, or poor body image.

Sex–therapy Stop start [Do-Redo method]
Men should stimulate the penis near the point of ejaculation and then stop for 30-60 seconds and again stimulate it. Repeat the process for five to six times.

Squeeze: - Stimulation to penis should be done near the point of ejaculation and then firmly squeeze around the penis just below the head. It has effect of preventing ejaculation.

Other than that these methods also can apply:

- Share your worries by communicating with your partner
- Spend more time with other form of sexual intimacy like kissing, cuddling, licking and massaging.
- Seek a doctor's help

Micturition/Voiding problems:

Micturition, also known as urination and voiding, is a process that allows the body to excrete excess water and get rid of some substances that are potentially harmful to the body. Excess water and other water-soluble substances exit the body in the form of urine. The act of micturition is supposedly a voluntary process but for some people, especially the very young and very old as well as those who are suffering from conditions affecting the brain and the spinal, it becomes uncontrollable. This is because micturition

is under the control of the nervous system, which serves as pathway through which the impulse to urinate travels.

Body organs that are involved in the micturition process are the kidneys, ureter, urinary bladder, and urethra. The kidneys are involved in the production of urine by filtering out excess water waste products. Two thin tube-like structures, called the ureters, connect each kidney to the urinary bladder. Urine that is formed in the kidneys will go down through the ureter into the urinary bladder for storage. Once an individual is ready to urinate, urine will flow out of the body through an opening in the urethra.

The micturition reflex, or the urge to urinate, often involves sending of signals between the urinary bladder, the spinal cord, and the brain. When the walls of the urinary bladder are stretched to their limit due to the presence of urine, nerves located in the bladder walls will send a signal that it is ready to release the accumulated fluid. This signal will travel to the spinal cord, and up to the brain. When the brain receives this signal, it will send down a command for the muscular walls of the bladder to contract and expel the urine out of the body. In circumstances where a person is stuck in traffic, the brain will normally relay a message to the bladder to delay the passing out of urine.

Symptoms of micturition problems include urinary retention, where an individual is unable to empty the bladder completely; urinary incontinence, where he is unable to hold back urination; and urinary frequency, where he experiences the urge for urination more often than normal. Urinary retention is the inability to empty the bladder. Chronic urinary retention is frequently asymptomatic - a patient is able to urinate, but may experience lower urinary tract symptoms (LUTS), related to storage and voiding difficulties. This is in contrast to acute urinary retention, a medical emergency, which is painful and the patient is unable to urinate despite a full bladder. Chronic urinary retention, whilst not immediately life-threatening, can lead to hydronephrosis and renal impairment and puts the patient at risk of acute-on-chronic retention, so requires diagnosis and treatment.

Other symptoms are anuria, where there is no urine output at all, and **oliguria**, where there is very low urine output. These are often due to the presence of underlying medical conditions such as **prostate problems**, pregnancy, and urinary tract infections. Patients with traumatic spinal cord injury, especially those resulting in the paralysis of muscles from the neck down, also often experience problems in micturition.

Clinical Tips for Treating Oliguria, Anuria

1. In oliguria, urine is produced and stored in the bladder. This is a relatively more manageable condition than anuria, in which urine is not formed and the bladder is empty. Therefore, accurate differentiation between these two related conditions is essential for proper treatment. Physical examination including observation, palpation and percussion is necessary to detect urine formation.

   Distention, discomfort on palpation, and solid resonance to percussion in the lower abdomen indicates formation of urine and a full bladder. The patient may express a desire to urinate but be unable to do so. Decreased urine output, lack of desire to urinate, a flat lower abdomen, and hollow resonance to percussion in the lower abdomen indicate lack of urine formatio in the bladder. This essentially corresponds to the Chinese term/disease Bi and is a medical emergency that needs to be treated in a hospital.

2. For anuria, auto-intoxication from metabolized waste can cause nausea, vomiting, dyspnea, tightness in the chest, systemic edema, constipation, irritability, delirium, convulsions, and loss of consciousness. This is an urgent condition requiring medical intervention.

3. Caution should be exercised when needling in the vacinity of the bladder if it is full and distended. Deep, vertical needling is contraindicated in such cases.

How is micturition/voiding dysfunction treated?

Treatment options are based on the underlying cause of the voiding dysfunction, severity of symptoms, and findings from the physical, laboratory, and medical test results. Treatment may consist of one or more of the following approaches. Your doctors will discuss which specific method(s) will be tried with your child.

Managing constipation. Proper management of constipation through the use of enemas, laxatives, and dietary fiber intake can reduce urinary wetting and urinary tract infections. Parents are encouraged to keep an elimination diary on the child. Over time, the stool softeners can be removed and the child remains on a high fiber intake.

Eliminating bladder irritants. Your doctor may recommend increasing your child's water intake to dilute the urine and eliminating caffeine, carbonated beverages, citrus juices, and chocolate - products thought to irritate the bladder and may make voiding uncomfortable for your child.

Treating urinary tract infections (UTI). A short course of antibiotics can be used in children with recurrent urinary tract infections.

Incorporating behavioral interventions. Behavioral interventions are tools and techniques children and their parents can use to gain control over voiding dysfunction. The goals of behavioral interventions are to help your child remain continent and empty the bladder effectively.

For nocturnal enuresis, an alarm system that rings when the bed gets wet can help the child respond to bladder sensations at night. The majority of research on bedwetting supports the use of urine alarms as the most effective treatment. Urine alarms are currently the only treatment associated with persistent improvement. The relapse rate is low, generally 5 to 10 percent, so that once a child's wetting improves, it almost always remains improved.

With daytime wetting and other types of voiding dysfunction,techniques such as increased water consumption, scheduled voiding (pee every 2 to 4 hours), high fiber intake, and Kegel exercises (contraction/relaxation) to strengthen pelvic floor muscles can be helpful. Relaxation and biofeedback may be used to help your child learn to relax as they void.

For some children, behavioral interventions are an adjunct (used in addition) to medications and other treatment methods. In other cases, these interventions are the primary treatment method. The behavioral interventions tried with your child will be individually tailored to his/her problems. They provide a means for helping your child learn to manage these problems independently.

Using anticholinergic medications. Anticholinergic medications, such as oxybutynin (Ditropan) or hyoscyamine (Levbid), are helpful in children with urge syndrome.

Surgery. Sometimes, though rarely, surgery is needed to correct an underlying anatomical problem that is the cause of the voiding dysfunction.

Primary/secondary enuresis:



Enuresis refers to a repeated inability to control urination. Use of the term is usually limited to describing individuals old enough to be expected to exercise such control.

History: Found evidence of mention in Egyptian medical texts as early as 1550 B.C.

Types of enuresis include:

- Nocturnal enuresis (bedwetting)
- Diurnal enuresis
- Mixed enuresis - Includes a combination of nocturnal and diurnal type. Therefore, urine is passed during both waking and sleeping hours.

Classification: 1. Primary enuresis refers to children who have never been successfully trained to control urination. This represents a fixation. 2. Secondary enuresis refers to children who have been successfully trained but revert back to wetting in a response to some sort of stressful situation. This represents a regression.

Current DSM-IV-TR Criteria: a. Repeated voiding of urine into bed or clothes (whether involuntary or intentional) b. Behavior must be clinically significant as manifested by either a frequency of twice a week for at least 3 consecutive months or the presence of clinically significant distress or impairment in social, academic (occupational), or other important areas of functioning. c. Chronological age is at least 5 years of age (or equivalent developmental level). d. The behavior is not due exclusively to the direct physiological effect of a substance (such as a diuretic) or a general medical condition (such as diabetes, spina bifida, a seizure disorder, etc.). All these criteria must be met in order to diagnose an individual.

Overall Prevalence and Characteristics: In the United States, approximately 15 to 20 percent of 5 year old children will develop symptoms related to disorder. Prevalence changes significantly with age. To be more specific, about 33 percent of 5 year-olds, 25 percent of 7 year olds, 15 percent of 9 year olds, 8 percent of 11 year olds, 4 percent of 13 year-olds, and 3 percent of 15 to 17 year-olds. Numbers show that diurnal enuresis is much less common. Overall, about 60 percent of those suffering are male. However,

this too depends on age. From ages 4 to 6, the number of boys and girls is about equal. However, the ratio changes so that by 11 years of age there are twice as many boys as girls. Incidence varies with social class with more incidences among those with low socioeconomic status. No evidence has been found related to ethnic differences.

Enuresis, more commonly called bed-wetting, is a disorder of elimination that involves the voluntary or involuntary release of urine into bedding, clothing, or other inappropriate places. In adults, loss of bladder control is often referred to as urinary incontinence rather than enuresis; it is frequently found in patients with late-stage Alzheimer's disease or other forms of dementia .

Description

Enuresis is a condition that has been described since 1500 B.C. People with enuresis wet their bed or release urine at other inappropriate times. Release of urine at night (nocturnal enuresis) is much more common than daytime, or diurnal, wetting. Enuresis commonly affects young children and is involuntary. Many cases of enuresis clear up by themselves as the child matures, although some children need behavioral or physiological treatment in order to remain dry.

There are two main types of enuresis in children. Primary enuresis occurs when a child has never established bladder control. Secondary enuresis occurs when a person has established bladder control for a period of six months, then relapses and begins wetting. To be diagnosed with enuresis, a person must be at least five years old or have reached a developmental age of five years. Below this age, problems with bladder control are considered normal.

Causes and symptoms

Symptoms

The symptoms of enuresis are straightforward—a person urinates in inappropriate places or at inappropriate times. The causes of enuresis are not so clear. A small number of children have abnormalities in the anatomical structure of their kidney or bladder that interfere with bladder control, but normally the cause is not the physical structure of the urinary system. A few children appear to have to have a lower-than-normal ability to concentrate urine, due to low levels of antidiuretic hormone (ADH). This hormone helps to regulate fluid balance in the body. Large amounts of dilute urine cause the bladder to overflow at night. For the majority of bedwetters, there is no single clear physical or psychological explanation for enuresis.

Causes in children

The fourth edition of the ***Diagnostic and Statistical Manual of Mental Disorders*** , fourth edition, text revision, or (*DSM-IV-TR*), does not distinguish between children who wet the bed involuntarily and those who voluntarily release urine. Increasingly, however, research findings suggest that voluntary and involuntary enuresis have different causes.

Involuntary enuresis is much more common than voluntary enuresis. Involuntary enuresis may be categorized as either primary or secondary. Primary enuresis occurs when young children lack bladder control from infancy. Most of these children have urine control problems only during sleep; they do not consciously, intentionally, or maliciously wet the bed. Research suggests that children who are nighttime-only bed wetters may have a nervous system that is slow to process the feeling of a full bladder.

Consequently, these children do not wake up in time to relieve themselves. In other cases, the child's enuresis may be related to a sleep disorder.

Children with diurnal enuresis wet only during the day. There appear to be two types of daytime wetters. One group seems to have difficulty controlling the urge to urinate. The other group consciously delays urinating until they lose control. Some children have both diurnal and nocturnal enuresis.

Secondary enuresis occurs when a child has stayed dry day and night for at least six months, then returns to wetting. Secondary enuresis usually occurs at night. Many studies have been done to determine if there is a psychological component to enuresis. Researchers have found that secondary enuresis is more likely to occur after a child has experienced a stressful life event such as the birth of a sibling, divorce or death of a parent, or moving to a new house.

Several studies have investigated the association of primary enuresis and psychiatric or behavior problems. The results suggest that primary nocturnal enuresis is *not* caused by psychological disorders. Bed-wetting runs in families, however, and there is strong evidence of a genetic component to involuntary enuresis.

Unlike involuntary enuresis, voluntary enuresis is not common. It is associated with such psychiatric disorders as oppositional defiant disorder , and is substantially different from ordinary nighttime bed-wetting. Voluntary enuresis is always secondary.

Causes in adults

Enuresis or urinary incontinence in elderly adults may be caused by loss of independent control of body functions resulting from dementia, bladder infections, uncontrolled diabetes, side effects of medications, and weakened bladder muscles. Urinary incontinence in adults is managed by treatment of the underlying medical condition, if one is present; or by the use of adult briefs with disposable liners.

Demographics

Enuresis is a problem of the young and is more common in boys than girls. At age five, about 7% of boys and 3% of girls have enuresis. This number declines steadily in older children; by age 18, only about 1% of adolescents experience enuresis. Studies done in several countries suggest that there is no apparent cultural influence on the incidence of enuresis in children. On the other hand, the disorder does appear to run in families; children with one parent who wet the bed as a child are five to seven times more likely to have enuresis than children whose parents did not have the disorder in childhood.

Diagnosis

Enuresis is most often diagnosed in children because the parents express concern to the child's doctor. The pediatrician or family physician will give the child a physical examination to rule out medical conditions that may be causing the problem, including structural abnormalities in the child's urinary tract. The doctor may also rule out a sleep disorder as a possible cause. In many cases the pediatrician can reassure the child's parents and give them helpful advice.

According to the American Psychiatric Association, to make a **diagnosis** of enuresis, a child must have reached the chronological or developmental age of five. Inappropriate urination must occur at least twice a week for three months; or the frequency of inappropriate urination must cause significant distress and

interfere with the child's school and/or social life. Finally, the behavior cannot be caused exclusively by a medical condition or as a side effect of medication.

Treatments

Treatment for enuresis is not always necessary. About 15% of children who have enuresis outgrow it each year after age six. When treatment is desired, a physician will rule out obvious physical causes of enuresis through a physical examination and medical history. Several different treatment options are then available.

Behavior modification

Behavior modification is often the treatment of choice for enuresis. It is inexpensive and has a success rate of about 75%. The child's bedding includes a special pad with a sensor that rings a bell when the pad becomes wet. The bell wakes the child, who then gets up and goes to the bathroom to finish emptying his bladder. Over time, the child becomes conditioned to waking up when the bladder feels full.

Once this response is learned, some children continue to wake themselves help from without the alarm, while others are able to sleep all night and remain dry. A less expensive behavioral technique involves setting an alarm clock to wake the child every night after a few hours of sleep, until the child learns to wake up spontaneously. In trials, this method was as effective as the pad-and-alarm system. A newer technique involves an ultrasound monitor worn on the child's pajamas. The monitor can sense bladder size, and sets off an alarm once the bladder reaches a predetermined level of fullness. This technique avoids having to change wet bed pads.

Other behavior modifications that can be used alone or with the pad-and-alarm system include:

- restricting liquids starting several hours before bedtime
- waking the child up in the night to use the bathroom
- teaching urinary retention techniques
- giving the child positive reinforcement for dry nights and being sympathetic and understanding about wet nights

Treatment with medications

There are two main drugs for treating enuresis. **Imipramine** , a tricyclic antidepressant, has been used since the early 1960s. It is not clear why this antidepressant is effective in treating enuresis when other antidepressants are not. Desmopressin acetate (DDAVP) has been widely used to treat enuresis since the 1990s. It is available as a nasal spray or tablet. Both imipramine and DDAVP are very effective in preventing bed-wetting, but have high relapse rates if medication is stopped.

Alternative therapies

Some success in treating bed-wetting has been reported using hypnosis. When hypnosis works, the results are seen within four to six sessions. **Acupuncture** and massage have also been used to treat enuresis, with inconclusive results.

Psychotherapy

Primary enuresis does not require **psychotherapy** . Secondary enuresis, however, is often successfully treated with therapy. The goal of the treatment is to resolve the underlying stressful event that has caused a relapse into bed-wetting. Unlike children with involuntary enuresis, children who intentionally urinate in inappropriate places often have other serious psychiatric disorders. Enuresis is usually a symptom of another disorder. Therapy to treat the underlying disorder is essential to resolving the enuresis.

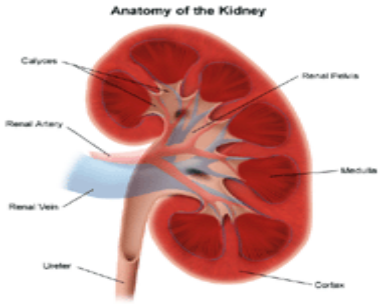
Prognosis

Enuresis is a disorder that most children outgrow. For those who do receive treatment, the overall success rate of behavioral therapy is 75%. The short-term success rate with drug treatments is even higher than with behavioral therapy. Drugs do not, however, eliminate the enuresis. Many children who take drugs to control their bed-wetting relapse when the drugs are stopped.

Prevention

Although enuresis cannot be prevented, one side effect of the disorder is the shame and social embarrassment it causes. Children who wet may avoid sleepovers, camp, and other activities where their bed-wetting will become obvious. Loss of these opportunities can cause a loss of self-esteem, social isolation, and adjustment problems. A kind, low-key approach to enuresis helps to prevent these problems.

End Stage Renal Disease (ESRD)



What is renal failure?

Renal failure refers to temporary or permanent damage to the kidneys that results in loss of normal kidney function. There are two different types of renal failure--acute and chronic. Acute renal failure has an abrupt onset and is potentially reversible. Chronic renal failure progresses slowly over at least three months and can lead to permanent renal failure. The causes, symptoms, treatments, and outcomes of acute and chronic are different.

Conditions that may lead to acute or chronic renal failure may include, but are not limited to, the following:

Acute Renal Failure	Chronic Renal Failure
Myocardial infarction. A heart attack may occasionally lead to temporary kidney failure.	**Diabetic nephropathy.** Diabetes can cause permanent changes, leading to kidney damage.
Rhabdomyolysis. Kidney damage that can occur from muscle breakdown. This condition can occur from severe dehydration, infection, or other causes.	**Hypertension.** Chronic high blood pressure (hypertension) can lead to permanent kidney damage.
Decreased blood flow to the kidneys for a period of time. This may occur from blood loss or shock.	**Lupus (SLE).** A chronic inflammatory/autoimmune disease that can injure the skin, joints, kidneys, and nervous system.
An obstruction or blockage along the urinary tract.	**A prolonged urinary tract obstruction or blockage.**
Hemolytic uremic syndrome. Usually caused by an E. coli infection, kidney failure develops as a result of obstruction to the small functional structures and vessels inside the kidney.	**Alport syndrome.** An inherited disorder that causes deafness, progressive kidney damage, and eye defects.
Ingestion of certain medications that may cause toxicity to the kidneys.	**Nephrotic syndrome.** A condition that has several different causes. Nephrotic syndrome is characterized by protein in the urine, low protein in the blood, high cholesterol levels, and tissue swelling.
Glomerulonephritis. A type of kidney disease that involves glomeruli. During glomerulonephritis, the glomeruli become inflamed and impair the kidney's ability to filter urine. Glomerulonephritis may lead to chronic renal failure in some individuals.	**Polycystic kidney disease.** A genetic disorder characterized by the growth of numerous cysts filled with fluid in the kidneys.
Any condition that may impair the flow of oxygen and blood to the kidneys such as cardiac arrest.	**Cystinosis.** An inherited disorder in which the amino acid cystine (a common protein-building compound) accumulates within specific cellular bodies of the kidney, known as lysosomes.
	Interstitial nephritis or pyelonephritis. An inflammation to the small internal structures in the kidney.

What is end-stage renal disease (ESRD)?

End-stage renal disease is when the kidneys permanently fail to work.

What are the symptoms of renal failure?

The symptoms for acute and chronic renal failure may be different. The following are the most common symptoms of acute and chronic renal failure. However, each individual may experience symptoms differently. Symptoms may include:

Acute: (Symptoms of acute renal failure depend largely on the underlying cause.)

- Hemorrhage

- Fever
- Weakness
- Fatigue
- Rash
- Diarrhea or bloody diarrhea
- Poor appetite
- Severe vomiting
- Abdominal pain
- Back pain
- Muscle cramps
- No urine output or high urine output
- History of recent infection (a risk factor for acute renal failure)
- Pale skin
- Nosebleeds
- History of taking certain medications (a risk factor for acute renal failure)
- History of trauma (a risk factor for acute renal failure)
- Swelling of the tissues
- Inflammation of the eye
- Detectable abdominal mass
- Exposure to heavy metals or toxic solvents (a risk factor for acute renal failure)

Chronic:

- Poor appetite
- Vomiting
- Bone pain
- Headache
- Insomnia
- Itching
- Dry skin
- Malaise
- Fatigue with light activity
- Muscle cramps
- High urine output or no urine output
- Recurrent urinary tract infections
- Urinary incontinence
- Pale skin
- Bad breath
- Hearing deficit
- Detectable abdominal mass
- Tissue swelling
- Irritability
- Poor muscle tone
- Change in mental alertness
- Metallic taste in mouth

The symptoms of acute and chronic renal failure may resemble other conditions or medical problems. Always consult your doctor for a diagnosis.

What is the treatment for acute and chronic renal failure?

Specific treatment for renal failure will be determined by your doctor based on:

- Your age, overall health, and medical history
- Extent of the disease
- Type of disease (acute or chronic)
- Underlying cause of the disease
- Your tolerance for specific medications, procedures, or therapies
- Expectations for the course of the disease
- Your opinion or preference

Treatment may include:

- Hospitalization
- Administration of intravenous (IV) fluids in large volumes (to replace depleted blood volume)
- Diuretic therapy or medications (to increase urine output)
- Close monitoring of important electrolytes such as potassium, sodium, and calcium
- Medications (to control blood pressure)
- Specific diet requirements

In some cases, patients may develop severe electrolyte disturbances and toxic levels of certain waste products normally eliminated by the kidneys. Patients may also develop fluid overload. Dialysis may be indicated in these cases.

Treatment of chronic renal failure depends on the degree of kidney function that remains. Treatment may include:

- Medications (to help with growth, prevent bone density loss, and/or to treat anemia)
- Diuretic therapy or medications (to increase urine output)
- Specific diet restrictions or modifications
- Dialysis
- Kidney transplantation

Psychological Interventions:

End-stage renal disease is progressive, and the disturbances it brings are progressive. Any intervention has to be tailored to the progress of the disease itself, with the individual's level of physical, psychological, and social functioning as the central focus. The concurrent physiologic, psychological, and social stresses demand cognitive effort from the patient in coping. Researchers suggest that patient adherence to a medical regime is significantly related to high social desirability and a shorter length of time on dialysis. Patients with ESRD are empowered for self-care in matters of drug administration and PD management, but adherence to treatment requirements must be a voluntary act of submission, with consent for the adjustment and adaptation to the illness and treatment. Psychosocial intervention is best started as early (at diagnosis) and demands continuous effort.

Nursing Implications: End-stage renal disease has a characteristically downward trajectory. Patients have to come to terms with their current physical condition. Psychosocial nursing interventions should attempt to facilitate adjustment to changes in the course of the illness and to normalize social interaction and lifestyle by preventing medical crises, controlling symptoms, and incorporating the PD treatment regimes into daily living. Knowledge can significantly minimize a patient's anxiety. It

is crucial that nurses have the skills to provide clear information, to help patients identify their goals in the course of treatment, and to assist with problem-solving for optimal physical functioning. Assessment: Assessment determines the patient's needs, identifies problems and potential problems, and collects information for a treatment plan so that appropriate support can be rendered. The assessment therefore focuses on the effect of the illness on the patient. Useful information includes the patient's lifestyle, patterns of daily living, personality, strengths and interests, normal coping patterns, understanding of the current illness, perception of treatment regimes, recent life stresses or changes, and major issues raised by the disease. By listening to the patient and the family in the course of discussion, nurses can identify the observable psychosocial interferences consequent to the disease and the needs for assistance. At the same time, information on the expected course and likely outcome of the disease can be provided.

Encouragement: The role of the health care professional is to encourage and, where possible, to enable patients to accept responsibility for their health and well-being and to fulfill their obligations within the family and society. As well as providing knowledge and clarifying misconceptions, nurses can encourage patients to accept the personal limitations consequent to the illness and its treatment. When a patient is encouraged to perform self-care, better self-esteem and power to maintain health are established. When open discussion and awareness of the mutual situation is encouraged between patients and their partners, positive and understanding attitudes are reinforced. The perception of emotional support has a documented association with better physical and mental health in dialysis patients.

Life Enhancement: Dependent patients may adapt to treatment regimes more easily, but excessive dependence can create extreme demands on caregivers and can impede rehabilitation. Some patients may achieve a secondary gain from the illness, and some may enjoy the role of "patient." Nurses can facilitate a patient's adaptations to treatment requirements by maximizing the patient's strengths and supporting the patient in the treatment environment at home, while making judicious use of available resources. Individualizing treatment and minimizing its complexity may encourage adherent behavior. Frequent assessment, education, motivation, reinforcement, encouragement, and teaching concerning self-management and self-monitoring will, at the very least, maximize the patient's comprehension of the illness and personal motivation for adherence— an especially important requisite for living with a chronic condition. Participation by patients in patient support networks, rehabilitation activities, physical exercise schedules, and educational programs may help individuals to establish new supportive relationships, to achieve social recognition and appreciation, and to overcome social isolation, which has been discovered to be associated with treatment noncompliance.

Compliance—adherence to treatment requirements in a therapeutic regimen—is a critical concept for chronic renal failure patients and for their caregivers alike. Compliance changes over time. Significant differences related to social support have been found, especially when the demographic variable of education is controlled.

SUPPORT TO FAMILY

Family members play an important role in the wellbeing of the PD patient. They should not be neglected in the process of patient care. A change in the pattern of family life (integrating the lives of family members more flexibly with the patient's life) may be necessary to meet the patient's PD needs. The patient and the family should be encouraged to share their feelings in a trusting relationship and to make flexible adjustments to cope with the course of the patient's illness. Previous studies have revealed that sadness, guilt, and loss were pervasive and prevalent in partners—a unique

perspective on the negative impact that dialysis can have on couples. The mobilization of community services is useful to reduce stressors.

Nursing Intervention: Local experience in the Renal Unit, Tuen Mun Hospital, Hong Kong SAR, illustrates practicable interventions. In our unit, for all adult ESRD patients who will start on PD, psychosocial intervention starts with a predialysis briefing and continues with subsequent encounters at the predialysis assessment, break-in education, teaching and learning sessions, telephone visits, home visits, and counseling sessions. Through the collaborative efforts of health care professionals and the patient support group, rehabilitation activities are coordinated with the goals of enhancing peer support and optimizing mental strengths, physical fitness, and social strengths. Activities include community networks, voluntary visits to new patients, a tuck shop, production of a CD for peer support, health talks, maintenance of a Web site (www.hk-doctor.com/kpa/), Tai Chi, gateball, overnight camps, outings, dinners, mah-jongg competitions, karaoke competitions, and participation in the rehabilitation programs organized by the Hong Kong Society of Nephrology. During the period February 1993 – August 2002, we followed 694 patients. Most maintained a positive attitude toward life and got as much out of each day as they possibly could by integrating PD with life. Four patients and 3 relatives were found to be experiencing depressive symptoms that necessitated psychiatric treatment. One patient and 3 relatives committed suicide.

SUPPORT TO STAFF

Dialysis professionals can have a major impact— positive or negative—on how patients deal with the life changes caused by PD. Prospects for renal rehabilitation increase if the dialysis staff convey a positive attitude and show belief in the patient's potential to achieve rehabilitation goals—particularly goals that the patient is able to control, such as adherence to treatment requirements, exercise, and self-care. However, not uncommonly, staff are working in emotionally-laden situations, such as those involving difficult behavior on the part of a patient or caregiver, deteriorating conditions in patients with whom a good rapport has been established, or failure of treatment. The experience of anxiety related to job demands and of spiritual distress related to an inability to find meaning or purpose in professional and personal life is frequently mentioned by staff. Renal team members can be given opportunities (in meetings or conferences, for example) to evaluate the causes of stress, to develop ideas, to share ideas with peers, and to create opportunities to honor or encourage members of the team. Some ways to divert attention from stress are to seek humor in experiences, to learn from patients to accept limitations while remaining professional in demeanor and presentation, and to take appropriate time away from work to rest and to play.

What is dialysis?

Dialysis is a procedure that is performed routinely on persons who suffer from acute or chronic renal failure, or who have ESRD. The process involves removing waste substances and fluid from the blood that are normally eliminated by the kidneys. Dialysis may also be used for individuals who have been exposed to or ingested toxic substances to prevent renal failure from occurring. There are two types of dialysis that may be performed, including the following:

- **Peritoneal dialysis.** Peritoneal dialysis is performed by surgically placing a special, soft, hollow tube into the lower abdomen near the navel. After the tube is placed, a special solution called dialysate is instilled into the peritoneal cavity. The peritoneal cavity is the space in the abdomen that houses the organs and is lined by two special membrane layers called the peritoneum. The dialysate is left in the abdomen for a designated period of time which will be determined by your doctor. The dialysate fluid absorbs the waste products and toxins through the peritoneum. The

fluid is then drained from the abdomen, measured, and discarded. There are three different types of peritoneal dialysis: continuous ambulatory peritoneal dialysis (CAPD), continuous cyclic peritoneal dialysis (CCPD), and intermittent peritoneal dialysis (IPD).

CAPD does not require a machine. Exchanges, often referred to as "passes," can be done three to five times a day during waking hours. CCPD requires the use of a special dialysis machine that can be used in the home. This type of dialysis is done automatically, even while you are asleep. IPD uses the same type of machine as CCPD, but treatments take longer. IPD can be done at home, but usually is done in the hospital.

Possible complications of peritoneal dialysis include an infection of the peritoneum, or peritonitis, where the catheter enters the body. Peritonitis causes fever and stomach pain. Your diet for peritoneal dialysis will be planned with a dietitian, who can help you choose meals according to your doctor's orders. Generally:

- You may have special protein, salt, and fluid needs.
- You may have special potassium restrictions.
- You may need to reduce your calorie intake, since the sugar in the dialysate may cause weight gain.

- **Hemodialysis.** Hemodialysis can be performed at home or in a dialysis center or hospital by trained healthcare professionals. A special type of access, called an arteriovenous (AV) fistula, is placed surgically, usually in your arm. This involves joining an artery and a vein together. An external, central, intravenous (IV) catheter may also be inserted, but is less common for long-term dialysis. After access has been established, you will be connected to a large hemodialysis machine that drains the blood, bathes it in a special dialysate solution which removes waste substances and fluid, then returns it to your bloodstream.

  Hemodialysis is usually performed several times a week and lasts for four to five hours. Because of the length of time hemodialysis takes, it may be helpful to bring reading material, in order to pass the time during this procedure. During treatment you can read, write, sleep, talk, or watch TV.

  At home, hemodialysis is done with the help of a partner, often a family member or friend. If you choose to do home hemodialysis, you and your partner will receive special training.

  Possible complications of hemodialysis include muscle cramps and hypotension (sudden drop in blood pressure). Hypotension may cause you to feel dizzy or weak, or sick to your stomach. Side effects are avoided by following the proper diet and taking medications, as prescribed by your doctor. A dietitian will work with you to plan your meals, according to your doctor's orders. Generally:
  - You may eat foods high in protein such as meat and chicken (animal proteins).
  - You may have potassium restrictions.
  - You may need to limit the amount you drink.
  - You may need to avoid salt.
  - You may need to limit foods containing mineral phosphorus (such as milk, cheese, nuts, dried beans, and soft drinks).
- When a person’s kidneys fail due to damage or disease, dialysis treatment is used as a replacement for kidney function. Dialysis is a procedure in which a person’s body is cleaned of impurities and toxins, a task which his or her kidneys would perform if they were still functional. People who undergo regular dialysis treatment are usually in end-stage renal failure and have no more than 10% to 15% kidney function remaining.

- Dialysis treatment is crucial for people suffering from kidney failure. Without functional kidneys, they can no longer remove salts, waste, and water, and their bodies cannot maintain safe levels of sodium, potassium, and other minerals. Dialysis also helps to control blood pressure, which can rise or fall dangerously due to an imbalance of salts and minerals.
- For some people, dialysis treatment is only a short-term measure, required as a result of acute kidney failure, or damage or disease that causes temporarily impaired kidney function. Others need regular dialysis as a result of chronic kidney failure. In these cases, the kidneys are permanently damaged, and only a successful kidney transplant will end the need for chronic dialysis.
- There are two types of dialysis treatment. These are called hemodialysis and peritoneal dialysis. Each procedure works slightly differently, but operates on the same principles of replacing kidney function by removing waste products from the blood.
- The process of hemodialysis uses an artificial kidney, called a hemodialyzer, to remove the waste products and fluids that build up in the blood. To allow blood to flow through the artificial kidney, a dialysis patient must undergo a minor surgical procedure that creates an access point in an arm or leg. For the average person, dialysis treatment occurs three times a week for around four hours per session. The actual frequency and time depends on the amount of kidney function an individual person has remaining, how quickly waste products build up in the blood, and other factors.
- The second type of dialysis is called peritoneal dialysis. Rather than removing the blood from the body in order to remove waste, the blood is cleaned while still inside the body. A doctor first creates an access point with a minor surgical procedure that places a catheter in the abdomen. At each peritoneal dialysis session, the catheter point is slowly filled with a solution called the dialysate, which then fills the interior of the abdominal cavity. Waste products in the blood filter through arteries and veins into the dialysate via osmosis.
- While dialysis can replace the work that the kidneys do, dialysis is not itself a cure for kidney failure; rather it is a treatment that manages the condition. Many people who require chronic dialysis can live fairly normal lives, apart from the need to undergo the treatment several times per week. As the dialysis procedure is improved, it is likely that people requiring this treatment will be able to live just as long as people with functional kidneys.

Primary and secondary infertility:

Infertility primarily refers to the biological inability of a person to contribute to conception. Infertility may also refer to the state of a woman who is unable to carry a pregnancy to full term. There are many biological causes of infertility, including some that medical intervention can treat.

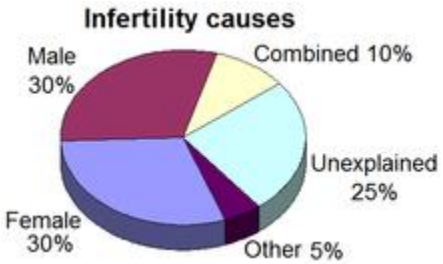
Women who are fertile experience a natural period of fertility before and during ovulation, and they are naturally infertile during the rest of the menstrual cycle. Fertility awareness methods are used to discern when these changes occur by tracking changes in cervical mucus or basal body temperature.

Definitions of infertility differ, with demographers tending to define infertility as childlessness in a population of women of reproductive age, while the epidemiological definition is based on "trying for" or "time to" a pregnancy, generally in a population of women exposed to a probability of conception. The time that needs to pass (during which the couple has tried to conceive) for that couple to be diagnosed with infertility differs between different jurisdictions. Existing definitions of infertility lack uniformity, rendering comparisons in prevalence between countries or over time problematic, and therefore data estimating the prevalence of infertility cited by various sources differs significantly. A couple that has tried unsuccessfully to have a child after a certain period of time (often a short period, but definitions vary) is sometimes said to be **subfertile**, meaning less fertile

than a typical couple. Both infertility and subfertility are defined as the inability to conceive after a certain period of time (the length of which vary), so often the two terms overlap.

Couples with **primary infertility** have never been able to conceive, while, on the other hand, **secondary infertility** is difficulty conceiving after already having conceived (and either carried the pregnancy to term or had a miscarriage). Secondary infertility is not present if there has been a change of partners (this follows tautologically from the convention of speaking of *couples*, rather than individuals, as being infertile; if there is a change of partners, then a new couple is created, with its own chances to be infertile.)

Causes



Data from UK, 2009.

This section deals with unintentional causes of sterility. For more information about surgical techniques for preventing procreation, see Sterilization (surgical procedure).

Causes in either sex

Factors that can cause male as well as female infertility are:

- Genetic factors
  - A Robertsonian translocation in either partner may cause recurrent spontaneous abortions or complete infertility.
- General factors
  - Diabetes mellitus, thyroid disorders, adrenal disease
- Hypothalamic-pituitary factors
  - Hyperprolactinemia
  - Hypopituitarism
  - The presence of anti-thyroid antibodies is associated with an increased risk of unexplained subfertility with an odds ratio of 1.5 and 95% confidence interval of 1.1–2.0.
- Environmental factors
  - Toxins such as glues, volatile organic solvents or silicones, physical agents, chemical dusts, and pesticides. Tobacco smokers are 60% more likely to be infertile than non-smokers.

German scientists have reported that a virus called Adeno-associated virus might have a role in male infertility, though it is otherwise not harmful. Mutation that alters human DNA adversely can cause infertility, the human body thus preventing the tainted DNA from being passed on.

Specific female causes
Further information: Female infertility

The following causes of infertility may only be found in females.

For a woman to conceive, certain things have to happen: intercourse must take place around the time when an egg is released from her ovary; the systems that produce eggs and sperm have to be working at optimum levels; and her hormones must be balanced.

Some women are infertile because their ovaries do not mature and release eggs. In this case synthetic FSH by injection or Clomid (Clomiphene citrate) via a pill can be given to stimulate follicles to mature in the ovaries.

Problems affecting women include endometriosis or damage to the fallopian tubes (which may have been caused by infections such as chlamydia).

Other factors that can affect a woman's chances of conceiving include being over- or underweight, or her age as female fertility declines sharply after the age of 35. Sometimes it can be a combination of factors, and sometimes a clear cause is never established.

Common causes of infertility of females include:

- ovulation problems (e.g. polycystic ovarian syndrome
- tubal blockage
- Pelvic inflammatory disease
- age-related factors
- uterine problems
- previous tubal ligation
- endometriosis
- advanced maternal age

Specific male causes
Further information: Male infertility

The main cause of male infertility is low semen quality. Another possible cause is a low sperm count. Infertility associated with viable, but immotile sperm may be caused by Primary Ciliary Dyskinesia.

Combined infertility

In some cases, both the man and woman may be infertile or sub-fertile, and the couple's infertility arises from the combination of these conditions. In other cases, the cause is suspected to be immunological or genetic; it may be that each partner is independently fertile but the couple cannot conceive together without assistance.

Unexplained infertility

In the US, up to 20% of infertile couples have unexplained infertility. In these cases abnormalities are likely to be present but not detected by current methods. Possible problems could be that the egg is not released at the optimum time for fertilization, that it may not enter the fallopian tube, sperm may not be

able to reach the egg, fertilization may fail to occur, transport of the zygote may be disturbed, or implantation fails. It is increasingly recognized that egg quality is of critical importance and women of advanced maternal age have eggs of reduced capacity for normal and successful fertilization. Also, polymorphisms in folate pathway genes could be one reason for fertility complications in some women with unexplained infertility.

Assessment

If both partners are young and healthy and have been trying to conceive for one year without success, a visit to the family doctor could help to highlight potential medical problems earlier rather than later. The doctor may also be able to suggest lifestyle changes to increase the chances of conceiving.

Women over the age of 35 should see their family doctor after six months as fertility tests can take some time to complete, and age may affect the treatment options that are open in that case.

A family doctor takes a medical history and gives a physical examination. They can also carry out some basic tests on both partners to see if there is an identifiable reason for not having achieved a pregnancy. If necessary, they refer patients to a fertility clinic or local hospital for more specialized tests. The results of these tests help determine the best fertility treatment.

Treatment

Treatment depends on the cause of infertility, but may include counselling, fertility treatments, which include in vitro fertilization. Treatment methods for infertility may be grouped as medical or complementary and alternative treatments. Some methods may be used in concert with other methods. Drugs used for women include Clomiphene citrate, Human menopausal gonadotropin, Follicle-stimulating hormone, Human chorionic gonadotropin, Gonadotropin-releasing hormone analogs, Aromatase inhibitor, Metformin.

At-home conception kit

In 2007 the FDA cleared the first at home tier one medical conception device to aid in conception. The key to the kit are cervical caps for conception. This at home [cervical cap] insemination method allows all the semen to be placed up against the cervical os for six hours allowing all available sperm to be placed directly on the cervical os. For low sperm count, low sperm motility, or a tilted cervix using a cervical cap aids conception. This is a prescriptive medical device, but not commonly prescribed by physicians.

At-home assessment

Prior to expensive fertility procedures, many women and couples turn to online sources to determine their chances of success. A take-home baby assessment can provide a best guess estimate compared with women who have succeeded with in vitro fertilization, based on variables such as maternal age, duration of infertility and number of prior pregnancies.

Medical treatments

Medical treatment of infertility generally involves the use of fertility medication, medical device, surgery, or a combination of the following. If the sperm are of good quality and the mechanics of the woman's reproductive structures are good (patent fallopian tubes, no adhesions or scarring), physicians may start

by prescribing a course of ovarian stimulating medication. The physician may also suggest using a conception cap cervical cap, which the patient uses at home by placing the sperm inside the cap and putting the conception device on the cervix, or intrauterine insemination (IUI), in which the doctor introduces sperm into the uterus during ovulation, via a catheter. In these methods, fertilization occurs inside the body.

If conservative medical treatments fail to achieve a full term pregnancy, the physician may suggest the patient undergo in vitro fertilization (IVF). IVF and related techniques (ICSI, ZIFT, GIFT) are called assisted reproductive technology (ART) techniques.

ART techniques generally start with stimulating the ovaries to increase egg production. After stimulation, the physician surgically extracts one or more eggs from the ovary, and unites them with sperm in a laboratory setting, with the intent of producing one or more embryos. Fertilization takes place outside the body, and the fertilized egg is reinserted into the woman's reproductive tract, in a procedure called embryo transfer.

Other medical techniques are e.g. tuboplasty, assisted hatching, and Preimplantation genetic diagnosis.

Ethics

There are several ethical issues associated with infertility and its treatment.

- High-cost treatments are out of financial reach for some couples.
- Debate over whether health insurance companies (e.g. in the US) should be required to cover infertility treatment.
- Allocation of medical resources that could be used elsewhere
- The legal status of embryos fertilized in vitro and not transferred in vivo. (See also Beginning of pregnancy controversy).
- Pro-life opposition to the destruction of embryos not transferred in vivo.
- IVF and other fertility treatments have resulted in an increase in multiple births, provoking ethical analysis because of the link between multiple pregnancies, premature birth, and a host of health problems.
- Religious leaders' opinions on fertility treatments.
- Infertility caused by DNA defects on the Y chromosome is passed on from father to son. If natural selection is the primary error correction mechanism that prevents random mutations on the Y chromosome, then fertility treatments for men with abnormal sperm (in particular ICSI) only defer the underlying problem to the next male generation.

Psychological impact

The consequences of infertility are manifold and can include societal repercussions and personal suffering. Advances in assisted reproductive technologies, such as IVF, can offer hope to many couples where treatment is available, although barriers exist in terms of medical coverage and affordability. The medicalization of infertility has unwittingly led to a disregard for the emotional responses that couples experience, which include distress, loss of control, stigmatization, and a disruption in the developmental trajectory of adulthood.

Infertility may have profound psychological effects. Partners may become more anxious to conceive, increasing sexual dysfunction. Marital discord often develops in infertile couples, especially when they are under pressure to make medical decisions. Women trying to conceive often have clinical depression

rates similar to women who have heart disease or cancer. Even couples undertaking IVF face considerable stress.

The emotional losses created by infertility include the denial of motherhood as a rite of passage; the loss of one's anticipated and imagined life; feeling a loss of control over one's life; doubting one's womanhood; changed and sometimes lost friendships; and, for many, the loss of one's religious environment as a support system.

Emotional stress and marital difficulties are greater in couples where the infertility lies with the man.

Social impact

In many cultures, inability to conceive bears a stigma. In closed social groups, a degree of rejection (or a sense of being rejected by the couple) may cause considerable anxiety and disappointment. Some respond by actively avoiding the issue altogether; middle-class men are the most likely to respond in this way.

In an effort to end the shame and secrecy of infertility, Redbook in October 2011 launched a video campaign, The Truth About Trying, to start an open conversation about infertility, which strikes one in eight women in the United States. In a survey of couples having difficulty conceiving, conducted by the pharmaceutical company Merck, 61 percent of respondents hid their infertility from family and friends. Nearly half didn't even tell their mothers. The message of those speaking out: It's not always easy to get pregnant, and there's no shame in that.

There are legal ramifications as well. Infertility has begun to gain more exposure to legal domains. An estimated 4 million workers in the U.S. used the Family and Medical Leave Act (FMLA) in 2004 to care for a child, parent or spouse, or because of their own personal illness. Many treatments for infertility, including diagnostic tests, surgery and therapy for depression, can qualify one for FMLA leave. It has been suggested that infertility be classified as a form of disability.

Fictional representation

Perhaps except for infertility in science fiction, films and other fiction depicting emotional struggles of assisted reproductive technology have had an upswing first in the latter part of the 2000s decade, although the techniques have been available for decades.Yet, the amount of people that can relate to it by personal experience in one way or another is ever growing, and the variety of trials and struggles is huge.

ROLE OF STRESS AND ANXIETY IN PSYCHODERMATOLOGICAL CONDITION

Unit-VII

WHAT IS PSYCHODERMATOLOGICAL CONDITION?

A psychodermatologic disorder is a condition that involves an interaction between the mind and the skin. Psycho dermatologic disorders fall into three categories: psycho physiologic disorders, primary psychiatric disorders and secondary psychiatric disorders. Psycho physiologic disorders (e.g., psoriasis and eczema) are associated with skin problems that are not directly connected to the mind but that react to

emotional states, such as stress. Primary psychiatric disorders involve psychiatric conditions that result in self-induced cutaneous manifestations, such as trichotillomania and delusions of parasitosis. Secondary psychiatric disorders are associated with disfiguring skin disorders. The disfigurement results in psychologic problems, such as decreased self-esteem, depression or social phobia. Most psychodermatologic disorders can be treated with anxiety-decreasing techniques or, in extreme cases, psychotropic medications.

VARIOUS PSYCHODERMATOLOGICAL CONDITIONS

Psoriasis

It is psycho-somatic condition, the upper layer of the skin, taken and forms scales, the underneath formed on the skin surface fail to fall, and the scalf is exposed to sun or heath form the stores variation in climate, metal skin. There is fat production of upper of epidarmeties of the skin.

Onset or exacerbation of psoriasis can be predated by a number of common stressors. Stress has been reported in 44% of patients prior to the initial flare of psoriasis, and recurrent flares have been attributed to stress in up to 80% of individuals. Early onset psoriasis before age 40 years may be more easily triggered by stress than late onset disease, and patients who self-report high levels of psychological stress may have more severe skin and joint symptoms. The most common psychiatric symptoms attributed to psoriasis include disturbances in body image and impairment in social and occupational functioning. Quality of life may be severely affected by the chronicity and visibility of psoriasis as well as by the need for lifelong treatment. Five dimensions of the stigma associated with psoriasis have been identified: (1) anticipation of rejection, (2) feelings of being flawed, (3) sensitivity to the attitudes of society, (4) guilt and shame, and (5) secretiveness. Depressive symptoms and suicidal ideation occur more frequently in severe psoriasis compared with controls. Depression may modulate itch perception, exacerbate pruritus, and lead to difficulties with initiating and maintaining sleep. In a study of 217 psoriasis patients with associated depression, 9.7% acknowledged a “wish to be dead,” and 5.5% reported active suicidal ideation.

Psoriasis also affects sexual functioning. In one study, 30% to 70% of patients reported a decline in sexual activity, and these patients more frequently reported joint pains, psoriasis affecting the groin region, scaling, and pruritus compared with patients without sexual complaints. These patients also had higher depression scores, greater tendency to seek the approval of others, and a marginally greater tendency to drink alcohol.

Psycho physiologically, neuropeptides such as vasoactive intestinal peptide and substance P may have a role in the development of psoriatic lesions. Imbalance of vasoactive intestinal peptide and substance P has been reported in psoriasis, and there may be an associated increase in stress-induced autonomic response and diminished pituitary-adrenal activity. Biofeedback, meditation, hypnosis-induced relaxation, behavioral techniques, symptom-control imagery training, and antidepressants are important adjuncts to standard therapies for this disease.

Atopic Dermatitis

Stressful life events preceding the onset of disease have been found in more than 70% of atopic dermatitis patients. Symptom severity has been attributed to interpersonal and family stress, and problems in psychosocial adjustment and low self-esteem have been frequently noted. Dysfunctional family dynamics may lead to lack of therapeutic response as well as to developmental arrest. Williams found that 45% of children with atopic dermatitis whose mothers received counseling were clear of lesions versus only 10% in the group receiving conventional (nonpsychosocial) therapy alone. Chronic infantile eczema has been

shown to respond when parental education was added to conventional treatment. Psychological interventions, such as brief dynamic psychotherapy, biofeedback, cognitive-behavioral therapy, relaxation techniques, and hypnosis, can be important adjuncts to treatment. Benzodiazepines and antidepressants are frequently used for treatment.

Acne Excoriee

The habitual act of picking at skin lesions, apparently driven by compulsion and psychological factors independent of acne severity, has been reported in the perpetuation of self-excoriation. Most patients with this disease are females with late onset acne. Psychiatric co morbidity of acne excoriee includes body image disorder, depression, anxiety, obsessive-compulsive disorder (OCD), delusional disorders, personality disorders, and social phobias. Immature coping mechanisms and low self-esteem have also been associated. Interesting gender differences have been observed in this disease. In men, self-excoriation is exacerbated by a coexisting depression or anxiety, while in women this behavior may be a manifestation of immature personality and serve as an appeal for help. SSRIs, doxepin, clomipramine, naltrexone, pimozide, and habit-reversal behavior therapy have been used in the treatment of this condition, and recent reports suggest that olan-zapine (2.5–5 mg daily for 2–4 weeks) may also be helpful.

Hyperhidrosis

With this disease, persistent sweating is brought on by emotional stimuli. These patients have social phobic and avoidance symptoms, sometimes leading to devastating consequences at work and social activities. States of tension, fear, and rage are common. The psychopathologic characteristics of patients with essential hyperhidrosis have been categorized into 3 groups: (1) objectifiable hyperhidrosis due to a psychosomatic disorder like atopic dermatitis; (2) objectifiable hyperhidrosis with secondary psychiatric sequelae like social phobia, anxiety, and depression as a consequence of chronic skin disease; and (3) delusional hyperhidrosis without any objective evidence of hyperhidrosis. This last category has been increasingly observed in patients with body dysmorphic disorder. Patients with hyperhidrosis have poorer coping ability and more emotional problems compared to patients with other dermatology problems. The SSRIs and benzodiazepines have been used with variable success in the treatment of hyperhidrosis associated with social anxiety disorder.

Urticaria

Severe emotional stress may exacerbate preexisting urticaria. Increased emotional tension, fatigue, and stressful life situations may be primary factors in more than 20% of cases and are contributory in 68% of patients. Difficulties with expression of anger and a need for approval from others are also common. Patients with this disorder may have symptoms of depression and anxiety, and the severity of pruritus appears to increase as the severity of depression increases. Cold urticaria may be associated with hypomania during winter and recurrent idiopathic urticaria with panic disorder. Doxepin, nor-triptyline, and SSRIs have been reported to be useful in the management of chronic idiopathic urticaria. Recurrent urticaria associated with severe anxiety disorders has been treated successfully with fluoxetine and sertra-line. Individual and group psychotherapies, stress management, and hypnosis with relaxation may also be useful in these patients.

Herpes Simplex Virus, Herpes Zoster, and Human Papillomavirus Infections

There is increasing evidence that stress has a role in recurrent herpetic infection. In one study, experimentally induced emotional stress led to herpes simplex virus reactivation. Other studies have demonstrated an inverse correlation between stress level and present CD4 helper/ inducer T lymphocytes, thus contributing to herpes virus activation and recurrences. It has also been suggested that stress-induced release of immunomodulating signal molecules (e.g., catecholamine, cytokines, and glucocorticoids) compromises the host's cellular immune response leading to reactivation of herpes simplex virus. Relaxation treatment and frontalis electromyography biofeedback may reduce the frequency of recurrences.

Herpes zoster has been associated with chronic child abuse, and severe psychological stress of any sort may depress cell-mediated immune response, predisposing children to the virus. Effects of hypnotherapy on common warts have been well documented in the literature. Feelings of depression, anger, and shame and the negative effects on sexual enjoyment and activity associated with human papillomavirus infections have also been reported.

Other disorders in which psychological factors are important in the genesis and course of disease include seborrheic dermatitis, aphthosis, rosacea, pruritus, and dyshidrosis.

PSYCHIATRIC DISORDERS WITH DERMATOLOGICAL SYMPTOMS

These disorders have received little emphasis in the psychiatry or dermatology literature, even though they may be associated with suicide and unnecessary surgical procedures. Most of these disorders occur in the context of somatoform disorder, anxiety disorder, factitious disorder, impulse-control disorder, or eating disorder.

Dermatitis Artefacta

A form of factitious disorder, dermatitis artefacta involves self-inflicted cutaneous lesions that the patient typically denies having induced. The condition is more common in women than in men (3:1 to 20:1). The lesions are usually bilaterally symmetrical, within easy reach of the dominant hand, and may have bizarre shapes with sharp geometrical or angular borders, or they may be in the form of burn scars, purpura, blisters, and ulcers. Erythema and edema may be present. Patients may induce lesions by rubbing, scratching, picking, cutting, punching, sucking, or biting or by applying dyes, heat, or caustics. Some patients inject substances, including feces and blood. Reported associated conditions include OCD, borderline personality disorder, depression, psychosis, and mental retardation. In one survey, dermatitis artefacta was found in 4% of 457 institutionalized children and adolescents with mental retardation. Physical and sexual abuse must be considered in the differential diagnosis as well as psychosocial stressors. Malignant transformations of dermatitis artefacta lesions sometimes resulting in death have been reported. Hypnosis can be useful in the evaluation of suspected cases. Direct confrontation of the patient should be avoided, and a supportive, nonjudgmental approach is the mainstay of management. Patients should be seen on an ongoing basis for supervision and support, whether or not lesions remain present. Relaxation exercises, antianxiety drugs, antidepressants such as SSRIs, and low-dose atypical antipsychotics might also be useful. In some case reports, an excellent clinical response has been observed with low-dose olanzapine.

Delusions of Parasitosis

DSM-IV-TR defines delusions of parasitosis as delusional disorder of somatic type. Patients believe that organisms infest their bodies; they often present with small bits of excoriated skin, debris, insects, or insect parts that they show as evidence of the infection. Pimozide, 1 to 10 mg/day, has been the treatment of choice in past; risperidone, trifluoperazine, haloperidol, chlorpromazine, and electroconvulsive therapy are among other treatments reported to be useful.

Trichotillomania

Trichotillomania is the recurrent pulling out of one's hair, resulting in noticeable hair loss. In the DSM-IV-TR, trichotillomania is classified as a disorder of impulse control; however, many dermatologists regard this behavior as quite compulsive. Childhood trauma and emotional neglect may play a role in the development of this disorder. Associated psychiatric conditions may include anxiety, depression, dementia, mental retardation, mood or adjustment disorder, co morbid substance abuse, and eating disorder. The patients experience an increasing sense of tension immediately before an episode of hair pulling and when attempting to resist the behavior; they feel relief of tension and sometimes a feeling of gratification after hair pulling. This dynamic highlights similarities and differences between trichotillomania and OCD and tic disorder. Specifically, the behavior may not be regarded as ego-dystonic, but it does share many emotional factors with compulsions. Interestingly, many treatments for tricho-tillomania are similar to those utilized in OCD. Thus, along with symptomatic treatment, fluoxetine, paroxetine, sertraline, clomipramine, lithium, buspirone, risperidone, cognitive-behavioral therapy, habit-reversal therapy, and hypnotherapy have been reported to be beneficial.

Obsessive-Compulsive Disorders

Patients usually present to dermatologists because of skin lesions resulting from scratching, picking, and other self-injurious behaviors. They typically have an increased level of psychiatric symptomatology compared with age and sex matched controls taken from the general population of dermatology patients, and many patients experience negative stigmatization in their daily life. Common behaviors include compulsive pulling of scalp, eyebrow, or eyelash hair; biting of the nails and lips, tongue, and cheeks; and excessive hand washing. In one report, the most common site involved was the face, followed by back and neck. Koo and Smith have found that OCD in child and adolescent dermatology patients most commonly presents as trichotillomania, onychotillomania, and acne excoriee. SSRIs, clomipramine, behavior modification, and psychodynamic psychotherapy have been reported to be effective.

Phobic States

Among the commonly feared conditions for people with phobic states are sexually transmitted diseases, cancer, acquired immunodeficiency syndrome (AIDS), and herpes. Phobias about dirt and bacteria may lead to repeated hand washing, resulting in irritant dermatitis. In some cases, patients mutilate themselves in an attempt to remove pigmented nevi that they believe are related to possible cancer. Cognitive-behavioral therapy plays a major role in treatment.

Dysmorphophobia

This condition is also called body dysmorphic disorder or dermatological non-disease. Patients with this condition are rich in symptoms but poor in signs of organic disease. Self-reported "complaints" or "concerns" usually occur in 3 main areas: face, scalp, and genitals. Facial symptoms include excessive redness, blushing, scarring, large pores, facial hair, and protruding or sunken parts of face. Other symptoms are hair loss, red scrotum, urethral discharge, and herpes and AIDS phobia. Strategies to relieve the anxiety due to the perceived defects may include camouflaging the lesions, mirror checking, comparison of "defects" with the same body parts on others, questioning/reassurance seeking, mirror avoidance, and grooming to cover up "defects." Women are more likely than men to be preoccupied with the appearance of their hips or their weight, to pick their skin, to camouflage defects with makeup, and to have co morbid bulimia nervosa. Men are more likely than women to be preoccupied with body build, genitals, and hair thinning and to be unmarried and to abuse alcohol. Patients with body image disorders, especially those involving the face, may be suicidal. Associated co morbidity in dysmorphophobia may include depression, impairment in social and occupational functioning, social phobias, OCD, skin picking, marital difficulties, and substance abuse. SSRIs, clomipramine, haloperidol, and cognitive-behavioral therapy have been used in this condition with variable success.

Neurotic Excoriations

As per DSM-IV-TR classification an impulse-control disorder not otherwise specified, neurotic excoriations are self-inflicted lesions that typically present as weeping, crusted, or lichenified lesions with postinflammatory hypopigmentation or hyperpigmentation. The preferred term for this disorder is *pathologic skin picking*. Usual sites are the extensor aspects of extremities, scrotum, and perianal regions. Repetitive scratching, initiated by an itch or an urge to excoriate a benign skin lesion, produces lesions. The behaviors of these patients sometimes resemble those with OCD.

Psychopathologically, patients with neurotic excoriations have personalities with compulsive and perfectionist traits. Common concurrent psychiatric diagnoses are OCD and other anxiety disorders, mood disorders, body dysmorphic disorders, substance abuse disorders, eating disorders, trichotillomania, compulsive-buying disorder, and personality disorders (obsessive-compulsive and borderline personality disorder). The extent and degree of self-excoriation has been reported to be a reflection of the underlying personality. Family- and work-related stress has also been implicated.

Phenomenological, there is an overlap between trichotillomania and pathologic skin picking; both conditions are similar in demographics, psychiatric comorbidities, and personality traits. In rare cases, there are medical complications, such as epidural abscess with subsequent paralysis, as a result of compulsive picking. SSRIs, doxepin, clomipramine, naltrexone, pimozide, olanzapine, benzodiazepines, amitriptyline, habit-reversal behavioral therapy, and supportive psychotherapy have been used in the treatment.

Neurotic Excoriations

In this disorder, there are cycles of stress leading to pruritus as well as of the pruritus contributing to stress. Psychologic stress and co morbid psychiatric conditions may lower the itch threshold or aggravate

itch sensitivity. Stress liberates histamine, vasoactive neuropeptides, and mediators of inflammation, while stress-related hemodynamic changes (e.g., variation in skin temperature, blood flow, and sweat response) may all contribute to the itch-scratch-itch cycle. Psychogenic pruritus has been noted in patients with depression, anxiety, aggression, obsessional behavior, and alcoholism. The degree of depression may correlate with pruritus severity. Habit-reversal training, cognitive-behavioral therapy, and anti-depressants may be beneficial.

DERMATOLOGIC DISORDERS WITH PSYCHIATRIC SYMPTOMS

This category includes patients who have emotional problems as a result of having skin disease. The skin disease in these patients may be more severe than the psychiatric symptoms, and, even if not life threatening, it may be considered "life ruining." Symptoms of depression and anxiety, work-related problems, and impaired social interactions are frequently observed. Psoriasis, chronic eczema, various ichthyosiform syndromes, rhinophymas, neurofibromas, severe acne, and other cosmetically disfiguring cutaneous lesions have grave effects on psychosocial interactions, self-esteem, and body image; major depression and social phobia may develop.

Alopecia Areata

Alopecia areata is a nonscarring type of hair loss on any hair-bearing area of the body. The influence of psychologic factors in the development, evolution, and therapeutic management of alopecia areata is well documented. Acute emotional stress may precipitate alopecia areata, perhaps by activation of overexpressed type 2 β corticotropin-releasing hormone receptors around the hair follicles, and lead to intense local inflammation. Release of substance P from peripheral nerves in response to stress has also been reported, and prominent substance P expression is observed in nerves surrounding hair follicles in alopecia are-ata patients. Substance P-degrading enzyme neutral en-dopeptidase has also been strongly expressed in affected hair follicles in the acute-progressive as well as the chronic-stable phase of the disorder. Comorbid psychiatric disorders are also common and include major depression, generalized anxiety disorder, phobic states, and paranoid disorder. A very strong family history and concurrent atopic and associated autoimmune diseases are common in alopecia areata patients. In a study with 31 patients, 23 patients (74%) had lifetime psychiatric diagnoses. Major depression and generalized anxiety disorder may occur in as many as 39% of patients, and those with patchy alopecia areata may be more likely to have generalized anxiety disorder. Increased rates of psychiatric diseases in first-degree relatives of patients with alopecia areata have also been noticed, including anxiety disorder (58%), affective disorder (35%), and substance abuse (35%).

Vitiligo

Vitiligo is a specific type of leukoderma characterized by depigmentation of the epidermis. It is associated with more psychosocial embarrassment than any other skin condition. In some studies, patients with vitiligo have been found to have significantly more stressful life events compared with controls, suggesting that psychologic distress may contribute to onset.Links between catecholamine-based stress, genetic susceptibility, and a characteristic personality structure have been postulated. Psychiatric morbidity is typically reported in approximately one third of patients, but, in one study, 56% of the sample had adjustment disorder and 29% had depressive disorders. Patients with vitiligo are frightened and embarrassed about their appearance, and they experience discrimination and often believe that they do not receive adequate support from providers. Younger patients and individuals in lower socioeconomic groups show poor adjustment, low self-esteem, and problems with social adaptation. Most patients with vitiligo report a negative impact on sexual relationships and cite embarrassment as the cause.

MISCELLANEOUS

This group includes disorders or symptoms that are not otherwise classified.

Cutaneous Sensory Syndromes

Patients with these syndromes experience abnormal skin sensations (e.g., itching, burning, stinging, and biting or crawling) that cannot be attributed to any known medical condition. Examples include glossodynia, vulvodynia, and chronic itching in the scalp. These patients often have concomitant anxiety disorder or depression.

Psychogenic Purpura Syndrome

This condition, also known as autoerythrocyte sensitization syndrome (Gardner-Diamond syndrome), is seen primarily in emotionally unstable adult females. These patients present with bizarre, painful, recurrent bruises on extremities, frequently after trauma, surgery, or severe emotional stress. The exact mechanism is unclear; however, hypersensitivity to extravasated red cells, autoimmune mechanisms, and increased cutaneous fibrinolytic activity has been implicated in the pathogenesis. These patients may have overt depression, sexual problems, feelings of hostility, obsessive-compulsive behavior, borderline personality disorder, and factitious dermatitis. Without a correct diagnosis, these individuals can receive treatments that are neither necessary nor effective. The diagnosis can be made in a patient who has atypical history and clinical picture of the syndrome and in whom a skin test with use of the patient's blood reveals a positive reaction.

Pseudopsychodermatologic Disease

These patients may have bizarre skin symptoms without obvious physical findings or a subclinical skin disease that has been eradicated or modified by scratching. Psychodermatologic disease can mimic other skin disorders, and medical disorders can mimic psychodermatologic conditions. Localized bullous pemphigoid lesions, for example, have been mistaken for dermatitis artefacta; multiple sclerosis, folliculitis, hypothyroidism, and vitamin B12 deficiency have been initially diagnosed as delusions of parasitosis.

Suicide in Dermatology Patients

Suicide has been reported in patients with longstanding debilitating skin diseases and must be considered when evaluating these patients. Even clinically mild to moderate severity skin disease may be associated with significant depression and suicidal ideation. Cotterill and Cunliffe reported suicides in 16 patients (7 men and 9 women) with body image disorder or severe acne.

PSYCHOSOCIAL IMPACT OF SKIN DISEASES:

Within the psycho dermatological literature, there is a great degree of consensus that skin disorder have a negative impact upon the psychological and emotional functioning of some patients. Indeed, research has provided evidence that such appearance-altering diseases can have profound behavioral, emotional and cognitive impact upon sufferers (Griffiths &Richards, 2001, Thompson & Kent, 2001). A brief overview of the most commonly researched psychological implication is thus provided.

Damaged skin often carries the connotation of contagion or lack of hygiene (Van Moffaert, 1992). Owing to a lack of health education and awareness in dermatology, some associate skin disease with such issues. This ignorance means that a skin disease patient may find that some people react negatively towards them or treat them differently because of the way they look. Consequently ,the suffer may experience distress, feel stigmatized and thus begin to avoid certain social activities that either involve the revealing of the lesion, such as swimming, or that involve potential intimacy with a third party, such as dating or physical display of affection.

In their qualitative study with vitiligo patients, Thompson et al. (2002) found that the central recurring theme in their interviews concerned perceived differences from previous appearance and form others. Common behavioral strategies used by these patients were concealment and avoidance, which were mostly utilized in order to avoid negative reactions from others. Moreover, acne patients have been shown to limit exposure through social avoidance and to conceal skin lesions(Kellett & Gilbert,2001).Psoriasis patients have also been found to engage in anticipatory and avoidance coping behaviors, Which are unrelated to the severity of their condition and this is hypothesized to relate to stigmatization and rejection (Griffiths& Richards,2001).Like previous work on disfigurement and social anxiety, skin disease patients use these dysfunctional behavioral strategies to manage the impression they make on others and their frequent use illustrates the overriding concerns about social exclusion(Thompson et al.,2002).

Feelings of anxiety, uncertainty and helplessness are often cited by dermatology patients as accompanying the diagnosis of their skin condition. Without the knowledge of when or how the condition will develop, the patient may be left wondering about what behaviors or actions might be contributing to its progression.Moreover,dermatology patients experience heightened self-consciousness,which,in turn, has negative implications for interpersonal interactions and relationships. Research has shown that self-consciousness is a common reaction amongst acne patients (Kellett & Gillbert,2001).Papadopoulos et al.(1999b) found a significantly high frequency in irrational, negative thoughts among vitiligo patients.

As well as affecting psychosocial functioning, negative psychosocial experiences may also affect the onset and progression of cutaneous conditions. Clinical observation has suggested that stress often precedes the onset or exacerbation of many dermatological conditions that share both psychosomatic and immunological components. Such as vitiligo, psoriasis and atopic dermatitis(Koblenzer,1983; Al-Abadie et al.,1994).For example ,emotional distress and stressful life events have been suggested as contributory factors in the onset of vitiligo(Papadopoulos et al.,1998).

ROLE OF STRESS AND ANXIETY IN PSYCHODERMATOLOGICAL CONDITION

Stress, emotional trauma, bereavement, divorce, redundancy, depression – all these have a psychological effect on the brain and the nervous system which, in turn, affects the skin. Actually having a chronic skin condition like eczema or psoriasis can, in itself, create psychological fluctuations via anxiety and depression which can make the skin condition worse. An estimated 20 per cent of psoriasis sufferers also have depression. During periods of anxiety or stress, the adrenal gland produces more of a hormone called cortisol, which affects the body’s immune system. This in turn can cause the skin’s own defenses to either

weaken, as in the case of eczema, or go into overdrive, as in the case of psoriasis. There is growing scientific evidence to prove that psychology and skin conditions are directly linked-

- A study published in the British Journal of Dermatology found that almost 40 cent of psoriasis sufferers recalled stressful events in the month before their condition got noticeably worse, though in some cases it could take just two days to bring on an attack.

- A study in the "The International Society of Dermatology" found that people with psoriasis reported a lower quality of life and higher than normal stress levels.

- A study in the American Journal of Pathology. Researchers found that the immune cells in skin can over-react when levels of stress rise, resulting in inflammatory skin diseases.

Stress may be the psychological item most commonly associated with skin disease, just as it is associated with a host of human ailments; perceived stress disturbs the dynamic equilibrium established between the nervous, endocrine and immune systems.[7] Stress is associated with increased pruritus, which is a central symptom of many common dermatoses, including atopic dermatitis, psoriasis, seborrheic eczema, prurigo nodularis, lichen planus, chronic urticaria, and pruritus of unknown cause.[7] Although it may be unclear precisely how stress, anxiety, or depression influences the pathogesis of a given dermatitis, there is a physiologic basis for understanding the relationship between these psychological states and the skin.

Stress has been indicated as a trigger in many dermatologic conditions, including atopic dermatitis, acne vulgaris, and chronic urticaria. With each of these conditions, one encounters both patients who experience a close chronologic association between stress and exacerbation of their skin disease, and patients for whom their emotional states seem to be unrelated to the natural course of their cutaneous disorder. These two groups are considered "stress responders" and "non-stress responders," respectively.

Just as in many dermatologic conditions, psoriasis appears to worsen with stress in a significant segment of patients. Studies report that the proportion of psoriasis patients who are stress-responders ranges from 37% to 78%.

Studies define stress along three general categories:

1) Major stressful life events (e.g., change of employment, major personal illness and financial problems)

2) Psychological or personality difficulties, and

3) Lack of social support.

Regardless of how stress is defined, studies consistently support a relationship between stress and psoriasis. Furthermore, a majority of patients consider stress to be the main cause for exacerbation of their psoriasis, ranking it above infections, trauma, medications, diet, or weather.

Stress sets off several physiological reactions in the body that can affect the skin. It causes the release of hormones like cortisol that thicken hair follicle cells and increase oil production—the perfect recipe for acne. Stress can also trigger neuropeptides, chemicals unleashed from nerve endings in the skin that leave it red or itchy, and encourage T cells (the skin's infection fighters) to overreact, making the skin turn over too quickly and flake or scale. Then there are the blood vessels: Under stress they become more reactive, either clamping shut (so skin looks pale or sallow) or opening too widely (causing the skin to flush).

Stress and other psychological factors trigger the formation and exacerbation of many dermatological diseases (Van Moffaert, 1992; Koo and Pham, 1992). Every person has a shock organ that is sensitive to stress, which is defined by environmental and genetic factors, and this shock organ is the skin in people who display dermatological symptoms under stress. Panconesi suggested naming dermatological diseases that are activated and whose symptoms are exacerbated by emotional stressors as "dermatological stress disorders" (2000). There is much research on the role of stress in dermatological diseases, and they are categorized as:

I. Environmental factors that cause stress;
II. Subjective experiences towards specific situations;
III. Biological responses to stress (Cohen,1995).

The idea that specific psychosomatic illnesses appear with specific conditions that cause stress is no longer valid today. Research has shown that the subjective experience of stress is more important than the actual stress and the conditions that cause stress. Alexander et al. (1968) reported that psychosomatic symptoms appear in the organs that innervate with the autonomic nervous system, that are formed as the result of lengthened physiological changes induced by unconscious repressed conflicts, and that specific unconscious conflicts result in specific psychosomatic illnesses. Currently, there exist some research findings supporting this idea.

Stress is the first reason that comes to mind, excluding hemophilus pylori and anti inflammatory medicine, in the formation of gastrointestinal ulcers and various studies have shown that gastrointestinal ulcers and hemorrhages occur more frequently in situations that cause social stress, such as earthquakes and economic crises.

This can be explained by the decrease in regional blood flow and increase in gastric acid secretion following stress (Levenstein, 1999). In the same way, in a controlled study with 1500 people following the great earthquake in Japan Kodama found that atopic eczema had increased when compared with normal controls (1999). As the rate of exacerbation of this illness in A and

B regions that were exposed to the earthquake was 38% and 34%, respectively, this rate was 7% in the non-exposed region. While the stress rates in the exposed regions after the earthquake were 63% and 48%, respectively for A and B regions, this rate was 19% in the non-affected region.

Biological responses to stress vary from person to person; vasoactive peptides, lymphokines, and chemical mediators are secreted after stress and inflammation develops as a result of their influence on the immune system. Experimental studies showed that the endocrine, immune, and nervous systems do not work autonomously and that there is a complex interaction between them (Ader, 1995).According to the 3-stepped stress model developed by Selye, psychosomatic dermatological diseases appear particularly during the adjustment process (1949).

Anxiety has been defined as a diffuse, vague and very unpleasant feeling of fear and apprehension. The anxious person worries a lot, particularly about unknown dangers. Anxiety is defined as a "tense emotional state" and is often marked by such physical symptoms as tension, tremor, sweating, palpitation and increased pulse. Anxiety is a psychological and physiological state characterized by somatic, emotional, cognitive, and behavioural components. It is the displeasing feeling of fear and concern. Dermatologists on average guessed that skin conditions such as herpes zoster, insect bites, bacterial infections, and contact dermatitis infrequently caused psychiatric morbidity, while patients actually reported depressive and anxiety disorder prevalence rates of more than 30 % (Sampogna and colleagues, 2003). Literature shows that skin disease sufferers experience increased levels of psychological and social

distress [Root, Kent and Al-Abadie, 1994), poor self body image and lower esteem than the general population [Papadopoulos, Bor & Legg, 1999].

PSYCHOLOGICAL APPROACHES TO TREATMENT FOR DERMATOLOGICAL CONDITIONS

The literature has documented psychological interventions for a number of cutaneous conditions, such as vitiligo, psoriasis, acne and atopic dermatitis, which have been suggested to be as effective for each of these types of disorders as classical medical procedures (Van Moffaert,1992;Papadopoulos & Bor,1999).For example, in their review of psychological therapies for the treatment of psoriasis, Winchell and Watts(1998) describe a case in which two psychiatric patients with psoriasis were given a suggestion that imipramine would have beneficial effects on their skin condition. Following this suggestion one of the patients experienced complete remission while the other improved significantly.

Psychological approaches, such as psychoanalysis and hypnosis (Gray & Lawlis, 1982) as well as behavioral (Wolpe, 1980) and cognitive–behavioral therapy (Papadopoulos et al., 1999b) have been used to treat people affected by skin disorders. Indeed, such interventions have been shown to produce clinically significant improvements in cutaneous conditions, such as atopic dermatitis (eczema), psoriasis, vitiligo and virus-mediated disease (VanMoffaert,1992) and have helped patients to improve their psychological well-being and quality of life (Cole et al., Papadopoulos et al.,1999a). Outlined below are the main therapeutic techniques used in dermatology settings.

Table – Approaches to treatment of dermatological conditions

	Behaviour therapy	**Cognitive behaviour therapy**	**Group therapy**	**Psychodynamic therapy**
Time frame	Here-and-now	Here-and-now	Here-and-now	Understanding the past, focuses on current relationships
Cost	Cost- effective	Cost- effective	Cost-effective	Expensive
Techniques	Systematic desensitization, Modeling, Relaxation, Habit-reversal training, Assertiveness and social skills training, Imagery	Problem-solving, Cognitive Restructuring, Guided imagery, Modelling	Psychoeducation, Social and assertiveness Skills training, Role-play	Analysis of transference and counter-transference, Hypnosis
Time	Short-term	Short-term	Short-term	Long-term

Efficacy	Psoriasis , Eczema , Vitiligo , Acne	Psoriasis , Eczema , Vitiligo , Acne	Psoriasis , Eczema , Vitiligo ,	Eczema

Psychological interventions of psycho dermatological conditions

Cognitive-behavioral Therapy:

Cognitive–behavior therapy (CBT) is a treatment approach that aims to change maladaptive ways of thinking, feeling and behaving through the use of cognitive and behavioral interventions. This model takes the view that it is not situations in and of themselves that are stressful, but rather the perception that one takes of them that makes them so. According to the cognitive model, the beliefs that patients hold about their condition often influence how they cope with and adapt to it. A common feature in the beliefs of people with emotional difficulties is that they have negative and irrational content. These perceptions are often the result of distortions in processing, such as 'cognitive errors' (Beck, 1976, 1993).

CBT focuses on examining and trying to challenge dysfunctional beliefs and appraisals which may be implicated in a person's low mood or avoidance of certain situations or behaviors'. Consequently, targeting cognitions and maladaptive behavior are the key areas of CBT interventions for facilitating change. According to this approach, beliefs are considered as hypotheses to be tested rather than assertions to be uncritically accepted. Therapist and client take the role of 'investigators 'and develop ways to test beliefs, such as 'Others do not like me because of my eczema' or 'I won't be happy anymore because of my vitiligo'. Success at challenging these beliefs involves providing evidence that they are erroneous, and underscored by anxiety and depression (Beck, 1993).

CBT has been successfully applied to various skin conditions. For example, Horne et al. (1989) used cognitive–behavioural therapy along with standard medical treatment in treating three patients suffering with atopic eczema. All three showed a post-treatment reduction in symptom severity, an increase in their ability to control the disorder and a decrease in their reliance on medication. Four controlled studies have also used a cognitive–behavioral approach with psoriasis patients (Price et al., 1991; Zacharie et al., 1996; Fortune et al., 2002; Fortune et al., 2004). Findings have shown adjunctive cognitive–behavioral interventions result in a reduction of psychological distress and in the clinical severity of the condition. Additionally, Papadopoulos et al. (1999b) compared two matched groups of vitiligo patients, one of which received CBT while the other received standard medical treatment alone. Results suggested that patients could benefit from CBT in terms of coping and living with vitiligo. There was also preliminary evidence to suggest that gains made through CBT influences the progression of the condition. Finally, Ehlers et al. (1995) employed CBT with patients with atopic dermatitis and found significant reductions in anxiety, frequency of scratching and itching as well as cortisone use.

Behaviour Therapy

Behaviour therapy incorporates applications derived from learning theory (classical and operant conditioning) and employs them to the treatment of persistent, maladaptive, learned habits. Among behavior therapy techniques are systematic desensitisation, assertiveness and social skills training, behavior analysis, relaxation training (e.g. autogenic and progressive muscle relaxation, biofeedback) habit-reversal training and imagery. The aim of these techniques is to progressively diminish maladaptive behavioural responses by repeatedly inhibiting the anxiety by means of competing responses (Wolpe, 1980). A behavior analysis is conducted where the clinician collects information about the relationship between stimuli and behavioral responses in order to understand the role of anxiety.

Diverse behavioral therapeutic strategies have been applied, either separately or in combination with other psychological techniques to dermatological conditions.

Systematic desensitisation is an appropriate technique for the treatment of dermatoses which feature anticipatory anxiety (Van Moffaert, 1992). The fear and apprehension that patients with skin disease may feel about themselves may be challenged by this technique. Through graded exposure, the patient enters situations that they may fear and avoid. The *habit- reversal technique* is a common strategy used to inhibit scratching and it has been reported to have some success with skin disorders, such as eczema and psoriasis (Ginsburg, 1995). It involves selfmonitoring for early signs and situational cues of scratching and practicing alternative responses, such as clenching the fists (Ehlers et al., 1995). *Relaxation* has beneficial effects on skin disorders because it reduces stress levels. It is a useful way to help patients prepare for anxiety-provoking situations or to cope with stressful social predicaments. Relaxation can be used on its own as a means to reduce anxiety or tension or can be paired with imagery. There are various different techniques, such as progressive muscle relaxation or autogenic relaxation training.

Imagery with skin disease patients is employed in order to help them cope with anxiety relating to their condition. Imagery is a useful technique for helping the patient to visualise the feared situation while in a relaxed state (Papadopoulos & Bor, 1999). *Assertiveness and social skills training* is appropriate for patients with cutaneous conditions that attract attention from others, such as staring or personal questions. Interventions focus on improving social skills and ways of expressing emotions, thus helping patients deal more effectively with the reactions of others
and learn a more positive mode of social functioning (Robinson et al., 1996).

Weinstein (1984) found that compared with patients receiving only medical treatment (psorasen plus ultraviolet light A, PUVA), both psychological treatment groups, one receiving progressive relaxation and guided imagery and the other meeting to discuss psychosocial concerns about psoriasis, were effective in reducing the signs and symptoms of psoriasis. Robinson et al. (1996) found a significant decrease in anxiety and an increase in confidence of facially disfigured people (among whom were people with acne and vitiligo) after a social skills workshop that aimed at improving social interaction skills. Additionally, Ehlers et al. (1995), in their controlled trial, used relaxation therapy with patients with atopic dermatitis and found significant improvement in the skin condition.

Stress Management Training:

Five skills from the core of almost all stress management program: self observation, cognitive restructuring, relaxation training, time management and problem-solving.

Self-Observation:

A daily diary format is used, with patients being asked to keep a record of how they responded to challenging or stressful events that occurred each day. A particular stress (e.g. argument with spouse) may precipitate a sign or symptom (e.g. pain in the neck).

Cognitive Restructuring:

Helping participants become aware of and change, their maladaptive thoughts, belief, and expectation .Patients are taught to substitute negative assumptions with positive assumptions.

Relaxation Exercises:

1. Relaxation Techniques:

Edmund Jacobson in 1938 developed a method called progressive muscle relaxation to teach relaxation without using instrumentation as is used in biofeedback. Patients were taught to relax muscle groups, such as those involved in " tension headaches" When they encountered and were aware of situations that caused tension in their muscles, the patients were trained to relax .This method is a type of systematic desensitization-a type of behavior therapy.

2. Hypnosis:

Early researchers in psychodermatology experimented with the use of hypnosis (Van Moffaert, 1992).Hypnosis brings about changes in physiological parameters, such as skin conductance, skin temperature and vasomotor reactions all of which can be decisive in the aetiology of skin diseases (Van Moffaert, 1992).Neurodermatitis, chronic urticaria and viral warts are skin diseases with which hypnosis has been successfully used (Barber, 1978).

3. Biofeedback:

Biofeedback can enhance the patient's awareness of tension and help them to relax, improving skin disorders that flare with stress or that have an autonomic nervous system aspect. Biofeedback of GSR can help reduce hyperhidrosis (excess sweating).Biofeedback of skin temperature by temperature –sensitive strip or by thermocouple can be used for relaxation, dyshidrosis and Reynaud's syndrome.108-110 HRV biofeedback can also help reduce the stress response that tends to exacerbate many inflammatory skin disorders. Hypnosis can produce relaxation and enhance the effects produced by biofeedback.

4. Time Management

Time-management methods are designed to help individuals restore a sense of balance to their lives. The first step in training in time-management skills is designed to enhance awareness of current patterns of time use. To accomplish this goal, individuals might be asked to keep a record of how they

spend their time each day, noting the amount of time spent in important categories, such as work, family, exercise, or leisure activities. Alternatively, they may be asked to list the important areas in their lives and, then, asked to provide two time estimates: (1) the amount of time they currently spend engaging in these activities and (2) the amount of time they would like to spend engaging in these activities. Frequently, a substantial difference is seen in the time individuals would like to spend on important activities and the amount of time they actually spend on such activities. With awareness of this difference comes increased motivation to make changes.

5. Problem-Solving

The final step is problem-solving in which patients basically try to apply the best solution to the problem situation and then review their progress with the therapist.

Group Therapy:

Group therapy is a mode of intervention that helps individuals with a common problem enhance their social functioning though group exercises. Group members are given the opportunity to share their experiences, feelings and difficulties in a safe atmosphere under the auspices of a group facilitator. Using a combination of instruction, modeling, role-play, feedback and open discussion, members of the group are encouraged to discover more about the interaction process. In most cases 6–12 clients meet with their therapist at least once a week for about 2 hours. Usually groups are organised around one type of problem (such as coping) or type of client (such as psoriasis patients).

Through group interaction, ineffective and immature ways of coping are discouraged, positive attitudes are fostered and feelings, such as loneliness and isolation, that many patients experience, diminish. Moreover, group members can bolster one another' self-confidence and self-acceptance, as they come to trust and value one another, and develop group cohesiveness. Group therapy allows participants to try out new skills in a supportive environment and members learn from one another. Thus this offers features not found in individual treatment.

Various approaches, such as social skills training to group therapy have been tried with patients with skin disorders (Robinson et al., 1996). Patients with chronic skin conditions, such as psoriasis or eczema are known to benefit from group therapy and such therapy has increased their confidence in coping with them disease (Ehlers et al., 1995; Seng & Nee, 1997; Fortune et al., 2002).

Overall, research suggests that psychological treatments lead to improvements in clinical severity of skin condition and reductions in psychological distress. This idea corroborates reports that psychological interventions are useful adjuncts to dermatological treatment in such cutaneous disorders and, when combined, can be considered an effective way of managing patients with such conditions. Skin disease appears to be a complex interplay of biological, psychological and social factors, and treatment should exist within the context of this interplay.

Specifically, CBT is gaining credibility as a psychological treatment in the management et al., 1999; Fortune et al., 2002, 2004). Findings have shown that cognitive techniques produced reductions in the

frequency of cognitions concerning itching, catastrophizing cognitions (Ehlers et al., 1995), and in beliefs about the consequences and the emotional causes of disease, and were maintained at 6 month and at 1-year follow-up. Research has also shown that relaxation is an important component of CBT and it has proved most effective in decreasing anxiety levels among dermatology patients (Ehlers et al., 1995; Zacharie et al., 1996). These data suggest that high anxiety levels often observed in these patients can be reduced with treatment and that treatment effects can be maintained even after a 1-year follow-up.

Counseling

- ✓ Information giving(education)
- ✓ Implications counseling
- ✓ Supportive counseling

Psychotherapeutic counselling: therapy

This focuses on healing, psychological adjustment, coping and problem resolution. Different theoretical approaches involve CBT, psychoanalytic therapy, behavior therapy, humanistic therapy and others.

How can therapy help in the treatment and management of dermatology patients?

Therapy can help dermatology patients to:

• come to terms with their conditions;

• explore treatment options and facilitate decision-making;

• examine difficulties they are experiencing with their condition and gain insight Into what factors maintain those difficulties;

• explore and challenge dysfunctional appraisals, beliefs and assumptions;

• identify useful coping strategies;

• facilitate social interaction skills;

• examine issues that may be indirectly linked to the skin condition;

• challenge and cope with anticipatory anxiety and depression.

Group therapy, especially social and assertiveness skills training can help dermatology Patients:

• encounter difficulties in social situations;

• discuss their problems with others who can empathise;

• develop a better understanding through the others' experiences of their condition;

• allow members to acquire and develop a variety of skills and put them in practice with other members;

• serve as a means of emotional and social support for skin patients.

DISCUSSION

At least one third of the patients seen in dermatology clinics present with a complaint that involves a significant psychological component. The prevalence of psychological disorders among patients with skin disease is about 25%-30%, with higher rates of over 30% among patients with acne, pruritus, urticaria, alopecia, and herpes simplex (Picardi, Abenia, Melchi, Puddu, & Pasquini, 2000), revealing the fact that skin and the psyche interact in many ways. Not only does the skin blanch in response to embarrassment, but many chronic skin disorders. Such as psoriasis and eczema, can be exacerbated by emotional stress. Psychological morbidity in addition to skin disease is associated with poorer quality of life (Sampogna et al., 2004).

About 30% of patients who have skin disorders are reported to have psychiatric disorders and psychosocial impairments (Kessler et al., 2005). Many skin disorders have negative psychosocial effects. Because the skin is the most accessible organ in the body, patients with serious psychopathologies frequently target their own skin to express aggressive or self-destructive tendencies. Moreover, disfigurement resulting from a skin disease may have profound psychosocial consequences not only by negatively affecting the patient's body image but also by leading to social stigma. Psychotropic drugs and non-pharmacologic psychotherapeutic interventions may be needed to counteract the negative effects of the skin disorder on the psyche (Fried, 2002). Often, patients who would benefit from referral to a psychiatrist or psychologist refuse to do so, leaving treatment, if any, in the hands of the dermatologists and their staff. A multimodal approach with both psychotropic drugs and nonpharmacologic psychotherapies is frequently better in treating complex psychosomatic (mind/emotions affecting the body/skin) and somato-psycho-social (body/skin affecting the mind/emotions and interpersonal interactions) problems.

It has been shown that there is a real need to address psychosocial issues surrounding disfiguring dermatological conditions. The impact skin disease has on the patients' well-being as well as his/her relationship to the outer world can be great and varied.

Although evidence was provided that psychological treatment can be efficacious in addressing them, still much of the research in this area is methodologically flawed.

Therefore, there is a great need for systematic evaluation to determine the treatment efficacy of different approaches to counseling and the development of psychological treatments, which will focus on the unique issues pertaining to dermatology patients. Future research should seek to examine the utility of different modalities for various skin conditions by employing designs that will compare them.

CONCLUSION

Skin disorders that produce change in the appearance of skin, hair, or nails, or changes in sensation can have a major impact on the psyche. Treating the skin disorder effectively often induces psychosocial improvement. The psyche through psychoneuroimmunoendocrine and behavioral mechanisms can in turn have a major impact on skin disorders. Treating the psyche effectively can improve many inflammatory skin disorders. The psychotropic drugs listed above are often somewhat effective by themselves. Adding non-drug psychocutaneous modalities such as biofeedback, cognitive –behavioral methods, hypnosis, often synergistically enhance the treatment response. Teaching patients to practice safe stress using these nonpharmacologic methods, supplemented if necessary with anxiolytic standard drugs or herbal alternatives, is important, since many inflammatory skin disorders are worsened by stress. Exercise, meditation, music therapy, progressive relaxation, self-hypnosis and other stress-reducing methods can not only help in achieving improvement in specific skin disorders, but also in enhancing overall health and resiliency.

Unit- VII Oncology

PSYCHOLOGICAL ISSUES ASSOCIATED WITH CANCER:

Cancer is a disease that develops when abnormal cells in your body divide and multiply without control. Normally, cells grow and divide to produce more cells only when your body needs them. But sometimes, cells keep dividing when new cells are not needed. These extra cells may form a mass called a tumor. Tumors can be either benign, which means not cancerous, or malignant, which is cancerous. Cancer cells can develop anywhere in the body and can spread to other body parts through your blood and lymph systems. Lymph is a clear fluid that carries blood cells that fight infection and disease throughout the body. Lymph travels through a system of vessels, much like blood vessels. (Lloyd et al., 2007).

PROBLEMS FACING BY CANCER PATIENTS & FAMILY:

The patient feels unable to prepare for treatment because she/he does not get enough information or does not understand it.
•The patient does not feel able to establish a relationship of trust with the professionals who treat them.
• The patient or family already has (had) other serious problems and stresses in their life that make significant demands on their coping resources.
• The patient does not have support from family or friends during the cancer journey.
• Problems in maintaining daily life functioning (e.g. work, finances, shopping, household tasks, looking after the children) are not addressed.

Other risk factors for a more complicated presentation of adjustment difficulties are:

In the patient/family:

• A history of psychiatric disorder
• Social isolation
• Dissatisfaction with past medical care
• Passive coping (i.e. not seeking information/ not talking to friends/family)

In the cancer:
• Limitation of daily life activities
• Disfigurement
• Poor prognosis/survival

In the treatment:
• Disfigurement
• Isolation (such as in e.g. bone marrow transplant or thyroid radiation therapy)
• Debilitating or distressing side effects
• Post Traumatic Stress Disorder

QUALITY OF LIFE:

The term quality of life (QOL) is used to evaluate the general well-being of individuals and societies. According to the World Health Organization (WHO), quality of life is defined as individual perception of life, values, objectives, standards, and interests in the framework of culture. Quality of life (QOL) typically involves the assessment of several dimensions: physical well-being, emotional well-being, social well-being, and functional well-being (Andersen, 1993). Cancer can produce many different symptoms, some subtle and some not at all subtle. Some symptoms of cancer affecting quality of life in patients would be cancer type and stage, as some types of cancer do not present any symptoms until they are in advanced stages, time since diagnosis, patient acceptance, and intensity of the disease and the level of psychological distress experienced by caregivers **(**Tavoli, 2008). A number of illness-related factors exist that can affect quality of life. The amount of distress experienced by an individual with cancer has been related to quality of life. Depression and anxiety symptoms are common in these patients and impair the patient's quality of life, comfort level and treatment compliance, which ultimately can affect the patient's survival.

DENIAL:

Denial is the patient's refusal to take on board the bad news. It is a complex concept that has different meanings in different contexts and serves a multitude of functions. When a person finds the challenges too overwhelming, denial may be the coping strategy that "works". Denial, in this situation, provides psychological protection. However, denial of the illness or of its severity for some patients can cause delayed diagnosis or compromised compliance with treatment. Furthermore, denial is used within an interpersonal context. Each person determines what information can be shared with others. Denial is not an all or nothing phenomenon.4 Most people use denial to one degree or another, denying some aspects of their illness at least some of the time. This denial may fluctuate widely on a day-to-day, or even a minute-to-minute basis. It is a dynamic process.

DENIAL IN CANCER PATIENTS:

Denying that it is really a cancer, and hope, despite all odds, that it will all be made well, or it will be discovered that it was a misdiagnosis - this sort of delusion has been observed to last for several years in some patients. Some others just accept "the death sentence" and die. Not all who die have given up. Not every dying person comes to terms with what is happening in a way that is clear to the outside observer.

HOW TO MANAGE DENIAL:

1. It is important for all therapists to understand their contribution to denial to ensure that the patient receives factual, clear information regarding the disease and its implication. We should make sure that the patient's denial is not due to lack of information, lack of understanding or lack of agreement with medical recommendations (Cousins, 1982; Shelp & Perl, 1985).
2. Distinguish between a fact being denied (e.g. diagnosis of cancer) and implications of the fact denied (e.g. cancer will not return). The former may interfere with necessary treatment; the latter may maintain morale in a difficult situation.

3. It should be addressed directly when denial compromises patients' safety, such as not reporting symptoms, or not complying with treatment. When denial inhibits actions of importance, for example, making decisions on treatment, planning realistically, and communicating honestly with loved ones, it is also important to intervene.

4. Emphasis to patients that they will not be abandoned. They will be supported and cared for, whether the medical news is good or bad.

GRIEF REACTION TO BODILY CHANGES:

Difficulties in this area may manifest themselves as:
- Significant emotional difficulty in accepting physical scarring, cosmetic prostheses or cosmetic results of reconstructive surgery aimed at restoring physical appearance;
- Loss of intimacy and sexuality;
- Loss of confidence and self-esteem, withdrawal and/or social isolation;
- Preoccupation with perceived physical defects, feelings of self-loathing, shame and/or inadequacy, anger, loss of confidence, anxiety and depression;
- Neglect of personal care; loss of appetite with associated weight loss *or* overeating and weight gain.

Interventions generally involve an assessment of mood, coping and adjustment behaviors, the grieving process and the personal meaning of the loss or change in physical appearance to the patient (and possibly the spouse and family). This may include an exploration of their personality, self-perceptions and self-worth, significant life roles and goals, and previous significant losses or changes. Interventions are aimed at facilitating the adjustment and grieving process. Interdisciplinary working with disciplines such as CNSs, Physiotherapists, Occupational Therapists, Speech and Language Therapists, Sexual Therapists or Palliative Care teams may be required.

RECURRENCE:

Cancer recurrence is defined as the return of cancer after treatment and after a period of time during which the cancer cannot be detected. (The length of time is not clearly defined.) The same cancer may come back where it first started or somewhere else in the body. For example, prostate cancer may return in the area of the prostate gland (even if the gland was removed), or it may come back in the bones. In either case it is a cancer recurrence.

SIDE EFFECT OF TREATMENT

Treatment for cancer is often extremely unpleasant. Surgical intervention is required for many cancers at some stage. Breast cancer was previously treated with radical mastectomy, a procedure dreaded by most women who underwent it. The mutilating effects of the surgery and the sense of loss of femininity have been described in numerous reports. Women fear a decline in their sexual attractiveness and rejection by their partners. They also complain of being lopsided. A further complication is the painful and visually embracing development of lymph edema in the arm on the affected side.

Chemotherapy has many adverse effects which induce psychological symptoms. In addition to the effects on fertility, which are described below, chemotherapy can cause intense anxiety as a result of the nausea and vomiting which are so commonly experienced, even though antiemetic drugs have reduced their frequency. Nausea and vomiting may be so severe during the first one or two treatments that a conditioned reflex is established. The patient becomes anxious and nauseated at the prospect of returning for further treatment and may then refuse to attend even though it is realized that treatment is necessary to improve the prognosis. Loss of hair is often a source of distress, especially for women. Chemotherapeutic drugs, such as vincristine and L-asparagines, can have a direct effect on cerebral function causing depression and delirium. Steroids, which are used in conjunction with chemotherapy, also have a well-recognized risk of inducing depression, mania or delirium.

Radiotherapy is better tolerated than chemotherapy. Nevertheless it can cause severe fatigue which can lead to depression. Irradiation to the brain appears to cause more profound fatigue than does irradiation to other areas. Cerebral atrophy is a recognized complication of brain irradiation and if sufficiently pronounced the patient shows clinical evidence of dementia.

Radiotherapy and chemotherapy both have harmful effects on gonad function and the consequences for sexual function and fertility can cause profound emotional distress, contributing to the development of a depressive disorder in the patient, partner or both.

Interventions generally involve an assessment of symptoms and their psychological and behavioural correlates, symptom management strategies such as relaxation, distraction, graded exposure/systematic desensitization, pacing, exercise regimes; and may involve rehabilitative management and modification of their psychological and behavioural triggers and consequences. Interdisciplinary working with disciplines such as Chemotherapy staff, CNSs, Physiotherapists, Occupational Therapists and Pain Management or Palliative Care teams may be required.

ASSESSMENT:

Group	Assessment	Intervention
All Health and Social Care Professionals	Recognition of Psychological Needs	Effective information giving, compassionate Communication and General Psychological Support
Health and Social Care Professionals with additional experience	Screening of Psychological Distress	Psychological techniques such as Problem Solving
Trained and accredited Professionals	Assessment of Psychological Distress and Diagnosis of some Psychopathology	Counselling and specific psychological interventions such as anxiety management and solution-focused therapy, delivered according to an explicit theoretical framework
Mental Health Specialists	Diagnosis of Psychopathology	Specialist psychological and psychiatric interventions such as psychotherapy, including cognitive behavioural therapy (CBT)

What is the role of psychologists working with cancer patients?

- coping with a sense that your future is uncertain;
- coping with how cancer affects your relationships;
- managing the demands of home and family life while caring for yourself;
- feelings of depression, anxiety and anger;
- feeling unhappy about the way you look;
- coping with adjustment, change and loss;
- coping with pain, discomfort, fatigue and treatment side-effects;
- coping with how cancer affects your sexual functioning;
- coping with how cancer affects self-esteem and sense of self.

CBT FOR CANCER PATIENTS:

- **Overgeneralization**: "Life is hopeless" V/s "At the moment; I take one day at a time".
- **Magnification**: "This has destroyed my life" V/s "I try to carry on my life as much as I can like I've always done".
- **All or Nothing thinking**: "Nothing can be done to help me" V/s "I may not be able to be cured, but I have plans for the future".
- **Selective Attention** on negative V/s "I'm making the most of my life as it is"
- **Negative Predictions**: ""Nothing I can do will make a difference" V/s "I have been doing things that I believe will improve my health like changing diet and exercising"
- **Mind-Reading**: Assuming V/s asking partner what he/she was thinking.
- **Guilty Thoughts**: "I should be able to clean the house, take care of the dogs, care for my kids, and find a cure for my cancer"

Personalization: "Everyone thinks this is my fault" V/s "I would like to make contact with others in the same boat".

OTHER THERAPY:

- Supportive expressive group therapy
- Couple therapy
- Family therapy
- Mindfulness
- Systematic desensitization

Unit-IX

HIV/AIDS

Acquired immune deficiency syndrome or **acquired immunodeficiency syndrome** (**AIDS**) is a disease of the human immune system caused by the human immunodeficiency virus (HIV). This condition progressively reduces the effectiveness of the immune system and leaves individuals susceptible to opportunistic infections and tumors.

HIV is transmitted through direct contact of a mucous membrane or the bloodstream with a bodily fluid containing HIV, such as blood, semen, vaginal fluid, preseminal fluid, and breast milk. This transmission can involve anal, vaginal or oral sex, blood transfusion, contaminated hypodermic needles, exchange between mother and baby during pregnancy, childbirth, breastfeeding or other exposure to one of the above bodily fluids.

AIDS is considered a pandemic. In 2009, the World Health Organization (WHO) estimated that there are 33.4 million people worldwide living with HIV/AIDS, with 2.7 million new HIV infections per year and 2.0 million annual deaths due to AIDS.

South Africa has the largest population of HIV patients in the world, followed by Nigeria and IndiaIndia has an estimated 2.5 million infections and an estimated adult prevalence of 0.36%.

Although treatments for AIDS and HIV can slow the course of the disease, there is no known cure or vaccine. Antiretroviral treatment reduces both the mortality and the morbidity of HIV infection, but these drugs are expensive and routine access to antiretroviral medication is not available in all countries. Due to the difficulty in treating HIV infection, preventing infection is a key aim in controlling the AIDS pandemic, with health organizations promoting safe sex and needle-exchange programmes in attempts to slow the spread of the virus.

Signs and symptoms

Main symptoms of AIDS.

X-ray of Pneumocystis pneumonia (PCP). There is increased white (opacity) in the lower lungs on both sides, characteristic of PCP

The symptoms of AIDS are primarily the result of conditions that do not normally develop in individuals with healthy immune systems. Most of these conditions are infections caused by bacteria, viruses, fungi and parasites that are normally controlled by the elements of the immune system that HIV damages.

These infections affect nearly every organ system.

People with AIDS also have an increased risk of developing various cancers such as Kaposi's sarcoma, cervical cancer and cancers of the immune system known as lymphomas. Additionally, people with AIDS often have systemic symptoms of infection like fevers, sweats (particularly at night), swollen glands, chills, weakness, and weight loss.

Tuberculosis (TB) is unique among infections associated with HIV because it is transmissible to immunocompetent people via the respiratory route, and is not easily treatable once identified.

Multidrug resistance is a serious problem. Tuberculosis with HIV co-infection (TB/HIV) is a major world health problem according to the World Health Organization: in 2007, 456,000 deaths among incident TB cases were HIV-positive, a third of all TB deaths and nearly a quarter of the estimated 2 million HIV deaths in that year. TB often presents atypically with extrapulmonary (systemic) disease a common feature. Symptoms are usually constitutional and are not localized to one particular site, often affecting bone marrow, bone, urinary and gastrointestinal tracts, liver, regional lymph nodes, and the central nervous system.

Gastrointestinal

Esophagitis is an inflammation of the lining of the lower end of the esophagus (gullet or swallowing tube leading to the stomach). In HIV-infected individuals, this is normally due to fungal (candidiasis) or viral (herpes simplex-1 or cytomegalovirus) infections. In rare cases, it could be due to mycobacteria.

Unexplained chronic diarrhea in HIV infection is due to many possible causes, including common bacterial and parasitic infections; and uncommon opportunistic .

In some cases, diarrhea may be a side effect of several drugs used to treat HIV, or it may simply accompany HIV infection, particularly during primary HIV infection.

Neurological and psychiatric

HIV infection may lead to a variety of neuropsychiatric sequelae, either by infection of the now susceptible nervous system by organisms, or as a direct consequence of the illness itself.

Toxoplasmosis is a disease caused by the single-celled parasite it usually infects the brain, but it can also infect and cause disease in the eyes and lungs.

It can cause fevers, headache, fatigue, nausea, and vomiting. Patients may also develop seizures and confusion; left untreated, it can be lethal.

Progressive multifocal leukoencephalopathy (PML) is a demyelinating disease, in which the gradual destruction of the myelin sheath covering the axons of nerve cells impairs the transmission of nerve impulses. It is caused by a virus called JC virus which occurs in 70% of the population in latent form, causing disease only when the immune system has been severely weakened, as is the case for AIDS patients.

AIDS dementia complex (ADC) is a metabolic encephalopathy induced by HIV infection and fueled by immune activation of HIV infected brain macrophages and microglia.

Prevalence is 10–20% in Western countries[1] but only 1–2% of HIV infections in India. This difference is possibly due to the HIV subtype in India. AIDS related mania is sometimes seen in patients with advanced HIV illness; it presents with more irritability and cognitive impairment and less euphoria than a manic episode associated with true bipolar disorder. Unlike the latter condition, it may have a more chronic course. This syndrome is less often seen with the advent of multi-drug therapy.

Tumors

Kaposi's sarcoma

Patients with HIV infection have substantially increased incidence of several cancers. This is primarily due to co-infection with an oncogenic DNA virus,

Sexual transmission

Sexual transmission occurs with the contact between sexual secretions of one person with the rectal, genital or oral mucous membranes of another. Unprotected sexual acts are riskier for the receptive partner than for the insertive partner, and the risk for transmitting HIV through unprotected anal intercourse is greater than the risk from vaginal intercourse or oral sex.

Blood products

This transmission route is particularly relevant to intravenous drug users, hemophiliacs and recipients of blood transfusions and blood products. Sharing and reusing syringes contaminated with HIV-infected blood represents a major risk for infection with HIV

Perinatal transmission

The transmission of the virus from the mother to the child can occur *in utero* during the last weeks of pregnancy and at childbirth. In the absence of treatment, the transmission rate between a mother and her child during pregnancy, labor and delivery is 25%.

WHO disease staging system

In 1990, the World Health Organization (WHO) grouped these infections and conditions together by introducing a staging system for patients infected with HIV-1.An update took place in September 2005. Most of these conditions are opportunistic infections that are easily treatable in healthy people.

- Stage I: HIV infection is asymptomatic and not categorized as AIDS
- Stage II: includes minor mucocutaneous manifestations and recurrent upper respiratory tract infections
- Stage III: includes unexplained chronic diarrhea for longer than a month, severe bacterial infections and pulmonary tuberculosis
- Stage IV: includes toxoplasmosis of the brain, candidiasis of the esophagus, trachea, bronchi or lungs and Kaposi's sarcoma; these diseases are indicators of AIDS.

1.Stigma

AIDS stigma exists around the world in a variety of ways, including ostracism, rejection, discrimination and avoidance of HIV infected people; compulsory HIV testing without prior consent or protection of confidentiality; violence against HIV infected individuals or people who are perceived to be infected with HIV. AIDS stigma has been further divided into the following three categories:

- *Instrumental AIDS stigma*—a reflection of the fear and apprehension that are likely to be associated with any deadly and transmissible illness.
- *Symbolic AIDS stigma*—the use of HIV/AIDS to express attitudes toward the social groups or lifestyles perceived to be associated with the disease.
- *Courtesy AIDS stigma*—stigmatization of people connected to the issue of HIV/AIDS or HIV-positive people.

Often, AIDS stigma is expressed in conjunction with one or more other stigmas, particularly those associated with homosexuality, bisexuality, promiscuity, prostitution, and intravenous drug use.

2.Myths

3. Economic impact

HIV and AIDS affects economic growth by reducing the availability of human capital.[185] Without proper nutrition, health care and medicine that is available in developed countries, large numbers of people suffer and die from AIDS-related complications. They will not only be unable to work, but will also require significant medical care.

Religion and AIDS

The topic of religion and AIDS has become highly controversial in the past twenty years, primarily because many prominent religious leaders have publicly declared their opposition to the use of condoms, which scientists feel is currently the only means of stopping the epidemic

General Pre-Test and Post-Test Information

Counseling before and after an HIV test is important because it provides critical information about HIV itself and about the testing process. While counseling services may not be available in all health care settings, many testing sites do offer these services. If you would like access to pre-test and post-test counseling, be sure to inquire about the availability of these services at your chosen test site. If they do not have them readily available, the staff may be able to direct you to alternate service providers who do.

Pre-test counseling sessions generally include the following:

- Information about the HIV test—what it tests for, what it might NOT tell you, and how long it will take you to get your results
- Information about how HIV is transmitted and how you can protect yourself from infection
- Information about the confidentiality of your test results
- A clear, easy-to-understand explanation of what your test results mean

The individual

Draw up the patient's sexual history to pinpoint behaviour that is at risk

- Are you having or have you had sexual relations with women? With men? With both?
- How old were you when you first had sex?
- How many sexual partners have you had during the past six to 12 months?
- Roughly how many sexual partners have you had, all told, during your life?
- What kinds of sexual relations do you have? Oral? Vaginal? Anal?
- For how long have you engaged in each type of sexual relation?
- Have you already had an STD? Which one?
- Do you use some method to prevent STDs and HIV?
- Do you use a condom with all of your partners? Always or occasionally?
- Have you ever had unprotected sex during a trip abroad or with a partner who had just come back from a trip?
- Do you have unprotected sex with casual partners?
- Do you have sex in exchange for drugs, money or gifts?

Assess personal obstacles to prevention

- Domineering partners
- Illiteracy
- Lack of awareness of risk
- Presence of threat or violence
- Deepseated fears, excessive anxiety
- Guilt, shame, and so on

Analyse obstacles in the event the test is refused and overcome them through counselling

- Fear of losing job, housing, insurance, financial supportb or friends, partners, custody of a child
- Fear of being indentified and of losing confidentiality
- Personal or cultural values or beliefs
- History of past or current sexual abuse

The test

- Choose the appropriate moment to conduct the test taking into consideration the diagnostic window *(three months after the last exposure)*
- Repeat the test, if need be *(six months after the last exposure)*
- Make a firm appointment for follow-up bearing in mind the time needed to obtain the result
- **Nominal test:** the patient's name is written on the test requisition
- **Non-nominal test:** the patient's name is replaced by a personal code, e.g. his initials, on the test requisition
- **Anonymous test:** the patient's name is replaced by a non- personalized code assigned by clinic staff on the test re quisition

The result

- Personally communicate the test result in the patient's presence
- Ascertain how well the patient under stands the test result
- If the result is negative, consider the possibility of another test six months after the latest exposure
- If the result is uncertain, conduct a second test six weeks after the first one
- If the result is positive, offer support and inform the patient of regular or emergency community support services
- Ensure follow-up and make the necessary appointments

Regardless of the outcome of the test, counselling on preventive measures is necessary to reinforce safe behaviour, overcome risk beha viour, alter certain living habits, reduce feelings of invulnerability, and so on:

  - counselling on sexual practices and consumption of drugs at reduced risk
  - counselling on conditions that hinder the adoption of preventive behaviour (alcohol, and so on)
  - counselling on specific situations: tatoos, body piercing, acupuncture, donations of sperm,

blood, tissues or organs, occupational exposure
- counselling on the notification of partners.

PRE- HIV TEST COUNSELLING for adolescents

When performing an HIV test on a child under the age of 14, your client is the parent or guardian. During the counselling session, the child should be occupied outside of the counselling room.

The parent/legal guardian of the child should be consulted in order to assess the emotional maturity of the child, as well as the presenting problem. Consulting the parent/legal guardian is also important in determining how much information can and should be given to the child. While it may be assumed that children are too young to be involved in the counselling process, research shows that children as young as fi ve may be mature enough to understand information on HIV/AIDS.

The issue of **consent** is very important when testing a child under the age of 14 for HIV. A parent/legal guardian must sign a consent form before the child can be tested for HIV. If a parent/guardian is not available, a number of other people can sign the consent form: either the superintendent of a hospital, or a police offi cer, or the court, or a child commissioner, or a statutory social worker. **Step-by-step guide** 5

• Introduce yourself and explain your role.

• **Confi dentiality** – explain to the parent/guardian that the counsellor will not talk to the child unless and until the parent would like this to happen. Let the parent/guardian take responsibility for talking to the child fi rst. Confi dentiality of the child's HIV test results must be assured. However, in certain situations the results may have to be disclosed (for example, as evidence in court).

Assessing the situation

• First and foremost, the **best interest of the child** must be considered.

• What is the rationale for HIV testing?

If there is a risk is through vertical transmission (that is, from mother to child), time must

be spent addressing the parent's concerns and the impact of HIV on the child's life. If the child has presented him/herself within 72 hours after being sexually abused, preventing HIV infection through post-exposure prophylaxis (PEP) should be a priority. The counsellor should decide together with the parent or legal guardian, what is in the best interest of the child; and whether, how and when to proceed with counselling and HIV testing.

• Due regard should be given to the emotional state of the child in deciding on whether to test for HIV. If the child is severely traumatised and a blood test would add to the trauma, a starter pack of **Post-exposure prophylaxis (PEP)** should be provided, and the HIV test carried out at a return visit.

• Establish what the parent/guardian knows about HIV/AIDS. Clarify any misconceptions and provide clear facts.

• Identify the child's level of risk of HIV infection.

The HIV test

Explain the testing procedure and the type of test that will be administered. The child will be given a Rapid HIV Test, which tests for HIV antibodies, not for the virus itself. If the test is positive, this result will be confi rmed with another test, an **ELISA** or Western Blot test.

Being Prepared

Discuss the ramifi cations of a positive or negative result.

What would the family actually do?

• Emotional coping

• How would family members' lives change?

• School and work-related issues

• Disclosure and support

• Quality of life

Disclosure

• Establish how much information the parent/guardian feels comfortable telling the child about HIV/AIDS. Information may only be given with the parent/guardian's consent.

• Discuss the advantages and disadvantages of disclosing the child's HIV status to

the child. Advantages of disclosure may include living positively with HIV through a healthy diet and support groups, empowering the child through participation, and educating the child to take precautions for protecting him/herself and others. However, disclosing to the child may be diffi cult for the parent/guardian, the child may not understand and may blame him/herself. Encourage disclosure by the parent/guardian to the child at a later stage.

• Emphasise that ongoing counselling may be required for the client (both child and parent/guardian) to come to terms with the child's results.

• To safeguard a client if he/she does not return for another visit, spend time discussing nutrition and medical options in case of the child being positive.

• Be aware that some children may engage in sexual relationships for fi nancial/survival reasons and may fi nd it diffi cult to negotiate safer sexual practices. Discuss as many of these as possible.

• After these issues have been discussed, the parent/guardian can decide if s/he wants to continue with the HIV test. Make it clear that this is an important decision and that the parent/guardian has the choice to postpone or not to do the test.

Pre-Test Counselling(UNICEF)

• Discuss how long it will take to receive the HIV test results as well as the procedure for giving these results to the client. Explain that the client will be contacted telephonically to set up an appointment to receive the results, irrespective of his/her HIV status. The counsellor should consider the logistical and/or fi nancial constraints on the client/family to determine the appropriateness of setting a follow-up appointment.

• Explore the consequences of the impact of testing, the advantages and disadvantages of HIV testing.

• Assess the child's understanding of HIV/AIDS. Fill in gaps and clarify misconceptions.

• Assess the level of risk.

• Address stigma and discrimination

• In case the client does not return for the results of the HIV test, address various nutrition and medical options.

CHECKLIST – PRE- HIV TEST COUNSELLING

This checklist may be photocopied and attached to the client's file for use after the counselling session to ensure that all issues were addressed.

What is the client's understanding of HIV?

How is HIV transmitted?

How is HIV diagnosed?

What HIV tests are available?

How does the test work?

What is the testing procedure?

What is the client's understanding of ARV drugs?

Review the implications of the test

Advantages and disadvantages of testing for HIV

How will the client cope with a positive result?

Explore psychosocial reactions

Anticipation of a positive result

Advantages and disadvantages of disclosing or not disclosing

Who will support the client directly after result?

This checklist may be photocopied and attached to the client's file for use after the counselling session to ensure that all issues were addressed.

Were the client's feelings arising from the HIV test result addressed?

A negative test result:

Feelings of relief, happiness, continued worry, disbelief

Give time for venting of feelings

Check what the client understands by the result

Understanding of the window period

Emphasise prevention of infection

Encourage re-testing after the window period

A positive test result:

Feelings of shock, anger, guilt, disbelief

Questioning whether there might be a mistake

Understanding of living positively with HIV

Advantages and disadvantages of disclosure

Plans for follow-up

Identify the client/s immediate concerns

Discuss how the client plans to spend the next few hours

Identify what support the client has

Identify difficulties/problems the client foresees and how he/she might address them.

Encourage the client to ask questions

Refer for further counseling

POST-TEST COUNSELLING (TEST NEGATIVE)

ıediately reveal that the test was negative

ɔw time for the relief and happiness to settle.

valuate whether the person could be in the window period -- withing 3-6 months of having got the infection during which period negative results of tests based on detection of antibodies are not conclusive. If there is reason for concern that this may be the case, schedule an appointment for a retest in about 3-4 months and ask the patient to abstain from sex (or at least from unprotected sex) during this interval

ssess the knowledge base and awareness of the patient

ınforce risk reduction information

ılore sustainable changes in behavior

POST-TEST MEETING FOR DISCLOSING A POSITIVE RESULT

ɛal test result directly but in a gentle tone and show empathy

ow time for ventillation of feelings. In majority of cases there is a emotional and physical breakdown. Allow the expression of grief through crying. Often there is denial -- it cannot happen to me, there must be a mistake in the test. Sometimes there is a silent acceptance but tumoil within.

e the patient has calmed down sufficiently to the point that you no longer fear that they may do themselves harm, reassure them of all possible help in terms of medical care, psychological support, and referrals to specialists.

edule a meeting one week later to begin therapy

POST-TEST COUNSELLING FOR A HIV POSITIVE RESULT

over very carefully the difference between HIV and AIDS

ss the benefits of a positive outlook to life in order to make the best use of the future and to stay productive.

ss the need for a good, healthy, and balanced diet. (Lots of fruits and vegetables. Avoid spicy and heavy foods. Take vitamin and mineral supplements.)

d for regular exercise that does not cause fatigue. 15-30 minutes of brisk walking.

p immune system strong. Avoid smoking, alcohol, drugs, and stress. Rest often to avoid fatigue.

ga and meditation help with maintaining a positive outlook and a healthy body

lain the need for them to minimize risky behavior to avoid transmitting the infection to another person

ourage them to continue being loving and caring parents, and to make provisions for children in advance

iew simple infection control measures in event of cuts, blood spills, hypodermic injections, etc

iew need To treat minor infections early and properly. Provide information on how to evaluate the morbidity and seriousness of opportunistic infections

iew the need for special attention to tuberculosis. Suggest regular screening for TB and for other opportunistic infections. Provide a referral service -- a list of sympatheic doctors with varied specialities in case of emergency.

iew recent developments in therapy and understanding of the disease. Offer hope for the future.

ld a support system consisting of friends and relatives. Encourage the patient to bring along one or more such persons to the counselling sessions. Develop a system of home based care for common infections and even for some of the simpler opportunistic infections.

ise confidentiality of status with associates at work and other casual associates. There is no need to advertise HIV status unless patient chooses to become an activist.

ıblish the need for counselling, and continue to build trust and to provide effective and regular counselling.

Psychosocial Issues

The progression of human immunodeficiency virus (HIV) infection from exposure to active acquired immunodeficiency syndrome (AIDS) involves a complex interplay between medical stages and psychosocial factors. Common initial reactions to the diagnosis of HIV infection are shock, denial, and anger. Depression can occur soon after diagnosis and often worsens with the onset of disease symptoms. The relative risk of persons with AIDS committing suicide is many times greater than that of the general population. Both patients and their families often experience feelings of acute suffering and prolonged grieving throughout the illness.

The “at Risk”, "Worried Well"

It is well known that exposure to HIV is greatest among individuals who are currently or have been in the past 10 years involved in high-risk behaviors. The behaviors that place individuals at greatest risk are unsafe sexual practices between homosexual men and intravenous drug abuse that involves sharing unclean needles

Individuals who are concerned about their past or present high-risk behaviors and have not yet been tested for the HIV infection have been described as the "worried well." Many of these people become preoccupied and increasingly anxious about their infection status and find it difficult to decide whether or when to be tested.

Crisis

Testing positive for HIV is usually very traumatic for patients. They frequently experience a crisis reaction that includes feelings of severe shock, a sense of unreality, and some form of denial. Feelings of anxiety and guilt can often be overwhelming. Depression may occur during this crisis period, especially if the patient has a premorbid history of depressive episodes. Common symptoms of depression characteristic of patients who test positive for HIV include feelings of helplessness and worthlessness. These are provoked by the terminal nature of the illness, by fear of social stigmatization, and loss of friends or family support. Personal fears are often compounded by the patient's direct knowledge of friends and acquaintances who have died of AIDS. This may lead the patient to suicidal ideation.

After the initial crisis period, most patients begin to reestablish a sense of personal equilibrium and accept their HIV-positive status. Depending on their personality and coping styles, patients will respond in various ways. It is important for primary care physicians to discuss "safer" sexual practices with their patients who are HIV positive and to refer them to local AIDS service organizations for support assistance.

Other patients will respond to their diagnosis by fighting back, vowing not to be a victim of the disease, and determining to become the first person to be cured of HIV or AIDS. In their struggle these patients frequently mobilize their strengthen and resources to take action (often political) against the disease, or they place intense and often unrealistic emphasis on nutrition, vitamin supplements, exercise, and "clean living."

Patients who have previously relied heavily on other people to emotionally support them in their daily life will] often respond by feeling increasingly helpless and dependent on their family, friends, and health care providers. When working with these patients, it is important for physicians to provide ongoing reassurance and constancy of care, carefully explaining major health changes to the patient as they occur.

Some patients are not able to accept the diagnosis and continue to deny their infection status. Denial can be a useful and adaptive defense for many patients with a fatal disease because it allows them to slowly and carefully filter into their conscious mind their own impending death.

Patients who demonstrate self-destructive destructive behaviors need to be closely monitored and either counseled in the office or referred for substances abuse or psychological assistance.

Research conducted on the three stages of HIV-related illness (asymptomatic, AIDS-related complex, and AIDS) shows that patients in the first two stages experience significantly higher levels of depression, mood disturbance, and anxiety than do those patients with AIDS.

Infected But Well

During the asymptomatic phase of HIV progression, when patients are infected but feeling well, there is often a change in life-style. Many individuals, primarily in the male homosexual population, have modified their sexual activity to include safer sexual practices, resulting in a significant reduction in the incidence of new cases of HIV infection in this population in the past several years.

Emergence of Symptoms

The onset of AIDS-related symptoms such as fatigue, weight loss, and diarrhea can trigger feelings of discouragement and despair. The awareness of symptoms often generates feelings of decreased self-control. Demoralization can occur as self-esteem declines. During this self-devaluing process, homosexual men may develop homophobic feelings of guilt. Anger and blame focused at lovers for provoking homosexual behavior is typical of such homophobic reactions.

The presentation of active AIDS may force the patient to publicly reveal his sexual orientation. If the patient is unable to accept his sexual identity, this may promote denial of the disease.

Physicians should be alert to the symptoms of depression in patients with HIV. Depression may manifest itself as forgetfulness, poor concentration, loss of interests, apathetic affect, and withdrawal. The symptoms may be difficult to differentiate from the AIDS dementia complex. Feelings of helplessness and worthlessness often result from the numerous personal losses experienced and the limited treatment and financial options available. Common fears are disfigurement, abandonment by lovers and family, and lack of self-control.

Risk for Suicide

Grieving

Health care providers can help patients with HIV infection, their lovers, and families with the mourning process, which often begins before death occurs. Lovers and families of persons with AIDs are at risk for complicated and prolonged bereavement. Persons with AIDS have often experienced the loss of numerous friends from the same disease. Grief reaction is often triggered by the physical impairment of

symptomatic AIDS. Active bereavement counseling by the health care provider can decrease the risk of suicide. Worden describes four common tasks in mourning:

1. Accepting the reality of the loss (includes physical loss)

2. Experiencing the pain of grief

3. Adjusting to an environment in which the deceased (or physical capability) is missing

4. Withdrawing emotional energy and reinvesting it in another relationship.

Preparation for Death

The final stage of adjustment is the preparation for death. Many patients will try to make bargains as their disease progresses.

Fatigue is a frequent symptom reported by persons living with HIV disease and one that affects all aspects of quality of life. To improve quality of care of persons with HIV disease, it is important to address all factors that contribute to fatigue

ISSUES FACED BY WOMEN WITH HIV/AIDS

Relationship Disruption

Women who share their HIV or AIDS diagnosis with family or friends risk stigmatization (including reactions of fear, shock, and blame), isolation (as a result of others' fears of casual transmission and the possibility of desertion), and potential loss of self-esteem (lack of confidence and self-blame). Keeping the diagnosis of HIV a secret may hinder a woman's ability to develop effective coping strategies and leave her vulnerable to fear, anger, and depression. These problems may worsen as the progression of HIV disease creates significant changes in behavior, attitudes, and physical appearance.

Women living with HIV often report a desire for support groups or individual therapy to deal with the depression, hopelessness, and anxiety surrounding the unpredictability of disease progression.

Family Caregiving Concerns

Traditional family caregiving responsibilities of HIV-infected women are often complicated by lack of social support and feelings of grief and loss

Psychosocial support addresses the ongoing psychological and social problems of HIV infected individuals, their partners, families and caregivers.

Why it is Important

- HIV infection affects all dimensions of a person's life: physical, psychological, social and spiritual. Counselling and social support can help people and their carers cope more effectively with each stage of the infection and enhances quality of life. With adequate support, PLWHA are more likely to be able to respond adequately to the stress of being infected and are less likely to develop serious mental health problems. Assessment and interventions may be aimed at the acutely stressful phase following notification of HIV infection, the ensuing adjustment period, and the process of dealing with chronic symptomatic HIV infection and disease progression through to death.
- HIV infection often can result in stigma and fear for those living with the infection, as well as for those caring for them, and may affect the entire family. Infection often results in loss of socio-economic status, employment, income, housing, health care and mobility. For both individuals and their partners and families, psychosocial support can assist people in making informed decisions, coping better with illness and dealing more effectively with discrimination. It improves the quality of their lives, and prevents further transmission of HIV infection.
- For people with HIV/AIDS who must adhere to TB treatment, long-term prophylaxis or antiretroviral therapy, on-going counselling can be critical in enhancing adherence to treatment regimens.

How it is Done

Developing guidelines and training for psychosocial care

- Psychosocial support should be specifically included in national guidelines for the management of HIV/AIDS.
- Training on the provision of psychosocial support should be incorporated into the curriculum for all health care providers.
- Guidelines for home care services can be developed and should include the provision of basic psychosocial care by community volunteers and family caregivers.
- Training courses for community volunteers can be organised and provided by health care workers.
- Training in professional disciplines (counselling, psychology, psychiatry) can be made available at the national and international level.
- Strategies for providing psychosocial support can be developed for specific groups (e.g. women, youth, men who have sex with men, injecting drug users, health care workers, etc.).

Developing services for HIV infected people, family members and health workers

- Assess what mental health and supportive services are available at both the governmental and non-governmental level that could be strengthened or replicated. Assess what additional services are needed.
- At the community level, groups can be started to provide for the needs of individual patients and affected family members. Psychosocial support services that address the specific needs of vulnerable groups can be especially important (e.g. women, children, etc.).
- A comprehensive support system linking and coordinating existing psychosocial services with each other and to health services needs to be established.
- At the hospital or clinic level, groups for health care workers in high HIV prevalence areas will be especially important for the support of nurses, physicians and other health care personnel who may be seeing large numbers of HIV infected patients. Supervision of health care workers will be key in assisting health workers to cope with issues related to HIV/AIDS.

- Peer groups (whether of heath workers working under stressful conditions, people living with HIV/AIDS, or family members of people with HIV/AIDS) can be a very effective way of providing psychosocial support.
- A comprehensive support system linking and coordinating existing psychosocial services with each other and to health services thus maximizing all resources needs to be set up.
- Building community capacities to provide counseling and support will ensure sustainability, continuity of interventions and community development.

Comprehensive palliative care is essential to the health and well-being of people living with HIV/AIDS (PLWHA) and is an integral part of the President's Emergency Plan for AIDS Relief (the Emergency Plan). Palliative care has traditionally been associated with terminal or end-of-life care. However, current thought and practice and Emergency Plan policy take the broader view that palliative care encompasses care provided from the time that HIV is diagnosed and throughout the continuum of HIV infection.

The Emergency Plan envisions a comprehensive, holistic, interdisciplinary approach to HIV care. It recognizes that different types and intensity of comprehensive palliative care interventions are needed, depending upon the stage and progression of disease and the needs of the individual and family.

Definition of Palliative Care

Palliative care, as stated in the *U.S. Five-Year Global HIV/AIDS Strategy*, aims to achieve optimal quality of life for PLWHA and their families and minimize suffering through mobilizing clinical, psychological, spiritual, and social care throughout the entire course of HIV infection. It also provides the routine monitoring that is essential to determining the optimal time to initiate ntiretroviral therapy (ART), and it continues during and after the initiation of treatment.

Types of Palliative Care Interventions

Palliative care for HIV-infected individuals becomes increasingly important as the disease progresses. The Emergency Plan can support all areas of comprehensive palliative care offered throughout the course of HIV disease but what individual country programs provide can vary in type, scope, and intensity, based on the progression of the disease, availability of anti-retrovirals (ARVs), and the needs of the individual and family.

The Emergency Plan can support

The Emergency Plan can support the following four categories of essential palliative care, where compatible with national guidelines:

Clinical care is generally provided by nurses, midwives, clinical officers, community and volunteer health workers, traditional healers and physicians. It includes a wide range of treatment and care including: routine, confidential HIV counseling and testing; routine follow-up to determine the optimal time to initiate ART; prevention and treatment of opportunistic infections (OIs) such as tuberculosis (TB); HIV prevention and behavior-change counseling, alleviation of HIV-related symptoms and pain; time-limited support for clinically malnourished PLWHA; and support for adherence to anti-retroviral therapy (ART). Illustrated below are types of interventions that Emergency Plan programs should provide, based on the presence or absence of clinical symptoms:

Asymptomatic clinical care is toward persons who are not experiencing specific signs or symptoms of HIV disease. These interventions include: routine clinical monitoring and assessments, including those that

assist in determining the optimal time to initiate ART (including laboratory and clinical evaluations); services to prevent TB, other OIs and malaria, such as the use of isoniazid prophylaxis, cotrimoxazole; impregnated mosquito bed nets (preferably long-lasting); safe-water systems; nutritional assessment and counseling; HIV prevention counseling, promotion of good personal and household hygiene; and the assessment and management of HIV-related psychosocial problems. Additional guidance will be forthcoming on the elements of basic preventive care for PLWHAs and their families.

Symptomatic clinical care is directed toward individuals who are experiencing progressive immunological impairment and related HIV symptoms. These interventions include: treatment of symptomatic illness; pain control (with opiods and non-opiods); prevention of TB, other OIs, and malaria using medications such as isoniazid prophylaxis, cotrimoxazole, and/or the use of (preferably long-lasting) impregnated mosquito bed nets; safe-water systems; nutritional assessment and counseling; promotion of good hygiene; the assessment and management of HIV-related psychosocial problems; basic nursing care (including but not limited to assessment and monitoring of symptoms and adherence to medications; assistance with bathing, mobility, mouth care and skin and wound care); preparation and support prior to and throughout ART; and time-limited nutrition rehabilitation and supplementation for clinically malnourished PLWHA.

Future guidance will provide details on food and nutrition programs the Emergency Plan can support. United States Government (USG) teams in Emergency Plan countries that are also part of the President's Malaria Initiative should coordinate closely and use both funding streams creatively to serve HIV-affected individuals in the distribution of (long-lasting) insecticide-impregnated bednets.

End-of-Life and bereavement care is directed toward the individual and family members in need of intensive management of symptoms and pain [using non-opioids and/or opioids and directed by the World Health Organization (WHO) analgesic ladder[1]]. They include culturally appropriate end-of-life care and bereavement interventions, as well as appropriate succession planning and referrals for orphans and vulnerable children.

Psychological care addresses the non-physical suffering of individuals and family members, and can include: mental health counseling; family care and support groups; support for disclosure of HIV status; bereavement care; development and implementation of culture- and age-specific initiatives for psychological care; and treatment of HIV-related psychiatric illnesses, such as depression and related anxieties.

Spiritual care addresses the major life events that cause people to question themselves, their purpose and their meaning in life. The interventions should be sensitive to the culture, religion(s) and rituals of the individual and community, and can include (but are not limited to): life review and assessment; counseling related to hopes and fears, meaning and purpose, guilt and forgiveness; and life-completion tasks.

Social care assists individuals and family members in maintaining linkages to and use of care, preventing HIV infection, and ensuring adherence to treatment. These can include: community-based support groups; community mobilization and leadership development of PLWHA; efforts to reduce stigma; legal services to assist with succession planning, inheritance rights, and legal documentation (such as a living will or power of attorney); assistance to secure government grants, housing, or health care; linkages to food support and income-generating programs; efforts to increase community awareness of HIV care, treatment, and prevention; and other activities to strengthen affected households and communities.

Prevention for HIV-infected persons has been shown to be both effective and efficient at preventing new infections Emergency Plan programs should incorporate prevention for positives into palliative care and treatment. Models to provide these interventions can include: interventions for sero-discordant couples, including confidential testing and ongoing counseling; community and clinic-based support groups; case-management and provider-delivered prevention messages focused on disclosure; partner testing; correct and consistent condom use for populations engaged in high-risk behavior and mutual fidelity.

Home-Based Care

Home-based programs deliver various types of HIV/AIDS care in the patient's home. Given the relative availability and affordability of home-based care programs for most resource-poor settings, these programs play a significant role in providing access to comprehensive palliative and supportive care for a large proportion of individuals and families affected by HIV disease. However, many home-based programs either do not include or are missing key palliative care interventions. The introduction of comprehensive care into home-based programs requires the training and education of medical providers (e.g. nurses, clinical officers, and physicians, including pediatricians and pediatric nurses) and community-care providers in the following areas:

- Clinical diagnosis and care, including pain, symptom and OI-assessment and -management.
- The delivery of medications (including pain medications) and other clinical interventions within the community and home.
- Basic nursing care, including client and household hygiene and promotion of disease prevention in the home.
- The use of established patient management protocols and standards.
- Procedures for referring patients for diagnostic, care, and treatment.
- Communication skills, including patient education in local languages on HIV/AIDS and HIV prevention messages, counseling on disclosure of HIV status, and grief, anxiety, and bereavement care.
- The establishment of interdisciplinary teams to address physical, psychosocial support, and spiritual needs of clients.
- Other standards and procedures for providing quality care.

Community-based Care

Community-based care is provided in a variety of community settings including free-standing outpatient clinics, day care centers, school or university-based clinics, community health centers, workplace clinics or stand-alone hospices. These delivery sites often provide a wide range of interventions, including primary care, management of acute and chronic medical conditions and supportive care. Emergency Plan programs should also link them with inpatient facilities, such as provincial or district hospitals.

Facility-Based Care

Hospital-based outpatient clinics and inpatient facilities provide direct and more advanced clinical care in the facility and are essential sites for the delivery of HIV palliative care. Both types of delivery sites provide access to health and basic social-service providers trained in the diagnosis and management of acute and chronic medical conditions and supportive care needs and are often linked to home- and community-based care providers for patient follow-up.

Hospice Care

Hospice is an approach to delivering end-of-life care, often provided in the home by trained nurses and or community care-givers, in community-based and hospital-based facilities, or in a free-standing hospice. Hospice care includes intensive end-of-life care, such as severe pain control with opioids and other medications[3], as well as support in the last months of life for individuals with a terminal illness. Hospice also provides intensive family and bereavement support.

Psychosocial interventions in HIV/AIDS

- Counseling services are important for helping clients deal with the testing process and learning their HIV status. These services need to focus on the psychological distress associated with an HIV-positive diagnosis and the reality of living with HIV/AIDS.
    - Counselors can address the fears associated with HIV, including fears of social discrimination and job loss.
    - Counselors can also talk with clients about how to diminish the risks of transmitting HIV to their partners or children.
    - Evidence from three countries in Africa and the Caribbean demonstrated that clients who sought and received voluntary counseling and testing, particularly those who tested positive for HIV, used condoms more often than those in control groups who simply received information.[6]
    - Counseling services should not only be available to those with HIV, but they should refer those at risk to testing services for sexually transmitted infections or harm-reduction, and provide condoms to those who are sexually active.

Since the stressors faced by PLWHAs are multiple and chronic in nature, a multidisciplinary team approach has to be adopted in order to cater for the various psychosocial needs of PLWHAs. Case management is a client centered form of care that connects clients with medical, psychosocial and supportive service. This form of service is an effective model that promotes adaptive coping of patients suffering from chronic diseases and HIV infection, in which a case manager takes up a facilitative role in case management.

Psychosocial assessment of newly diagnosed HIV-infected patients

The initial counselling and assessment given to newly diagnosed patients is of paramount importance. The objective of the intake assessment is to understand the needs and current functioning in each aspect of life, such as occupational functioning, social functioning, and source of supports. After this initial assessment, an agenda is set up to prioritise areas of needs and services required. Referrals to other professionals, including, clinical psychologist, social worker and dietitian may be necessary, when needs are identified in the interview. Knowledge about HIV infection and their current health status is reinforced, myths and misunderstandings are clarified, and realistic expectations are set with patients. Other relevant information, such as that on mental health, substance use, skills in independent living, are also provided. A supportive and nonjudgemental attitude of the health care workers during the first encounter with the newly diagnosed HIV positive patients is crucial in order to build rapport and establish partnership of care between each other. Patients are encouraged to discuss openly and frankly with the health care workers about their concerns and worries.

Psychological assessment and intervention

Research in Hong Kong shows similar findings as the West that depressed mood is very common among PLWHAs.[2] Although they may not be afflicted with clinical affective disorders, many of them experience significant levels of distress. Some PLWHAs would need psychological assessment and intervention for evaluation of mood, and management of stress. Psychological intervention helps patients deal with various mood and anxiety problems and learn adaptive coping skills. Psychological intervention can be in the form of individual treatment and group treatment. Individual treatment involves individualised treatment plan for the issue a person and the psychologist identified during assessment. On the other hand, group treatment delivers intervention in the form of group with a specific topic and theme.

PLWHAs support groups and caregivers led support groups

Patient support groups serve important functions for medically ill patients. These groups provide a forum to share feelings and experiences with each other, share information on treatment and resources, thereby lessening feelings of isolation and being neglected.[9] PLWHAs are given an opportunity to discuss HIV-related issues openly within the support groups, which may otherwise not be available in other contexts of daily life. In Hong Kong, patient support groups are formed at specialist clinics where patients receive medical treatment. Various non-governmental organizations (NGOs) also form groups and provide mutual support among patients.

Important therapy ingredient-

- **Reducing stress**
- **Improving coping**
- **Building social support**
- **Providing healthy environment for expressing feelings**

Cognitive-behavioural group treatment

- Conceptualize the process of cognition and the thought processes with respect to an HIV/AIDS-related status.
- Discover how to use some specific cognitive– behavioral strategies to counter thoughts that disturb clients with an HIV-positive or unknown status.
- Discover how to teach clients skills to cope with negative thoughts .

Strategies-

- Assessment
- Identifying negative thoughts
- Challenging negative thoughts
- Substituting negative thoughts with positive ones

Assessment-

- Issues related to HIV status
  —Whether symptomatic or asymptomatic

  —Stage and course of illness

- Issues related to the client's personality, and coping and adjustment abilities
- Issues related to the attitude and support of the family
- Issues related to the severity of distress related to the infection

Substituting NAT with positive ones-

- Redefining
- Generating alternatives
- Examining the evidence
- Decatastrophizing
- Re-attribution procedures
- Thought-recording
- Cognitive rehearsal

Behavioural techniques-

- Relaxation procedure
- Activity scheduling
- Behavioral task assignments
- Role-playing
- Social skills training

Supportive Techniques-

- Problem-solving
- Emotional catharsis and ventilation
- Environmental manipulation
- Externalization of interests
- Creative art therapy
- Spiritual therapies/developing a philosophy of life

Memory defects-

- Use a diary, calendars and notebooks
- Use reminders, stick papers
- Make notes for the doctor, counselor, phone calls, etc.
- Make lists for important things to be checked when leaving the house and for household-related activities
- For medication: Use an alarm clock as a reminder
- Use a cassette tape recorder, cordless bell and noise-activated key chain

Attention –related problems-

- Decrease the overload of information
- Do one task at a time
- Break large complex tasks down into small steps
- Decrease the distracters, e.g. television
- Talk/meet people one at a time

Cognitive behavioral stress management-

- 10 structured modules, group based
- Health education

- Increasing awareness of thoughts, emotions, and physical responses
- Teaching cognitive techniques
- Coping-skills training
- Interpersonal skills training
- Relaxation techniques
- Enhancing support resources

Relaxation technique-

- Diaphragmatic (slow, deep) breathing
- Autogenic training
- Progressive muscle relaxation
- Relaxation imagery
- Mindfulness meditation

Cognitive-behavioural group treatment is widely applied to patients of various mental and medical problems. It has been reported that cognitive-behavioural group could be effective in improving the quality of life and mood in HIV-infected. Typically, these interventions involve groups of 6-8 participants which are led by one or two psychotherapist. These groups usually meet weekly for 2-3 months. Most cognitive behavioural groups involve cognitive restructuring of maladaptive thoughts, provision of knowledge on stress responses, relaxation exercises, health behaviour change, and discussion on constructive coping skills. The aims are to reduce distress of participants and to promote their efficacy in dealing with stress related to HIV illness.

Couple and Family Counselling

- When culturally and socially appropriate and legal, counselling a couple together so that they can decide together to be tested and to return for results is often an effective strategy.

- One must pay close attention to the cultural and family dynamics between the two partners, which will provide information about counselling techniques that may be helpful.

Counselling on drug adherence

. Secondary prevention - counselling on safer sex

The social interface

The ultimate objective of psychosocial care is not only for prolonging survival of patients, but to assist them to attain quality of life and to enable them to reintegrate back to society. Enhancing self-care ability is crucial to facilitate normal living.

It is important to note that HIV-related social stigmas have never disappeared.[14] Social stigma can be a form of chronic stress for HIV/AIDS patients. In addition, fear of stigmatisation and discrimination would stop people from disclosing their diagnosis to friends and family members, while the burden and stressors of HIV are all kept to the infected individuals. Some people develop internal stigmas.

Antiretroviral drugs are medications for the treatment of infection by retroviruses, primarily HIV. When several such drugs, typically three or four, are taken in combination, the approach is known as **Highly Active Antiretroviral Therapy**, or **HAART**. The American National Institutes of Health and other organizations recommend offering antiretroviral treatment to all patients with AIDS.

Hope theory & its application:

- **Hollerman and Snyder (1991), found that hope was a more successful predictor of Problem focused coping than negative affectivity and optimism.**
- **Hope also may have an impact on behaviors related to HIV infection and AIDS; more specifically, hope may be implicated in such behaviors as sexual practices, use of sexually related protection, testing behavior, and treatment adherence.**

- Because Hope is a construed as the thought process involving a sense of agency and pathways for desired goals, it basically is a cognitive construct (Snyder et al., 1991).

- Given that a focal point of hope theory involves the thoughts about the attainment of goals, goal blockages are an obvious threat to fulfilling those goals.

- Perceived successful goal pursuits lead to positive emotions, and perceived unsuccessful goal pursuits should lead to negative emotions (Snyder et al., 1991).

- So, promoting hope in a person would be the goal of therapy.

Unit-X

Pain

Pain Management: Psychological Therapy

Psychological therapy may be part of your pain treatment plan.

When you are in pain, it is natural to feel angry, sad, hopeless, and depressed. Pain can alter your personality, disrupt your sleep, and interfere with your work and relationships. But, it doesn't have to. Psychological treatment provides a safe, non-drug method to treat your pain directly by reducing high levels of physiological stress that often aggravate pain. Psychological treatment also helps improve the indirect consequences of pain by helping you learn how to cope with the problems associated with pain.

The most common psychological treatments are:

- **Talk therapy:** Talk therapy offers the support and counseling of a psychiatrist or psychologist.
- **Relaxation training:** Deep relaxation has been associated with healing and pain reduction.
- Stress management **:** Stress can make pain worse. Stress management treatment can help you understand the relationship between stress and pain and teach you ways to reduce stress and ease pain.
- **Pain coping skills training:** By learning how to accommodate your life to pain, you can improve your quality of life significantly.

Psychological treatment can be considered for any intense and recurrent pain problem in conjunction with other pain management treatments. Your health care team can help you decide which treatments may be right for you.

Psychological Intervention

Patients suffering from chronic pain or other stress-related illnesses often struggle with their emotional well-being. Psychological counseling plays an important role in pain management. Our treatment program often involves individual therapy with a health psychologist who can address:

- Acceptance
- Activity avoidance
- Anger
- Anxiety
- Attitude changes
- Behavioral changes
- Communication skills
- Depression
- Emotional distress
- Grief
- Guilt
- Hostility
- Maladaptive cognitions
- Pain conceptualization
- Self-esteem
- Self-reliance
- Sexual dysfunction
- Sleep disturbance
- Social withdrawal
- Substance abuse

Electromyography (EMG) and thermal biofeedback training is used to help patients increase their ability to control voluntary physiological activities, giving them the ability to manage the body's reaction to pain and stress. Relaxation strategies such as visual imagery, diaphragmatic breathing, autogenic relaxation, progressive muscle relaxation and mini-relaxation techniques are taught.

Pain Managemen

Psychological/Mind-Body Therapies

Psychological factors are important contributors to the intensity of pain and to the disability associated with chronic pain. Pain and stress are intimately related. There may be a vicious cycle in which pain

causes stress, and stress, in turn, causes more pain. Mind/body approaches address these issues and provide a variety of benefits, including a greater sense of control, improved coping skills, decreased pain intensity and distress, changes in the way pain is perceived and understood, and increased sense of well being and relaxation. These approaches may be very valuable for adults and children with pain (Rusy, 2000).

For pain specialists, the therapies that are psychological, or focused on the relationship between mind and body, are considered mainstream; for many others in the health professions, and for the public at large, the same therapies might be considered complementary/alternative. The evidence in support of these approaches is very strong and how they are labeled is less important than their acceptance as necessary treatments.

Cognitive-Behavioral Therapy (CBT)

CBT has proven to be effective in reducing pain and disability when it is used as part of a therapeutic strategy for chronic pain. CBT addresses the psychological component of pain, including attitudes and feelings, coping skills, and a sense of control over one's condition. It can provide educational information and diffuse feelings of fear and helplessness. It can help a patient look at ways in which their attitudes contribute to inaccurate and unrealistic expectations, and can help them find a more realistic and balanced view of the problem. CBT may include training in various types of relaxation approaches, which can help people in chronic pain lower their overall level of arousal, decrease muscle tension, control distress, and decrease pain, depression and disability (Barkin, 1996). Relaxation imagery or pain-management imagery may also be taught to promote relaxation and changes in pain intensity or quality (Fernandez, 1989). Finally, CBT may include the teaching of life skills and coping skills that can assist the patient in productive problem solving and the prevention or minimization of future pain episodes.

CBT has been found to be effective as part of a treatment regimen for a variety of pain conditions including episodic migraine and chronic daily headache (Lake, 2001), chronic musculoskeletal pain (Haigh, 1999), pain in the well elderly (Manetto, 1996), chronic cancer pain (Thomas, 2000), rheumatoid arthritis and osteoarthritis (Bradley, 2002), fibromyalgia (Berman, 1999), myofascial temporomandibular disorders (Sherman, 2001), chronic low back pain (van Tulder, 2001), carpal tunnel syndrome pain (Feuerstein, 1999), and chronic pelvic pain (Reiter, 1998). It has been suggested to benefit patients with chronic fatigue syndrome, irritable bowel syndrome (Kroenke, 2000), and anxiety (Ketterer, 1999). Although research into the use of CBT in children is in the early stages, it holds promise for reducing pain-related distress in children (Chen, 2000).

Biofeedback

Biofeedback is the use of electronic monitoring instruments to provide patients with immediate feedback on heart rate, blood pressure, muscle tension, or brain wave activity. This allows the patient to learn how to influence these bodily responses through conscious control and regulation. Electromyographic (EMG) biofeedback, for example, can teach patients how to relax a particular muscle or how to achieve more generalized relaxation for stress reduction. Biofeedback has been shown to be effective in the management of migraine headaches, fibromyalgia, temporomandibular disorders, and rheumatoid arthritis (Barkin, 1996), Raynaud's disease, tension headaches (DePalma, 1997), headaches in children (Serration, 2000) and the pain associated with irritable bowel syndrome (Leafy, 1998). Many other applications of biofeedback continue to be studied.

Hypnosis

Hypnosis is a highly relaxed, trance-like state in which the conscious or rational part of the brain is temporarily tuned out through a focus on relaxation and non-attention to distracting thoughts. During hypnosis, changes like those found in meditation can occur, such as a slowing of the pulse and respiration, and an increase in alpha brain waves. The person may become more open to specific suggestions and therapeutic goals such as pain reduction. In the post-suggestion phase, the continued use of the new behavior after the hypnosis session is reinforced.

Medical hypnosis has been shown to be helpful in reducing both acute and chronic pain (Holyroid, 1996). A National Institutes of Health panel found hypnosis to be effective in alleviating the pain associated with cancer (NIH, 1996). A 1997 review of the literature on hypnosis in pain control, which evaluated all controlled scientific studies comparing hypnosis to other psychological interventions for pain, showed hypnosis to be equally or more effective in reducing suffering and possibly even reducing pain sensation (Holyroid, 1996). Other studies have shown effectiveness of hypnosis for pain associated with burns, cancer, and rheumatoid arthritis (DePalma, 1997) (Sellick, 1998) and pain and anxiety reduction related to surgery (Lang, 2000). Clinical evidence supports the use of hypnosis in reducing pain in a wide variety of acute and chronic pain conditions for a substantial number of patients.

Imagery

Imagery is the use of imagined pictures, sounds, or sensations for generalized relaxation or for specific therapeutic goals, such as the reduction of pain. These images can be initiated by the patient or guided by a practitioner. The sessions in which imagery is used can be individual or group.

More than half the studies of imagery for pain--postoperative pain, cancer pain, chronic low back pain, burns, and migraine headache--report significant relief from the procedure (Eller, 1999). In a review of laboratory research on coping strategies for pain control, imagery was the most effective in relieving pain (Eller, 1999). Many of these studies, however, evaluate imagery together with other interventions such as hypnosis, cognitive-behavioral therapy and relaxation techniques. Nonetheless, the limited evidence suggests the usefulness of guided imagery in reducing the sensory and emotional components of pain. The Agency for Health Care Policy and Research recommends the use of imagery for reduction of pain intensity and distress for cancer pain and for the management of mild to severe acute pain (AHCPR, 1992).

Relaxation Therapies

Relaxation therapies include a range of techniques such as autogenic training, various forms of meditation, progressive muscle relaxation, deep breathing, and paced respiration. The goal of these therapeutic approaches is overall relaxation and stress reduction. Practice can produce a set of physiologic changes that result in slowed respiration, lowered pulse and blood pressure, an increase in alpha wave brain activity, and possibly even reduction in the body's inflammatory response mechanism (Lutgendorf, 2000). This can have a positive impact on health and improve symptoms in many acute and chronic illnesses and conditions, including pain.

Relaxation training may decrease pain, depression and disability (Barkin, 1996). According to the 1996 National Institutes of Health report on the treatment of chronic pain and insomnia, there is strong evidence for the effectiveness of relaxation techniques in reducing chronic pain in a variety of medical conditions. Effects may include reduced pain and muscle tension, reduced anxiety and insomnia, and increased activity level (Good 1996, Carroll, 1998; Mandle, 1996).

Meditation is a specific type of relaxation intervention that also can have an effect on pain. Results

suggest that meditation can result in a higher tolerance to pain; decreased anxiety, stress and depression; increased activity levels; decreased use of pain-related medications; and increased levels of self-esteem (Mills, 1981; Kabat-Zinn, 1985; Harmon, 1999). The meditative technique based on increased awareness and staying in the moment, called mindfulness meditation, has been successfully used in treatment programs to reduce pain and improve mood in patients with chronic pain from a variety of conditions, including facial pain, coronary and noncoronary chest pain, gastrointestinal pain, low back, neck and shoulder pain and headache (Kabat-Zinn, 1982).

Prayer

Prayer is not usually considered a mind-body or a psychological approach, but it is worthwhile considering it in this context of mind/body treatments. Changes in the concept of health and illness, a broadening view of healing and curing, and interest in other cultural systems of medicine have created a growing openness to the spiritual dimensions of health. There are many anecdotal reports of the power of prayer and its effects on health. This has led to a substantial number of scientific research studies. As of 1992, there were at least 131 controlled trials that evaluated the effect of prayer on humans, microorganisms, plants, cancer cells and animals; 56 of these studies found significant effects (Targ, 1997). In human trials, prayer had a positive effect on levels of stress, depression, hypertension and substance abuse, illness-related complications, and life satisfaction (Targ, 1997). In a systematic review of randomized trials of distant healing, 57% of 23 trials showed a positive treatment effect (Astin, 2000). A recent study of the effects of direct and distant healing on chronic pain, however, showed no specific effect after eight weekly 30 minute treatments (Abbot 2001). This is an area that clearly warrants further stud

The role of psychological interventions in the management of patients with chronic pain

Abstract

Chronic pain can be best understood from a biopsychosocial perspective through which pain is viewed as a complex, multifaceted experience emerging from the dynamic interplay of a patient's physiological state, thoughts, emotions, behaviors, and sociocultural influences. A biopsychosocial perspective focuses on viewing chronic pain as an illness rather than disease, thus recognizing that it is a subjective experience and that treatment approaches are aimed at the management, rather than the cure, of chronic pain. Current psychological approaches to the management of chronic pain include interventions that aim to achieve increased self-management, behavioral change, and cognitive change rather than directly eliminate the locus of pain. Benefits of including psychological treatments in multidisciplinary approaches to the management of chronic pain include, but are not limited to, increased self-management of pain, improved pain-coping resources, reduced pain-related disability, and reduced emotional distress – improvements that are effected via a variety of effective self-regulatory, behavioral, and cognitive techniques. Through implementation of these changes, psychologists can effectively help patients feel more in command of their pain control and enable them to live as normal a life as possible despite pain. Moreover, the skills learned through psychological interventions empower and enable patients to become active participants in the management of their illness and instill valuable skills that patients can employ throughout their lives.

Keywords: chronic pain management, psychology, multidisciplinary pain treatment, cognitive behavioral therapy for pain

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Introduction

Pain is a ubiquitous human experience. It is estimated that approximately 20%–35% of adults experience chronic pain.[1,2] The National Institute of Nursing Research reports that pain affects more Americans than diabetes, heart disease, and cancer combined.[3] Pain has been cited as the primary reason to seek medical care in the United States.[4] Furthermore, pain relievers are the second most commonly prescribed medications in physicians' offices and emergency rooms.[5] Further solidifying the importance of adequate assessment of pain, the Joint Commission on the Accreditation of Healthcare Organizations issued a mandate requiring that pain be evaluated as the fifth vital sign during medical visits.[6]

The International Association for the Study of Pain (IASP) defines pain as "an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage".[7] The IASP's definition highlights the multidimensional and subjective nature of pain, a complex experience that is unique to each individual. Chronic pain is typically differentiated from acute pain based on its chronicity or persistence, its physiological maintenance mechanisms, and/or its detrimental impact on an individual's life. Generally, it is accepted that pain that persists beyond the expected period of time for tissue healing following an injury or surgery is considered chronic pain. However, the specific timeframe constituting an expected healing period is variable and often difficult to ascertain. For ease of classification, certain guidelines suggest that pain persisting beyond a 3–6 month time window is considered chronic pain.[7] Nevertheless, classification of pain based solely on duration is a strictly practical and, in some instances, arbitrary criterion. More commonly, additional factors such as etiology, pain intensity, and impact are considered alongside duration when classifying chronic pain. An alternative way to characterize chronic pain has been based on its physiological maintenance mechanism; that is, pain that is thought to emerge as a result of peripheral and central reorganization. Common chronic pain conditions include musculoskeletal disorders, neuropathic pain conditions, headache pain, cancer pain, and visceral pain. More broadly, pain conditions may be primarily nociceptive (producing mechanical or chemical pain), neuropathic (resulting from nerve damage), or central (resulting from dysfunction in the neurons of the central nervous system).[8]

Unfortunately, the experience of pain is frequently characterized by undue physical, psychological, social, and financial suffering. Chronic pain has been recognized as the leading cause of long-term disability in the working- age American population.[9] Because chronic pain affects the individual at multiple domains of his/her existence it also constitutes an enormous financial burden to our society. The combined direct and indirect costs of pain have been estimated to range from $125 billion to $215 billion, annually.[10,11] The widespread implications of chronic pain include increased reports of emotional distress (eg, depression, anxiety, and frustration), increased rates of pain-related disability, pain-related alterations in cognition, and reduced quality of life. Thus, chronic pain can be best understood from a biopsychosocial perspective through which pain is viewed as a complex, multifaceted experience emerging from the dynamic interplay of a patient's physiological state, thoughts, emotions, behaviors, and sociocultural influences.

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Pain management

Given the widespread prevalence of pain and its multi-dimensional nature, an ideal pain management regimen will be comprehensive, integrative, and interdisciplinary. Current approaches to the management of chronic pain have increasingly transcended the reductionist and strictly surgical, physical, or pharmacological approach to treatment. Current approaches recognize the value of a multidisciplinary treatment framework that targets not only nociceptive aspects of pain but also cognitive-evaluative, and motivational-affective aspects alongside equally unpleasant and impacting sequelae. The interdisciplinary management of chronic pain typically includes multimodal treatments such as combinations of analgesics, physical therapy, behavioral therapy, and psychological therapy. The multimodal approach more adequately and comprehensively addresses pain management at the molecular, behavioral, cognitive-affective, and functional levels. These approaches have been shown to lead to superior and long-lasting subjective and objective outcomes including pain reports, mood, restoration of daily functioning, work status, and medication or health care use; multimodal approaches have also been shown to be more cost-effective than unimodal approaches.[12,13] The focus of this review will be specifically on elucidating the benefits of psychology in the management of chronic pain.

Patients will typically initially present to a physician's office in the pursuit of a cure or treatment for their ailment/acute pain. For many patients, depending on the etiology and pathology of their pain alongside biopsychosocial influences on the pain experience, acute pain will resolve with the passage of time, or following treatments aimed at targeting the presumed cause of pain or its transmission. Nonetheless, some patients will not achieve resolution of their pain despite numerous medical and complementary interventions and will transition from an acute pain state to a state of chronic, intractable pain. For instance, research has demonstrated that approximately 30% of patients presenting to their primary-care physician for complaints related to acute back pain will continue to experience pain and, for many others, severe activity limitations and suffering 12 months later.[14] As pain and its consequences continue to develop and manifest in diverse aspects of life, chronic pain may become primarily a biopsychosocial problem, whereby numerous biopsychosocial aspects may serve to perpetuate and maintain pain, thus continuing to negatively impact the affected individual's life. It is at this point that the original treatment regimen may diversify to include other therapeutic components, including psychological approaches to pain management.

Psychological approaches for the management of chronic pain initially gained popularity in the late 1960s with the emergence of Melzack and Wall's "gate-control theory of pain"[15] and the subsequent "neuromatrix theory of pain".[16] Briefly, these theories posit that psychosocial and physiological processes interact to affect perception, transmission, and evaluation of pain, and recognize the influence of these processes as maintenance factors involved in the states of chronic or prolonged pain. Namely, these theories served as integral catalysts for instituting change in the dominant and unimodal approach to the treatment of pain, one heavily dominated by strictly biological perspectives. Clinicians and patients alike gained an increasing recognition and appreciation for the complexity of pain processing and maintenance; consequently, the acceptance of and preference for multidimensional conceptualizations of pain were established. Currently, the biopsychosocial model of pain is, perhaps, the most widely accepted heuristic approach to understanding pain.[17] A biopsychosocial perspective focuses on viewing chronic pain as an illness rather than disease, thus recognizing that it is a subjective experience and that treatment approaches are aimed at the management, rather than the cure, of chronic pain.[17] As the utility of a broader and more comprehensive approach to the management of chronic pain has become evident, psychologically-based interventions have witnessed a remarkable rise in popularity and recognition as adjunct treatments. The types of psychological interventions employed as part of a multidisciplinary pain treatment program vary according to therapist orientation, pain etiology, and patient characteristics. Likewise, research on the effectiveness of psychologically based interventions for chronic pain has shown variable, albeit promising, results on key variables studied. This overview will briefly describe frequently employed psychologically based treatment options and their respective effectiveness on key outcomes.

Current psychological approaches to the management of chronic pain include interventions that aim to achieve increased self-management, behavioral change, and cognitive change rather than directly eliminate the locus of pain. As such, they target the frequently overlooked behavioral, emotional, and cognitive components of chronic pain and factors contributing to its maintenance. Informed by the framework offered by Hoffman et al[18] and Kerns et al,[19] the following frequently employed psychologically-based treatment domains are reviewed: psychophysiological techniques, behavioral approaches to treatment, cognitive behavioral therapy, and acceptance-based interventions.

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Psychophysiological techniques

Biofeedback

Biofeedback is a learning technique through which patients learn to interpret feedback (in the form of physiological data) regarding certain physiological functions. For instance, a patient may use biofeedback equipment to learn to recognize areas of tension in their body and subsequently learn to relax those areas to reduce muscular tension. Feedback is provided by a variety of measurement instruments that can yield information about brain electrical activity, blood pressure, blood flow, muscle tone, electrodermal activity, heart rate, and skin temperature, among other physiological functions in a rapid manner. The goal of biofeedback approaches is for the patient to learn how to initiate physiological self-regulatory processes by achieving voluntary control over certain physiological responses to ultimately increase physiological flexibility through greater awareness and specific training. Thus a patient will use specific self-regulatory skills in an attempt to reduce an undesired event (eg, pain) or maladaptive physiological reactions to an undesired event (eg, stress response). Many psychologists are trained in biofeedback techniques and provide these services as part of therapy. Biofeedback has been designated as an efficacious treatment for pain associated with headache and temporomandibular disorders (TMD).[20] A meta-analysis of 55 studies revealed that biofeedback interventions (including various biofeedback modalities) yielded significant improvements with regard to frequency of migraine attacks and perceptions of headache management self-efficacy when compared to control conditions.[21] Studies have provided empirical support for biofeedback for TMD, albeit more robust improvements with regard to pain and pain-related disability have been found for protocols that combine biofeedback with cognitive behavioral skills training, under the assumption that a combined treatment approach more comprehensively addresses the gamut of biopsychosocial problems that may be encountered as a result of TMD.[22]

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Behavioral approaches

Relaxation training

It is generally accepted that stress is a key factor involved in the exacerbation and maintenance of chronic pain.[16,23] Stress can be predominantly of an environmental, physical, or psychological/emotional basis, though typically these mechanisms are intricately intertwined. The focus of relaxation training is to reduce tension levels (physical and mental) through activation of the parasympathetic nervous system and through attainment of greater awareness of physiological and psychological states, thereby achieving reductions in pain and increasing control over pain. Patients can be taught several relaxation techniques and practice them individually or in conjunction with one another, as well as adjuvant components to

other behavioral and cognitive pain management techniques. The following are brief descriptions of relaxation techniques commonly taught by psychologists specializing in the management of chronic pain.

Diaphragmatic breathing

Diaphragmatic breathing is a basic relaxation technique whereby patients are instructed to use the muscles of their diaphragm as opposed to the muscles of their chest to engage in deep breathing exercises. Breathing by contracting the diaphragm allows the lungs to expand down (marked by expansion of abdomen during inhalation) and thus increase oxygen intake.[24]

Progressive muscle relaxation (PMR)

PMR is characterized by engaging in a combination of muscle tension and relaxation exercises of specific muscles or muscle groups throughout the body.[25] The patient is typically instructed to engage in the tension/relaxation exercises in a sequential manner until all areas of the body have been addressed.

Autogenic training (AT)

AT is a self-regulatory relaxation technique in which a patient repeats a phrase in conjunction with visualization to induce a state of relaxation.[26,27] This method combines passive concentration, visualization, and deep breathing techniques.

Visualization/Guided imagery

This technique encourages patients to use all of their senses in imagining a vivid, serene, and safe environment to achieve a sense of relaxation and distraction from their pain and pain-related thoughts and sensations.[27]

Collectively, relaxation techniques have generally been found to be beneficial in the management of a variety of types of acute and chronic pain conditions as well as in the management of important pain sequelae (eg, health-related quality of life).[28–31] Relaxation techniques are usually practiced in conjunction with other pain management modalities, and there is considerable overlap in the presumed mechanisms of relaxation and biofeedback, for instance.

Operant behavior therapy

Operant behavior therapy for chronic pain is guided by the original operant conditioning principles proposed by Skinner[32] and refined by Fordyce[33] to be applicable to pain management. The main tenets of the operant conditioning model as it relates to pain hold that pain behavior can eventually evolve into and be maintained as chronic pain manifestations as a result of positive or negative reinforcement of a given pain behavior as well as punishment of more adaptive, non-pain behavior. If reinforcement and the ensuing consequences occur with sufficient frequency, they can serve to condition the behavior, thus increasing the likelihood of repeating the behavior in the future. Therefore, conditioned behaviors occur as a product of learning of the consequences (actual or anticipated) of engaging in the given behavior. An example of a conditioned behavior is continued use of medication – a behavior that results from learning through repeated associations that taking medication is followed by removal of an aversive sensation (pain). Likewise, pain behaviors (eg, verbal expressions of pain, low activity levels) can be become conditioned behaviors that serve to perpetuate chronic pain and its sequelae. Treatments that are guided by operant behavior principles aim to extinguish maladaptive pain behaviors through the same learning principles that these may have been established by. In general, treatment components of operant behavior

therapy include graded activation, time contingent medication schedules, and use of reinforcement principles to increase well behaviors and decrease maladaptive pain behaviors.

Graded activation

Psychologists can implement graded activity programs for chronic pain patients who have vastly reduced their activity levels (increasing likelihood of physical deconditioning) and subsequently experience high levels of pain upon engaging in activity. Patients are instructed to safely break the cycle of inactivity and deconditioning by engaging in activity in a controlled and time-limited fashion. In this manner, patients can gradually increase the length of time and intensity of activity to improve functioning. Psychologists can oversee progress and provide appropriate reinforcement for compliance, correction of misperceptions or misinterpretations of pain resulting from activity, where appropriate, and problem-solve barriers to adherence. This approach is frequently embedded within cognitive-behavioral pain management treatments.

Time-contingent medication schedules

A psychologist can be an important adjunct healthcare provider in overseeing the management of pain medications. In some cases, psychologists have the opportunity for more frequent and in-depth contact with patients than physicians and thus can serve as valuable collaborators of an integrated multidisciplinary treatment approach. Psychologists can institute time-contingent medication schedules to reduce the likelihood of dependence on pain medications for attaining adequate control over pain. Furthermore, psychologists are well equipped to engage patients in important conversations regarding the importance of proper adherence to medications and medical recommendations and problem-solve perceived barriers to safe adherence.

Fear-avoidance

The fear-avoidance model of chronic pain is a heuristic most frequently applied in the context of chronic low back pain (LBP).[34] This model draws largely from the operant behavior principles described previously. In essence, the fear-avoidance model posits that when acute pain states are repeatedly misinterpreted as danger signals or signs of serious injury, patients may be at risk of engaging in fear-driven avoidance behaviors and cognitions that further reinforce the belief that pain is a danger signal and perpetuate physical deconditioning. As the cycle continues, avoidance may generalize to broader types of activity and result in hypervigilance of physical sensations characterized by misinformed catastrophic interpretations of physical sensations. Research has shown that a high degree of pain catastrophizing is associated with maintenance of the cycle.[35] Treatments aimed at breaking the fear-avoidance cycle employ systematic graded exposure to feared activities to disconfirm the feared, often catastrophic, consequences of engaging in activities. Graded exposure is typically supplemented with psychoeducation about pain and cognitive restructuring elements that target maladaptive cognitions and expectations about activity and pain. Psychologists are in an excellent position to execute these types of interventions that closely mimic exposure treatments traditionally used in the treatment of some anxiety disorders.

Though specific graded exposure treatments have been shown to be effective in the treatment of complex regional pain syndrome type I (CRPS-1)[36] and LBP[37] in single-case designs, a larger-scale randomized controlled trial comparing systematic graded exposure treatment combined with multidisciplinary pain program treatment with multidisciplinary pain program treatment alone and with a wait-list control group found that the two active treatments resulted in significant improvements on outcome measures of pain intensity, fear of movement/injury, pain self-efficacy, depression, and activity level.[38] Results from this trial suggest that both interventions were associated with significant treatment effectiveness such that the

graded exposure treatment did not appear to result in additional treatment gains.[38] A cautionary note in the interpretation of these results highlights that the randomized controlled trial (RCT) included a variety of chronic pain conditions that extended beyond LBP and CRPS-1 and did not exclusively include patients with high levels of pain-related fear; the interventions were also delivered in group formats rather than individual formats. Although in-vivo exposure treatments are superior at reducing pain catastrophizing and perceptions of harmfulness of activities, exposure treatments seem to be as effective as graded activity interventions in improving functional disability and chief complaints.[39] Another clinical trial compared the effectiveness of treatment-based classification (TBC) physical therapy alone to TBC augmented with graded activity or graded exposure for patients with acute and sub-acute LBP.[40] Outcomes revealed that there were no differences in 4-week and 6-month outcomes for reduction of disability, pain intensity, pain catastrophizing, and physical impairment among treatment groups, although graded exposure and TBC yielded larger reductions in fear-avoidance beliefs at 6 months.[40] Findings from this clinical trial suggest that enhancing TBC with graded activity or graded exposure does not lead to improved outcomes with regard to measures associated with the development of chronic LBP beyond improvements achieved with TBC alone.[40]

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Cognitive-behavioral approaches

Cognitive-behavioral therapy (CBT) interventions for chronic pain utilize psychological principles to effect adaptive changes in the patient's behaviors, cognitions or evaluations, and emotions. These interventions are generally comprised of basic psychoeducation about pain and the patient's particular pain syndrome, several behavioral components, coping skills training, problem-solving approaches, and a cognitive restructuring component, though the exact treatment components vary according to the clinician. Behavioral components may include a variety of relaxation skills (as reviewed in the behavioral approaches section), activity pacing instructions/graded activation, behavioral activation strategies, and promotion of resumption of physical activity if there is a significant history of activity avoidance and subsequent deconditioning. The primary aim in coping skills training is to identify current maladaptive coping strategies (eg, catastrophizing, avoidance) that the patient is engaging in alongside their use of adaptive coping strategies (eg, use of positive self-statements, social support). As a cautionary note, the degree to which a strategy is adaptive or maladaptive and the perceived effectiveness of particular coping strategies varies from individual to individual.[41] Throughout treatment, problem-solving techniques are honed to aid patients in their adherence efforts and to help them increase their self-efficacy. Cognitive restructuring entails recognition of current maladaptive cognitions the patient is engaging in, challenging of the identified negative cognitions, and reformulation of thoughts to generate balanced, adaptive alternative thoughts. Through cognitive restructuring exercises, patients become increasingly adept at recognizing how their emotions, cognitions, and interpretations modulate their pain in positive and negative directions. As a result, it is presumed that the patients will attain a greater perception of control over their pain, be better able to manage their behavior and thoughts as they relate to pain, and be able to more adaptively evaluate the meaning they ascribe to their pain. Additional components sometimes included in a CBT intervention include social skills training, communication training, and broader approaches to stress management. Via a pain-oriented CBT intervention, many patients profit from improvements with regard to their emotional and functional well-being, and ultimately their global perceived health-related quality of life.

CBT interventions are delivered within a supportive and empathetic environment that strives to understand the patient's pain from a biopsychosocial perspective and in an integrated manner. Therapists see their role as "teachers" or "coaches" and the message communicated to patients is that of learning to better manage their pain and improve their daily function and quality of life as opposed to aiming to cure

or eradicate the pain. The overarching goal is to increase the patients' understanding of their pain and their efforts to manage pain and its sequelae in a safe and adaptive manner; therefore, teaching patients to self-monitor their behavior, thoughts, and emotions is an integral component of therapy and a useful strategy to enhance self-efficacy. Additionally, the therapist endeavors to foster an optimistic, realistic, and encouraging environment in which the patient can become increasingly skilled at recognizing and learning from their successes and learning from and improving upon unsuccessful attempts. In this manner, therapists and patients work together to identify patient successes, barriers to adherence, and to develop maintenance and relapse-prevention plans in a constructive, collaborative, and trustworthy atmosphere. An appealing feature of the cognitive behavioral approach is its endorsement of the patient as an active participant of his/her pain rehabilitation or management program.

Research has found CBT to be an effective treatment for chronic pain and its sequelae as marked by significant changes in various domains (ie, measures of pain experience, mood/affect, cognitive coping and appraisal, pain behavior and activity level, and social role function) when compared with wait-list control conditions.42 When compared with other active treatments or control conditions, CBT has resulted in notable improvements, albeit smaller effects (effect size ~ 0.50), with regard to pain experience, cognitive coping and appraisal, and social role function.42 A more recent meta-analysis of 52 published studies compared behavior therapy (BT) and CBT against treatment as usual control conditions and active control conditions at various time-points.43 This meta-analysis concluded that their data did not lend support for BT beyond improvements in pain immediately following treatment when compared with treatment as usual control conditions.43 With regard to CBT, they concluded that CBT has limited positive effects for pain disability, and mood; nonetheless, there are insufficient data available to investigate the specific influence of treatment content on selected outcomes.43 Overall, it appears that CBT and BT are effective treatment approaches to improve mood; outcomes that remain robust at follow-up data points. However, as highlighted by several reviews and meta-analyses, a critical factor to consider in evaluating the effectiveness of CBT for the management of chronic pain is centered on issues of effective delivery, lack of uniform treatment components, differences in delivery across clinicians and treatment populations, and variability in outcome variables of interest across research trials.13 Further complicating the interpretation of effectiveness findings are patient characteristics and additional variables that may independently affect treatment outcome.

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Acceptance-based approaches

Acceptance-based approaches are frequently identified as third-wave cognitive-behavioral therapies. Acceptance and commitment therapy (ACT) is the most common of the acceptance-based psychotherapies. ACT emphasizes the importance of facilitating the client's progress toward attaining a more valued and fulfilling life by increasing psychological flexibility rather than strictly focusing on restructuring cognitions.44 In the context of chronic pain, ACT targets ineffective control strategies and experiential avoidance by fostering techniques that establish psychological flexibility. The six core processes of ACT include: acceptance, cognitive defusion, being present, self as context, values, and committed action.45 Briefly, acceptance encourages chronic pain patients to actively embrace pain and its sequelae rather than attempt to change it, in doing so encouraging the patient to cease a futile fight directed at the eradication of their pain. Cognitive defusion (deliteralization) techniques are employed to modify the function of thoughts rather than to reduce their frequency or restructure their content. In this manner, cognitive defusion may simply alter the undesirable meaning or function of negative thoughts and thus decrease the attachment and subsequent emotional and behavioral response to such thoughts. The core process of being present emphasizes a non-judgmental interaction between the self and private thoughts and events. Values are utilized as guides for electing behaviors and interpretations that are

characterized by those values an individual strives to instantiate in everyday life. Finally, through committed action, patients can realize behavior changes aligned with individual values. Thus, ACT utilizes the six core principles in conjunction with one another to take a holistic approach toward increasing psychological flexibility and decreasing suffering. Patients are encouraged to view pain as inevitable and accept it in a nonjudgmental manner so that they can continue to derive meaning from life despite the presence of pain. The interrelated core processes exemplify mindfulness and acceptance processes and commitment and behavior change processes.[45]

Results of research on the effectiveness of ACT-based approaches for the management of chronic pain are promising, albeit still warranting further evaluation. A RCT comparing ACT with a waitlist control condition reported significant improvements in pain catastrophizing, pain-related disability, life satisfaction, fear of movements, and psychological distress that were maintained at the 7 month follow-up.[46] A larger trial reported significant improvements for pain, depression, pain-related anxiety, disability, medical visits, work status, and physical performance.[47] A recent meta-analysis evaluating acceptance-based interventions (ACT and mindfulness-based stress reduction) in patients with chronic pain found that, in general, acceptance-based therapies lead to favorable outcomes for patients with chronic pain.[48] Specifically, the meta-analysis revealed small to medium effect sizes for pain intensity, depression, anxiety, physical wellbeing, and quality of life, with smaller effects found when controlled clinical trials were excluded and only RCTs were included in the analyses.[48] Other acceptance-based interventions include contextual cognitive-behavioral therapy and mindfulness-based cognitive therapy, though empirical research on the effectiveness of these therapies for the management of chronic pain is still in its infancy.

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Expectations

An important and vastly overlooked common underlying element of all treatment approaches is consideration of the patient's expectation for treatment success. Despite the numerous advances in the formulation and delivery of effective multidisciplinary treatments for chronic pain, relatively little emphasis has been placed on recognizing the importance of expectations for success and on focusing efforts on enhancement of patients' expectations. The recognition that placebo for pain is characterized by active properties leading to reliable, observable, and quantifiable changes with neurobiological underpinnings is currently at the vanguard of pain research. Numerous studies have confirmed that, when induced in a manner that optimizes expectations (via manipulation of explicit expectations and/or conditioning), analgesic placebos can result in observable and measurable changes in pain perception at a conscious self-reported level as well as a neurological pain-processing level.[49,50] Analgesic placebos have been broadly defined as simulated treatments or procedures that occur within a psychosocial context and exert effects on an individual's experience and/or physiology.[51] The current conceptualization of placebo emphasizes the importance of the psychosocial context within which placebos are embedded. Underlying the psychosocial context and ritual of treatment are patients' expectations. Therefore, it is not surprising that the placebo effect is intricately embedded in virtually every treatment; as such, clinicians and patients alike will likely benefit from recognition that therein lies an additional avenue by which current treatment approaches to pain can be enhanced.

It has been proposed that outcome expectancies are core influences driving the positive changes attained through the various modes of relaxation training, hypnosis, exposure treatments, and many cognitive-oriented therapeutic approaches. Thus, a sensible approach to the management of chronic pain capitalizes on the power of patients' expectations for success. Regrettably, too often, health care providers neglect to directly address and emphasize the importance of patients' expectations as integral factors contributing to

successful management of chronic pain. The zeitgeist in our society is that of mounting medicalization of ailments fueling the general expectation that pain (even chronic pain) ought to be eradicated through medical advancements. These all too commonly held expectations leave many patients disillusioned with current treatment outcomes and contribute to an incessant search for the "cure". Finding the "cure" is the exception rather than the rule with respect to chronic pain conditions. In our current climate, where chronic pain afflicts millions of Americans annually, it is in our best interest to instill and continue to advocate a conceptual shift that instead focuses on effective management of chronic pain. A viable and promising route to achieving this is to make the most of patients' positive (realistic) expectations and educate pain patients as well as the lay public (20% of whom will at some future point become pain patients) on what constitutes realistic expectations regarding the management of pain. Perhaps, this can occur initially through current, evidence-based education regarding placebo and nonspecific treatment effects such that patients can correct misinformed beliefs they may have previously held. Subsequently clinicians can aim to enhance patients' expectations within treatment contexts (in a realistic fashion) and minimize pessimistic expectations that deter from treatment success, therefore, learning to enhance their current multidisciplinary treatments through efforts guided at capitalizing on the improvements placebo can yield, even within an "active treatment". Psychologists can readily address these issues with their patients and help them become advocates of their own treatment success.

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Emotional concomitants of pain

An often challenging aspect of the management of chronic pain is the unequivocally high prevalence of comorbid emotional distress. Research has demonstrated that depression and anxiety disorders are upward to three times more prevalent among chronic pain patients than among the general population.[52,53] Frequently, pain patients with psychiatric comorbidities are labeled "difficult patients" by healthcare providers, possibly diminishing the quality of care they will receive. Patients with depression have poorer outcomes for both depression and pain treatments, compared with patients with single diagnoses of pain or depression.[54,55] Psychologists are remarkably suited to address most of the psychiatric comorbidities typically encountered in chronic pain populations and thus improve pain treatment outcomes and decrease the emotional suffering of patients. Psychologists can address key symptoms (eg, anhedonia, low motivation, problem-solving barriers) of depression that readily interfere with treatment participation and emotional distress. Moreover, irrespective of a psychiatric comorbidity, psychologists can help chronic pain patients process important role transitions they may undergo (eg, loss of job, disability), interpersonal difficulties they may be encountering (eg, sense of isolation brought about by pain), and emotional suffering (eg, anxiety, anger, sadness, disappointment) implicated in their experience. Thus, psychologists can positively impact the treatment course by reducing the influence of emotional concomitants that are addressed as part of therapy.

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Conclusion

Benefits of including psychological treatments in multidisciplinary approaches to the management of chronic pain are abundant. These include, but are not limited to, increased self-management of pain, improved pain-coping resources, reduced pain-related disability, and reduced emotional distress-improvements that are effected via a variety of effective self-regulatory, behavioral, and cognitive techniques. Through implementation of these changes, a psychologist can effectively help patients feel more in command of their pain control and enable them to live as normal a life as possible despite pain. Moreover, the skills learned through psychological interventions empower and enable patients to become

active participants in the management of their illness and instill valuable skills that patients can employ throughout their lives. Additional benefits of an integrated and holistic approach to the management of chronic pain may include increased rates of return to work, reductions in health care costs, and increased health-related quality of life for millions of patients throughout the world.

The nature of pain

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- Managing chronic pain: How psychologists can help with pain management (PDF, 273KB)

Pain is an all-too-familiar problem and the most common reason that people see a physician. Unfortunately, alleviating pain isn't always straightforward. At least 100 million adults in the United States suffer from chronic pain, according to the Institute of Medicine. The American Academy of Pain Medicine reports that chronic pain affects more Americans than diabetes, heart disease and cancer combined.

Pain serves an important purpose by alerting you to injuries such as a sprained ankle or burned hand. Chronic pain, however, is often more complex. People often think of pain as a purely physical sensation. However, pain has biological, psychological and emotional factors. Furthermore, chronic pain can cause feelings such as anger, hopelessness, sadness and anxiety. To treat pain effectively, you must address the physical, emotional and psychological aspects.

Medical treatments, including medication, surgery, rehabilitation and physical therapy, may be helpful for treating chronic pain. Psychological treatments are also an important part of pain management. Understanding and managing the thoughts, emotions and behaviors that accompany the discomfort can help you cope more effectively with your pain — and can actually reduce the intensity of your pain.

Seeing a psychologist about pain

Psychologists are experts in helping people cope with the thoughts, feelings and behaviors that accompany chronic pain. They may work with individuals and families through an independent private practice or as part of a health care team in a clinical setting. Patients with chronic pain may be referred to psychologists by other health care providers. Psychologists may collaborate with other health care professionals to address both the physical and emotional aspects of the patient's pain.

When working with a psychologist, you can expect to discuss your physical and emotional health. The psychologist will ask about the pain you experience, where and when it occurs, and what factors may affect it. In addition, he or she will likely ask you to discuss any worries or stresses, including those related to your pain. You also may be asked to complete a questionnaire that allows you to record your own thoughts and feelings about your pain.

Having a comprehensive understanding of your concerns will help the psychologist begin to develop a treatment plan.

For patients dealing with chronic pain, treatment plans are designed for that particular patient. The plan often involves teaching relaxation techniques, changing old beliefs about pain, building new coping skills and addressing any anxiety or depression that may accompany your pain.

One way to do this is by helping you learn to challenge any unhelpful thoughts you have about pain. A psychologist can help you develop new ways to think about problems and to find solutions. In some cases, distracting yourself from pain is helpful. In other cases, a psychologist can help you develop new ways to think about your pain. Studies have found that some psychotherapy can be as effective as surgery for relieving chronic pain because psychological treatments for pain can alter how your brain processes pain sensations.

A psychologist can also help you make lifestyle changes that will allow you to continue participating in work and recreational activities. And because pain often contributes to insomnia, a psychologist may also help you learn new ways to sleep better.

Progressing and improving

Most patients find they can better manage their pain after just a few sessions with a psychologist. Those who are experiencing depression or dealing with a long-term degenerative medical condition may benefit from a longer course of treatment. Together with your psychologist, you will determine how long treatment should last. The goal is to help you develop skills to cope with your pain and live a full life.

Stress and chronic pain

Having a painful condition is stressful. Unfortunately, stress can contribute to a range of health problems, including high blood pressure, heart disease, obesity, diabetes, depression and anxiety. In addition, stress can trigger muscle tension or muscle spasms that may increase pain. Managing your emotions can directly affect the intensity of your pain.

Psychologists can help you manage the stresses in your life related to your chronic pain.

Psychologists can help you learn relaxation techniques, such as meditation or breathing exercises to keep stress levels under control. Some psychologists and other health care providers use an approach called biofeedback, which teaches you how to control certain body functions.

In biofeedback, sensors attached to your skin measure your stress response by tracking processes like heart rate, blood pressure and even brain waves. As you learn strategies to relax your muscles and your mind, you can watch on a computer screen as your body's stress response decreases. In this way, you can determine which relaxation strategies are most effective, and practice using them to control your body's response to tension.

Stress is an unavoidable part of life, but managing your stress will help your body and your mind and lessen your pain.

Tips for coping with pain

Consider the following steps that can be helpful in changing habits and improving your sleep:

- **Stay active.** Pain — or the fear of pain — can lead people to stop doing the things they enjoy. It's important not to let pain take over your life.
- **Know your limits.** Continue to be active in a way that acknowledges your physical limitations. Make a plan about how to manage your pain, and don't push yourself to do more than you can handle.

- **Exercise.** Stay healthy with low-impact exercise such as stretching, yoga, walking and swimming.
- **Make social connections.** Call a family member, invite a friend to lunch or make a date for coffee with a pal you haven't seen in a while. Research shows that people with greater social support are more resilient and experience less depression and anxiety. Ask for help when you need it.
- **Distract yourself.** When pain flares, find ways to distract your mind from it. Watch a movie, take a walk, engage in a hobby or visit a museum. Pleasant experiences can help you cope with pain.
- **Don't lose hope.** With the right kind of psychological treatments, many people learn to manage their pain and think of it in a different way.
- **Follow prescriptions carefully.** If medications are part of your treatment plan, be sure to use them as prescribed by your doctor to avoid possible dangerous side effects. In addition to helping you develop better ways to cope with and manage pain, psychologists can help you develop a routine to stay on track with your treatment.

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Unit-XI

Terminally ill

Introduction- Health psychology is the field within psychology devoted to understanding psychological influences on how people stay healthy, why they become ill, and how they respond when they do get ill. health psychologist study both such issues and promote interventions to help people stay well or get over-illness psychological factors appear to impact upon the development and progression of chronic diseases such as coronary heart disease,cancer, diabetes, arthritis and HIV/AIDS similarly ,psychosocial intervention have been shown to improve the quality of life of patients with established disease and seem to influence biological processes thought to ameliorate disease progression.

Terminal illness- Terminal illness is a medical term popularized in the 20th century to describe a disease that cannot be cured or adequately treated and that is reasonably expected to result in the death of the patient within a short period of time. This term is more commonly used for progressive diseases such as cancer or advanced heart disease than for trauma. In popular use, it indicates a disease which will eventually end the life of the sufferer.

A patient who has such an illness may be referred to as a terminal patient, terminally ill or simply terminal. Often, a patient is considered to be terminally ill when the life expectancy is estimated to be six months or less, under the assumption that the disease will run its normal course. The six-month standard is arbitrary, and best available estimates of longevity may be incorrect. Consequently, though a given patient may properly be considered terminal, this is not a guarantee that the patient will die within six months. Similarly, a patient with a slowly progressing disease, such as AIDS, may not be considered terminally ill because the best estimates of longevity were greater than six months. However this does not

guarantee that the patient will not die unexpectedly early. In general, physicians slightly overestimate survival so that, for example, a person who is expected to live for about six weeks would likely die around four weeks. Glare P,Virik K, Jones M et al. (2003).

- **Stages of terminally illness**

Each patient reacts differently to the news of carrying a terminal illness such as cancer. In general, almost all patients go through various stages of acceptance when a disease like cancer has been diagnosed. Pass, OM.; Morrison, T. (2006).

The first stage is disbelief. Most people are shocked that it could happen to them, there is extreme anxiety especially about the unknown. Shock, despair and anger are common. There is also guilt that perhaps the person has done something wrong to receive such a diagnosis.(University of Virginia Health System. "Coping with terminal cancer" 2010-02-09) some individuals use humor as a psychological defense mechanism; others become helpless and often start to bargain. This first stage usually lasts from a few days to a few weeks.

The second stage is depression which is usually a reaction to the diagnosis. The depression is mild to moderate in intensity and needs family support. Only in rare cases is any type of medical therapy required.

Duration of depression often can last several weeks but soon fades and the person goes into the final stage of ACCEPTANCE.

- **“Euthanasia"**

History

According to the historian N. D. A. Kemp, the origin of the contemporary debate on euthanasia started in 1870. (Nick Kemp ,7 September 2002). Nevertheless, euthanasia was debated and practiced long before that date. Euthanasia was practiced in Ancient Greece and Rome: for example, hemlock was employed as a means of hastening death on the island of Kea, a technique also employed in Marseilles and by Socrates in Athens. Euthanasia, in the sense of the deliberate hastening of a person's death, was supported by Socrates, Plato and Seneca the Elder in the ancient world, although Hippocrates appears to have spoken against the practice, writing "I will not prescribe a deadly drug to please someone, nor give advice that may cause his death" (nothing there is some debate in the literature about whether or not this was intended to encompass euthanasia). (Mystakidou, et. Al,2005) Stolberg, Michael (2007), Gesundheit, et.al (2006).

- ✓ **Definition**

The word "euthanasia" was first used in a medical context by Francis Bacon in the 17th century, to refer to an easy, painless, happy death, during which it was a "physician's responsibility to alleviate the 'physical sufferings' of the body." Bacon referred to an "outward euthanasia"—the term "outward" he used to distinguish from a spiritual concept—the euthanasia "which regards the preparation of the soul." (Francis Bacon: the major works By Francis Bacon, Brian Vickers).

“Euthanasia is the "painless inducement of a quick death".(Kohl, Marvin ,1974).

"The painless killing of a patient suffering from an incurable and painful disease or in an irreversible coma".(The definition offered by the Oxford English Dictionary).

"A mode or act of inducing or permitting death painlessly as a relief from suffering"

(Marvin Khol and PaulKurtz's)

- ✓ **Classification of euthanasia**

Euthanasia may be classified according to whether a person gives informed consent into three types: voluntary, non-voluntary and involuntary.

There is a debate within the medical and bioethics literature about whether or not the non-voluntary (and by extension, involuntary) killing of patients can be regarded as euthanasia, irrespective of intent or the patient's circumstances. In the definitions offered by Beauchamp & Davidson and, later, by Wreen1998, consent on the part of the patient was not considered to be one of their criteria, although it may have been required to justify euthanasia. (Wreen, Michael ,1988) However, others see consent as essential.

Voluntary euthanasia- Euthanasia conducted with the consent of the patient is termed voluntary euthanasia. Active voluntary euthanasia is legal in Belgium, Luxembourg and the Netherlands. Passive voluntary euthanasia is legal throughout the U.S. per Cruzan v. Director, Missouri Department of Health. When the patient brings about his or her own death with the assistance of a physician, the term assisted suicide is often used instead. Assisted suicide is legal in Switzerland and the U.S. states of Oregon, Washington and Montana.

Non-voluntary euthanasia- Euthanasia conducted where the consent of the patient is unavailable is termed non-voluntary euthanasia. Examples include child euthanasia, which is illegal worldwide but descriminalized under certain specific circumstances in the Netherlands under the Groningen Protocol.

Involuntary euthanasia- Euthanasia conducted against the will of the patient is termed involuntary euthanasia.

Passive and active euthanasia- Voluntary, non-voluntary and involuntary euthanasia can all be further divided into passive or active variants.

Passive euthanasia entails the withholding of common treatments, such as antibiotics, necessary for the continuance of life.

Active euthanasia entails the use of lethal substances or forces, such as administering a lethal injection, to kill and is the most controversial means. A number of authors consider these terms to be misleading and unhelpful.(Harris, NM. Oct 2001).

Beginnings of the contemporary euthanasia debate

In the mid-1800s, the use of morphine to treat "the pains of death" emerged, with John Warren recommending its use in 1848. A similar use of chloroform was revealed by Joseph Bullar in 1866. However, in neither case was it recommended that the use should be to hasten death.

In 1870 Samuel Williams, a schoolteacher, initiated the contemporary euthanasia debate through a speech given at the Birmingham Speculative Club in England, which was subsequently published in a one-off publication entitled Essays of the Birmingham Speculative Club, the collected works of a number of members of an amateur philosophical society.(Rachels January 1975)Williams' proposal was to use chloroform to deliberately hasten the death of terminally ill patients: Euthanasia may also lead to counterexamples: such definitions may encompass killing a person suffering from an incurable disease for

personal gain (such as to claim an inheritance), and commentators such as Tom Beauchamp & Arnold Davidson have argued that doing such would constitute "murder simpliciter" rather than euthanasia.

Some element incorporated into many definitions is that of intentionality – the death must be intended, rather than being accidental, and the intent of the action must be a "merciful death".

Michael Wreen argued that "the principal thing that distinguishes euthanasia from intentional killing simpliciter is the agent's motive: it must be a good motive insofar as the good of the person killed is concerned"

A view mirrored by Heather Draper, who also spoke to the importance of motive, arguing that "the motive forms a crucial part of arguments for euthanasia, because it must be in the best interests of the person on the receiving end."

Definitions such as that offered by the House of Lords Select Committee on Medical Ethics take this path, where euthanasia is defined as "a deliberate intervention undertaken with the express intention of ending a life, to relieve intractable suffering."

Beauchamp & Davidson also highlight Baruch Brody's "an act of euthanasia is one in which one person ... Kills another person for the benefit of the second person, who actually does benefit from being killed".

Draper argued that any definition of euthanasia must incorporate four elements: an agent and a subject; an intention; a causal proximity, such that the actions of the agent lead to the outcome; and an outcome. Based on this, she offered a definition incorporating those elements, stating that euthanasia "must be defined as death that results from the intention of one person to kill another person, using the most gentle and painless means possible, that is motivated solely by the best interests of the person who dies." Prior to Draper, Beauchamp & Davidson had also offered a definition which includes these elements, although they offered a somewhat longer account and one that specifically discounts fetuses in order to distinguish between abortions and euthanasia.

In discussing his definition, Wreen noted the difficulty of justifying euthanasia when faced with the notion of the subject's "right to life". In response, Wreen argued that euthanasia has to be voluntary, and that "involuntary euthanasia is, as such, a great wrong". Other commentators incorporate consent more directly into their definitions. For example, in a discussion of euthanasia presented in 2003 by the European Association of Palliative Care (EPAC) Ethics Task Force, the authors offered: "Medicalized killing of a person without the person's consent, whether nonvoluntary (where the person in unable to consent) or involuntary (against the person's will) is not euthanasia: it is murder. Hence, euthanasia can be voluntary only." Although the EPAC Ethics Task Force argued that both non-voluntary and involuntary euthanasia could not be included in the definition of euthanasia, there is discussion in the literature about excluding one but not the other.

Nick Kemp (2002) wrote in favour of euthanasia, as did Annie Besant, the essayist and reformer who later became involved with the National Secular Society, considering it a duty to society to "die voluntarily and painlessly" when one reaches the point of becoming a 'burden'. (Ian Dowbiggin ,March 2007) Popular Science also analyzed the issue in May 1873, assessing both sides of the argument. Nevertheless, Kemp notes that at the time, medical doctors did not participate in the discussion; it was "essentially a philosophical enterprise... Tied inextricably to a number of objections to the Christian doctrine of the sanctity of human life".(Nick Kemp ,7 September 2002).

Death

Definition

1. Apparent death

"Apparent death is the cessation of life as indicated by the absence of all vital functions."

2. Legal death

"The total absence of activity in the brain and central nervous system, the cardiovascular system, and the respiratory system as observed and declared by a physician. Mosby's Medical Dictionary, 8th edition. © 2009, Elsevier.Www.advintl.com.

3.MedicallyDeath

A state defined in the US by the Uniform Determination of Death Act, 1981, as that which occurs in an individual who has sustained either (1) Irreversible cessation of circulatory or respiratory functions, or (2) Irreversible cessation of all functions of the entire brain, including the brain stem Segen's Medical Dictionary. © 2012 Farlex, www.omegastunguns.com

4.Spiritual Death- Spiritual Death is the separation of the soul (spirit) from God.

Spiritual death is "a state of being in which the human soul is separated from God and has not been enlivened by his Spirit."

5.Religious death - When the human's brain has stopped functioning completely, drugs and a respirator are keeping his heart beating and his or her lungs inflating."Usually, the same way it has traditionally been defined in all cultures: by a lack of vital signs. Most world religions lack a clear doctrinal statement that certifies when, exactly, the moment of death can be said to have occurred. For most of human history, there was no need for one since prior to the invention of life-support equipment, the absence of circulation or respiration was the only way to diagnose death. This remains the standard of death in most religions.

Facing death -Facing his or her own death is probably most difficult task of an individual's development. Death brings loss of everyone and everything that a person loves. Every aspect of life that one may have taken for granted suddenly seems intensely valuable the certainty of its loss is acutely painful (Kubler – Ross,1969).

One's feeling about death depend partly upon how one conceives of death in the first place,and particularly upon how much one fears it. Those with a strong religious orientation are often very much at peace in the last days of their world view hold that there is an afterlife during which they will be happy .often fear of death reflects an individual's difficulty in facing the incompleteness of his or her life. death may seem like a premature and unfair revoking of one's potential (Kubler –Ross,1969)

Stage of dying

Researchers in the field of thanatology (the study of death and dying) have learned a great deal about how people react psychologically to terminal illness (Kastenbaum,1977;Kastenbaum &Aisenberg,1972;kubler-Ross,1969). When it becomes clear death is likely in the near future ,individual cope emotionally.Dr.Kubler –Ross outlined five stages that many, although not all ,dying individual go through. This stages do not occur in any particular order, and the patient may experience a shifting back and forth between stages.

1. The first stage is "denial", in which the person simply refuses to accept that his or her illness is terminal.

2.The second stage is "anger" ,and the individual may lash out at anyone from doctors to family to God

3.The third stage is bargaining ,in which the person tries to " buy " his or her health ,either by working out an elaborate framework through which death can be cheated ,or by begging higher power for healing .

4. The fourth stage is "depression",typified by withdrawal and resignation to the fate of death.

5. The fifth and final stage is "acceptance" ,in which the person no longer tries to fight against the inevitable ,but is not depressed or angry.

Death anxiety- Death anxiety is the morbid, abnormal or persistent fear of death or dying. The British National Health Service defines death anxiety as a feeling of dread, apprehension or solicitude (anxiety) when one thinks of the process of dying, or ceasing to be or what happens after death. It is also referred to as thanatophobia (fear of death) and necrophobia (fear of death or the dead). Lower ego integrity, more physical problems, and more psychological problems are predictive of higher levels of death anxiety in elderly people.

Types death anxiety

1. death anxiety

Predatory death anxiety is the most basic form of death anxiety, with its origins stemming from the first unicellular organisms' set of adaptive resources.

2. Predation death anxiety;

Predation death anxiety is a form of death anxiety that arises, often occurring unconsciously rather than consciously, when an individual physically and/or mentally harms another.

3. Existential death anxiety;

Existential death anxiety is the basic knowledge and awareness that natural life is short. It is said that existential death anxiety directly correlates to language; that is, language has created the basis for this type of death anxiety through communicative and behavioural changes. However, existential death anxiety, unlike predatory death anxiety, does not involve episodes of psychological or physical harm.

Personal Meanings of Death

Humans develop meanings and associate them with objects and events in their environment. These meanings that we associate to certain things are what provoke certain emotions within an individual. People tend to develop personal meanings of death and those meanings could accordingly be negative or positive for the certain individual. If they are positive, then the consequences of those meanings can be comforting to the individual. If negative they can cause emotional turmoil in the individual when faced with the death of someone or when faced with death itself. Depending on the certain meaning one has associated with death, the consequences will vary accordingly whether they are negative or positive meanings.(Cicirelli, V. G.,1998).

Relationship between adult attachment and Death Anxiety

There has been much literature that supports the existence of a correlation between one's state of coping skills, mental health, emotions and cognitive reactions to stressful events, and a one's ability to regulate affect concerning one's Death Anxiety. A series of tests determined that significantly high levels of Death Anxiety tend to occur in close relationships with an intimate partner (moreso amongst females than males).

Gender, Age, and Death Anxiety

The connection between death anxiety and the sex one belongs to appears to be strong. Studies show that females tend to have more death anxiety than males. Thorson and Powell (1984) did a study to investigate this connection, and they sampled men and women from 16 years of age to over 60. The Death Anxiety Scale showed higher mean scores for women than for men. Moreover, researchers believe that age and culture could be major influences in why women score higher on death anxiety scales than men.(Harrawood, et.al (2008).

It is during the years of young adulthood (20 to 40 years of age) that death anxiety most often begins to become prevalent. However, during the next phase of life, the middle age adult years (40-64 years of age), death anxiety peaks at its highest levels when in comparison to all other age ranges throughout the lifespan. Surprisingly, levels of death anxiety then slump off in the old age years of adulthood (65 years of age and older). This is in contrast with most people's expectations, especially regarding all of the negative connotations younger adults have about the elderly and the aging process (Kurlychek&Trenner, 1982).

Bereavement and grieving

The death of love one can be emotionally devastating and one of the most stressful life events a person can experience recovery may be painful . Understanding the effects of death on survivors is essential if medical professional are to intervene in an effective manner to help survivors cope. Health care professionals are typically involved in the process of death, either through the care they provide during terminal illness or through resuscitation attempts. They also typically inform the family in the case of accidents or sudden deaths. The extent of health professional's contributions in helping survivors cope, of course, depends upon the nature of his or her relationship to the patient's family members and other loved ones. Health professionals, including psychologist, can provide information, education and emotional support .they can help recognize clinically patterns of reactions and emotional support. They can help recognize clinically abnormal pattern of reaction and adjustment in those who have suffered a loss so that individuals who need help can get it .

During a patient terminal illness, health care staff can effectively establish themselves as resource person for the family and family and friends of the adjustment of those to be left behind. Clarification of the patient's physical situation, accurate information about the cause of the condition, as wells on going updates of the patient's care can be extremely important. physicians can arrange for extended visiting hours so that patients and family can have privacy to talk openly with one another .thus, it is important for healthcare professional to understand the experience of survivors as to learn ways to help them cope with their own physical and emotional reactions.

Before examining the grief experience in more detail, let us define some important terms that we will be using in this section.

- ✓ **Bereavement** is the fact of loss through death.a person who survives the death of loved one is bereaved.
- ✓ **Bereavement reactions** are psychological, physical, or behavioral responses to bereavement.

- ✓ **Grief** is the feeling or affective state associated with the condition of bereavement.

- ➢ **The manner in which physicians break bad news to patient can affect grieving.**

The grief reaction can be overwhelming to a person ,both psychologically and physically .not only person's mind affected ,but his or her biological homeostasis is interrupted as well .during the period of acute grief, the bereaved person's movements are often slowed down ,and he or she may have slumped posture . Sometimes there is alternating agitation, restlessness, and increased motor activity .during period of despair the bereaved person give up favorite activity and avoiding socializing. Some may engage in excessive smoking and drinking and other health –compromising behaviors, particularly if they had been users of such substances before the death. These and other risk-taking behavior likely represent in attempt to defend against the painful feelings of grief rather than to accept and work through them. it is important to note that although grief and depression may appear and similar in term of the individual 's behavior they are quite different an important way .both involve sadness and "vegetative "signs such as appetite and sleep disturbances ,but depression also involved felling of unworthiness or worthlessness, negative belief and thoughts about the self ,and withdrawal from social networks.

- ✓ **The biological effects of bereavement and grieving**

The process of grieving sometimes results in outcomes that threaten the bereaved individual 's sense of stability and forestall his or her recovery.

For example,a person who has been bereaved may express psychological pain through physical symptoms. Acute grief is often associated with physical complaints including pain, gastrointestinal disturbances, and vegetative symptom such as loss of energy, and sleep and appetite disturbances. Beyond these expected physical complaints, some bereaved individual

- ✓ **Psychological experience**

-shock,denial,Numbness ,anger ,Guilt Waves of distress,crying,Aimless activity,Agitation, anxiety ,Appetite and sleep disturbances

- ✓ **Cognitive changes**

-hallucination,dreams,difficulty in concentration,sympathetic nervous system activation

disturbed autonomic regulation

- ✓ **Immune system changes**

-Decreased natural killer cell activity,reduced lymphocyte production

- ✓ **Cardiovascular changes**

Cardiovascular regulation disturbances,cardiac arrhythmia,exacerbation of hypertension

-sudden cardiac death

- ✓ **Changes in adrenal functioning**

increase in adrenal activity,Stresshormone secretion

- ✓ **Social experience**

-Social isolation, withdrawal ,increased health risk behavior

PSYCHOSOCIAL AND BIOBEHAVIORAL ASPECT OF TERMINALLY ILLNESS

Quality of life

When studying psychosocial aspect of chronically ill, we need to study the concept of "quality of life" it is the subjective experience that can be best rated by the patient. Few valid and reliable technique for measurement of quality of life have been developed .Sickness Impact Profile(SIP) measures quality of life on the basis of functioning in three categories physical, psychosocial, and other(sleep ,eating ,work reactions) using another widely used measure, the index of activities of daily living (ADL),independent scores for six functions bathing, dressing ,toileting ,mobility, continence, and feeding can be obtained. The reasons for studying quality of life can be summarized as:

1. Quality of life measures can help opinion witch particularly problem (sexual functioning depression) are likely to emerge for the patients particularly diseases such information would be helpful in anticipating the kind of interventions that might be required.

2. It is important to assess the impact of treatments on quality of life it helps assess whether the treatment is more harmful than the diseases itself (e.g. cancer),to assess the impact of unpleasant therapies and to compare varied therapies.

3. Use in policy level decision making it helps in maximizing the likelihood of long-term survival with the highest quality of life possible health policy makers to compare the impact of different chronic diseases on health care costs and to assess the cost –effectiveness of different intervention.

Emotional responses to chronic illness:

Sufferers from chronic illnesses must integrate the patient role into their lives psychologically adapt to their disorders. Immediately after the diagnosis, patient is often in state of crisis marked by physical, social and psychological disequilibria. The habitual ways of coping may fail them resulting in experience of intense feelings of disorganization, anxiety fear and other emotions .since diagnosis changes the plans an individual had made for his future life, it tend to be disorienting and uprooting for the patient. each moment may bring to mind another plan that must now be modified in light of the diagnosis of the illness .the patient may be consurned by increasing worries ,appear unresponsive, preoccupied by wrong problems and unable to understand the scope and limitations of the treatments that will be require. Disorientation is usually accompanied by denial and anxiety, which may translate in to depression as the disease progresses.

1. Denail

It is defense mechanism , the subconscious blocking out of the full realization of the reality and implication of the disorder .denial of the implications of symptom may act as a deterrent to seeking proper treatment and may have adverse effects if it interferes with the ability to take in necessary information to that will be part of patient's treatment or self –management program .how can mask the terror associated with the chronic disease ,until the patient more accustomed to the diagnosis and better able to sort out realistically the restrictions that it will pose . among patient who do not have to follow prescribed treatment regimens ,denial can help protect them psychologically, making it possible for them to resume their normal lives quickly

2. Anxiety

Anxiety is reliably tied to illness events (a twinge of chest pain for CVD patient raises concern over another heart attack), such as awaiting test result or obtaining and can interfere checkups (is course of diseases worsening) since anxiety is intrinsically distressing and can interfere with good functioning .it is a problem radiotherapy ,glucose control and other treatment procedures are likely to be adversely affective due to anxiety while anxiety directly attributable to the disease.

4. Depression the most common reaction to a chronic disease .it may be delayed reaction to chronic illness because it often takes time for patient to understand full implication of their condition. Depress patient are less motivated to undergo rehabilitation ,less likely to restore lives to prediagnosis level more likely to common suicide catastrophize overgeneralize and negatively interpret their situation. assessing and diagnosing overgeneralize and negative interpretation their situation .assessing and diagnosing depression is often difficult and therefore goes unattended more often than not it is expected to occur and therefore conveniently ignored severity of illness extent of pain and disability ,social stress, lack of social support ,self –denial etc. exacerbate depression in chronically ill patient.

MANAGEMENTFOR TERMINAL ILL PATIENTS

Counseling for terminal ill patients

There are some obvious ways in which counseling a terminally ill patient is very different from counseling a healthy individual, or even someone who is ill but expects to recover . In the later cases, long term plans can be made ,and focus of counseling tends to be on the future . In case of terminal illness, the counseling relationship is likely to be cut short by the patients conditions , and it is impossible to focus on long term goals. Instead ,issues center around completing " unfinished business" and coming to view one's life as having been worthwhile. Patients may find it hard to view their lives as meaningful if there are things in their lives left undone , or if they have failed to accomplish tasks that are important to them .the therapist's work involves helping the patient to see the value in his or her life and accomplishments.

One framework for caring for terminally ill patients proposes eight important elements , which are frames as "8 Cs"(Pusari,1998) they are described in detail in table .these elements are proposed to be essential for anyone who is attempting to provide holistic care to a dying individual –care that takes into account not just the person's physical needs, but also his or her emotional ,spiritual and cultural needs.Elements of care for the terminally ill patients

1. Compassion: Having a deep sense of caring for the terminally ill individual and keeping his or her best interest at the forefront

2. Competence: utilizing appropriate methods of care and conducting them efficiently.

3. Confidence: having a belief in oneself as an effective caretaker and a belief in the meaningfulness of the patients

4. Conscience: embodying a sense of integrity when dealing with patient.

5. Commitment: demonstrating that one will be there over the course of the illness and that one will not flee from difficulties brought about by the process of dying .

6. Courage : having the strength to face head –on the difficult issue a related to the patient's condition.

7. Culture :taking social and cultural factors related to the process of dying into account

8. Communication: dealing openly with the patient regarding his or her illness and prognosis ;being a good listener as well as an honest respondent.

Hospice care- In 1975 in the united states , a powerful grassroots movement began that argued for and helped to promote an alternative to aggressive hospital care for the terminally ill. This alternative is called hospice . The number of hospices in the united states has now grown to several thousand, and they provide care not only for the dying patient but also his or her family. The he atmosphere is homelike and patient have considerable autonomy in making decisions about their care .Attention given not only to the requirements for physical care but also to the emotional ,social ,and spiritual needs of the patient and family .

Hospice care is available in this country in inpatient setting as well as in home care (which involves visits from the staff daily or several time s a week) Multidisciplinary terms of physicians ,nurses, social workers,psychiatrist, psychologist and physical therapist provide hospice services . Many hospices also provide some bereavement intervention,in preparation for the death as well as after the death ,in an effort to promote adjustment among loved ones of the deceased .in all hospices ,a great deal of attention is paid to the quality of the patient's life. Family members are encouraged to speaks openly with terminally ill loved one. There is an effort, in a supportive atmosphere, to do so .Although the technical level of care is comparable, the hospices environment is much less stressful for the patient than is a tradition hospitals (Adkins,1984;Kane et al 1985) while home care can be emotionally the most satisfying for the patient, studies do show that even with help from a home –based hospices program, home care place tremendous stress on the other members of the family (Aneshensel et al.1993) hospices, which combine a family –like felling with the support of a hospital or skilled nursing facility may ,for many be the best choice for palliative care.

Bereavement counseling

AIDS –related bereavement may differ from traditional model in at least four ways (Kain,1997)

1. First many people who die of AIDS do so at a young age.Walker (1991) notes that although survivors may have anticipated an HIV –positive person's death, it still remains out of synchrony with the expected life –cycle of some one young .

2.Second ,HIV-related bereavement is also distinguished from other type of grief by the stigma associated with AIDS. This stigma may prevent those who survive from freely mourning or acknowledging the cause of a friend or loved –one 's death.

3.Third ,AIDS bereavement can become complicated when the grieving person has served as primary caregiver to the deceased this is especially the case for caregivers who had a difficult time feeling entitled to respite from attending to the needs of the dying person t see their own personal affairs.From many people ,survivor guilt may prevent them from fully engaging in the bereavement process. Particularly in the gay community,survivors may feel guilt about being HIV –negative in light of the suffering of their peers .children may also experience survival guilt in response to the death of parent or sibling.

4.forurth ,when survivors are themselves HIV-positive ,worry about their own health status may confound bereavement .clinical and counseling psychologist may need to address some or all of these issues in helping their patients come to terms with a friend or loved one's death.

- Coping strategies and chronic illness
- Patient's beliefs about chronic diseases

1.belife about the nature of the illness.

2.belife about the causes of the illness.

3.belife about the controllability of the illness

- Interventions for psychological issues after chronic illness

1.pharmacologicalintrvantions

2.individual therapy

3.brief psychotherapeutic intervantions

4.relaxation and excercise

5.social support intervantions

6.support groups.

- Rehabilitation and chronic illness

1.physical rehabilitation

2.vocational issues

3.social interaction problems

4.personal issues

A) the physical self –body image

B) the achiving self

C)the social self

D)the private self

5.child as patient

6.proffessionals involved in rehabilitation of the chronically ill.s

Terminal illness is a disease that cannot be cured or adequately treated and that is reasonably expected to result in the death of the patient within a short period of time. This term is more commonly used for progressive diseases such as cancer or advanced heart disease than for trauma. In popular use, it indicates a disease that eventually ends the life of the sufferer.

A patient who has such an illness may be referred to as a **terminal patient**, **terminally ill** or simply **terminal**. Often, a patient is considered terminally ill when their estimated life expectancy is six months or less, under the assumption that the disease will run its normal course. The six-month standard is arbitrary,[1] and best available estimates of longevity may be incorrect. Consequently, though a given patient may properly be considered terminal, this is not a guarantee that the patient will die within six months. Similarly, a patient with a slowly progressing disease, such as AIDS, may not be considered terminally ill because the best estimates of longevity were greater than six months. However this does not guarantee that the patient will not die unexpectedly early. In general, physicians slightly overestimate the survival time of terminally ill cancer patients, so that, for example, a person who is expected to live for about six weeks would likely die around four weeks.[2]

Stages

Different patients react to the news that they have a terminal illness in different ways. In general, almost all patients go through various stages of acceptance when a disease like cancer has been diagnosed.[3]

The first stage is disbelief. Most people are shocked that it could happen to them, there is extreme anxiety especially about the unknown. Shock, despair and anger are common. There is also guilt that perhaps the person has done something wrong to receive such a diagnosis.[4] Some individuals use humor as a psychological defense mechanism; others become helpless and often start to bargain. This first stage usually lasts from a few days to many months.

The second stage is depression, which is usually a reaction to the diagnosis. The depression is mild to moderate in intensity and needs family support. Only in rare cases is any type of medical therapy required.

Duration of depression often can last several weeks to throughout the illness. The goal is to help the person go into the final stage of acceptance.

Management

Main article: End-of-life care

By definition, there is no cure or adequate treatment for terminal illnesses. However, some kinds of medical treatments may be appropriate anyway, such as treatment to reduce pain or ease breathing.

Some terminally ill patients stop all debilitating treatments to reduce unwanted side effects. Others continue aggressive treatment in the hope of an unexpected success. Still others reject conventional medical treatment and pursue unproven treatments such as radical dietary modifications. Patients' choices about different treatments may change over time.[5]

Palliative care is normally offered to terminally ill patients, regardless of their overall disease management style, if it seems likely to help manage symptoms such as pain and improve quality of life. Hospice care, which can be provided at home or in a long-term care facility, additionally provides emotional and spiritual support for the patient and loved ones. Some alternative medicine approaches,

such as relaxation therapy,[6] massage,[7] and acupuncture[8] may relieve some symptoms and other causes of suffering.

Caregiving

For the person with a terminal illness, a caregiver is often needed. The caregiver may be a nurse, licensed practical nurse or a family member. The individual may require assistance from a caregiver to receive medications for pain and to control symptoms of nausea or vomiting. The caregiver can assist the individual with daily living activities and assist with movement. Caregivers provide assistance with food and psychological support and ensure that the individual is comfortable.

The patient's family may have questions and most caregivers can provide information to help ease the mind. Doctors generally do not provide estimates for fear of instilling false hopes or obliterate an individual's hope.[9] In most cases, the caregiver works along with physicians and follows professional instructions. Caregivers may call the physician or a nurse if the individual:

- experiences excessive pain.
- is in distress or having difficulty breathing.
- has difficulty passing urine or is constipated.
- has fallen and appears hurt.
- is depressed and wants to harm themselves.
- refuses to take prescribed medications, raising ethical concerns best addressed by a person with more extensive formal training.
- or if the caregiver does not know how to handle the situation.

Most caregivers become the patient's listeners and let the individual express fears and concerns without judgment. Caregivers reassure the patient and honor all advance directives. Caregivers respect the individual's need for privacy and usually hold all information confidential.[10]

Refusal of nutrition and hydration

People who feel they are near the end of their life often refuse food and/or water. Published studies indicate that "within the context of adequate palliative care, the refusal of food and fluids does not contribute to suffering among the terminally ill", and might actually contribute to a comfortable passage from life: "At least for some persons, starvation does correlate with reported euphoria."[11]

Dying

Patients, healthcare workers, and recently bereaved family members often describe a "good death" in terms of effective choices made in a few areas:[12]

- Assurance of effective pain and symptom management.
- Education about death and its aftermath, especially as it relates to decision-making.
- Completion of any significant goals, such as resolving past conflicts.

People who are terminally ill may not always follow recognizable stages of grief. For example, a person who finds strength in denial may never reach a point of acceptance or accommodation and may react negatively to any statement that threatens this defense mechanism. Other people find comfort in arranging their financial and legal affairs or planning their funerals.

Unit – XIII: Contemporary Issues

Research and development in health psychology

The issues of research design and measurement are central to the conduct of optimal research in the study of health, illness, and medical care, and it is on scientific research evidence that sound interventions and clinical application are based. A solid understanding of the details of design and measurement serves an important purpose for the health psychologist. His or her understanding and evaluation of existing research findings on any particular topic are always dependent upon the ability to evaluate critically the methodology used to arrive at those finding.

Conducting research on the psychology of health, illness, and medical care can be difficult because of the nature of the topics studied in this field. The issues are not simply and easily defined, causality is often difficult to determine, and people's expectations can influence the phenomena under study. The people affected by illness (for example, the patient, the physician, and the family) are embedded within a complex social structure, and absolute precision in answering research questions can often be very difficult to achieve. Let's look at one example. The connection between our responses to stressful situations and our health suggests that we may have some control over what happens to us (mind over matter, if you will). We hope that if we have the right thoughts and emotions, we can triumph over disease. Such hopes may lead people to accept these possibilities as facts and the trends as clear causes. But the issue is not nearly that simple. When diagnosed with a cancerous tumor, it is not enough to simply "will" the tumor to disappear. If it were that easy, we would not need medical technology at all! There is evidence that maladaptive responses to stress may have the power to make us sick. Maladaptive responses may include punishing ourselves emotionally when something goes wrong instead of figuring out what we can do to fix it, as well as having feeling of self-blame for bad situations and a sense of hopelessness for the future (). Such thinking might, via complex mechanisms, lead to physical problem and emotional depression. There is some evidence, though not conclusive, that such thinking can lead to depression of the responses of the immune system (). Such findings have prompted the popular press to offer such convincing and provocative headlines as "Worrying Can Make You Sick." Some have touted relaxation as the sole means to eliminate cancerous tumors. These claims have been premature, however. Before an illness phenomenon can be completely understood, a great deal must be learned about the complex mechanisms involved.

Another very important point we must make is that, in some realms of health psychology, causality simply cannot ever be determined by the methods available. The only scientific method that can determine a cause-effect relationship is the "true experiment". A true experiment may be impossible to carry out, however, because of its prohibitive difficulty or lack of ethical justification. Researchers may need to settle for much less direct methods of studying the complex ways in which thoughts influence health. These less direct approaches leave a great deal of room for alternative explanations, and, as a result, controversies arise.

Despite the complexities of research, clear errors are avoidable, however. Common pitfalls can be overcome with awareness and thoughtfulness in research design.

The most common research flaws and considers how certain methodologies can lead us to correct conclusions and can form the basis for wise clinical decisions. Two most common research designs in health psychology

Until recently, nearly all psychologists have dealt fairly exclusively with issues or problems concerning "the mind." In the field of psychology, "illness" has traditionally referred to cognitive and emotional

phenomena such as schizophrenia, anxiety, or depression. In the past forty years, however, psychologists have studied "illness" as it relates to the physical body. Health psychologists have begun to recognize the potentially far-reaching implications of their work for the achievement and maintence of physical health and well-being. They have become particularly aware of the interplay of theory, research, and practice in the prevention and treatment of disease and in the understanding of illness. As healthcare itself has evolved, opportunities for psychologists to contribute to the healthcare process are constantly increasing (Cummings, 1996; Newman & Reed 1996). The common types of illness today are quite different from those experienced a century ago. Further, the ways in which people obtain medical care have also changed dramatically. These changes have been brought about by tremendous advances in the biological sciences and in technology, as well as in response to various social pressures. In turn, these changes have expanded the field of healthcare and have created professional niches for health psychologists to fill.

Early Research Literature

Throughout history, a few physician educators have had opinions about the need to understand sthe patient's psychology in order to care for his or her physical needs. In the fourth century BC, for example, Hippocrates wrote about how the physician's demeanor can engender a patient's trust and encourage a patient's will to live: "The patient, though his condition is perilous, may recover his health simply through his contentment with the goodness of the physician" (Hippocrates, 1923). In 1904, Sir William Osler, a famous medical educator, lectured to medical students that they should listen to the patient because the patient's own words may reveal the diagnosis (Osler, 1904). Another famous physician, Frederick Shattuck (1907) argued that medicine is an art as well as a science.

In the 1950s, article appeared in the research literature with titles such as "Choosing and Changing Doctors" (Gray & Cartwright, 1953) and "Why Do People Detour to Quacks?" (Cobb, 1954). These early articles, written by medical practitioners as well as by psychologists and sociologists, attempted to deal with the complex questions that surrounded the rather widespread dissatisfaction of patients with their medical care professionals. Such papers appeared in the literature in the fields of medicine, psychiatry, nursing, psychology, sociology, pharmacy, hospital administration, medical economics, social work, and anthropology. They pointed to the need for medical professionals to recognize and deal effectively with the psychological as well as the physical aspects of clinical care.

The psychology of physical health and illness, as a field of study, was initially limited in several ways. First, the literature was quite sparse. Few journals carried more than one or two articles a year on the psychological and social aspects of physical illness. Second, the early research was not programmatic (i.e., systematically based on previous work). Researchers tended to carry out isolated studies and failed to build a body of knowledge about a particular topic by improving on the mistakes of earlier work. Third, much of the early work was primarily clinically based. The work of psychoanalysts on the expression of emotional illness through physical disorder is an example. Often, research was done using a clinical population, and then the results were generalized to nonclinical groups. Fourth, the early work was time-bound. That is, instead dealt with prevailing social conditions. The early work on the psychology of health and illness were rather unsophisticated. As a result, in many cases finding remained inconclusive and clinical applications were tentative and sometimes incorrect.

Current Research Literature

The late 1970s saw tremendous progress in research on the psychology of health, illness, and medical care. Psychologists across the country who were interested in medical issues formed an organization. Scholars, researchers, and clinicians who had once been isolated in their work on health issues found the opportunity to share ideas and encouragement. In 1978, the Division of Health Psychology (38) was formed in the American Psychological Association. Division 38 is devoted to promoting health maintenance and illness prevention, as well as to identifying correlates of health, illness, and dysfunction

through specific scientific, educational, and professional contributions from the field of psychology (Matarazzo, 1980). Scientists and professionals within Division 38 come from many different types of backgrounds, but all have in common an interest in promoting health.

Currently, there are thousands of psychologists who keep in touch with each other's work through newsletters, journals and meetings. Other organizations have also sprung up, such as the Society of Teachers of Family Medicine and the Society of Behavioral Medicine. All of these organizations have facilitated dialogue among teachers, researchers, and clinicians and have provided outlets such as The Journal of Behavioral Medicine, The Annals of Behavioral Medicine, and Health Psychology for the publication of research. Such journals help to centralize the literature on the psychological aspects of health, illness, and medical treatment. Psychologists working in the field now compile their work into edited books (such as the International Encyclopedia of the social and Behavioral Sciences (Smelser & Baltes, in press).

In the 1980s and 1990s, we saw tremendous growth in this exciting field. With thousands of psychologists focused on issues of health, illness, and medical treatment, there has been a virtual explosion of information regarding both effective psychological care for ill patients and specific psychological interventions that assist in people's attempts to stay healthy. The inauguration of the field of health psychology has encouraged both the development of theory on which to build future research and systematic examinations from which to development of theory on which to build future research and Psychoneuroimmunology emerged as an important factor in health psychology research. Psychoneuroimmunology combines behavioral science, immunology, and neuroscience and attempts to better understand how psychological factors influence health and disease through the immune system (Kemeny et al., 1992). This represents a distinct change in focus from traditional immunological research, which looked at the immune system as an autonomous system that could not be affected by psychological processes. Take, for example, a colony of monkeys that lives in a psychological laboratory. The laboratory technicians have noticed that these monkeys seem to have more than the usual number of illnesses. They sanitize the cages thoroughly, clean the animals' fur, and begin paying special attention to the quality of the food the animals continue to get sick. A traditional immunological approach would say that animals have some kind of low-grade infection that is making them sick. A psychoneuroimmunological approach, however, would suggest that perhaps there is something about the social environment that is influencing that is influencing the monkey's health. For instance, there may be an unusual level of stress in the colony because of overcrowding or because some of the males are particularly aggressive and fight a great deal. These things may, in turn, impair the functioning of the immune systems of the monkeys, making them more susceptible to illness. It is becoming increasingly clear that all bodily systems. Mind and body are not separate and independent entities, but rather function together to maintain homeostasis. As this view becomes more popular in the scientific community, the contributions of health psychology become more and more important.

The Changing Nature of Medical Practice

Health psychology is continually responsive to changes in health trends and medical practice. For example, as the population has aged, the significance of prostate cancer as a cause of mortality, especially among older men, has become evident. Understanding the impact of prostate disease on a patient's psychosexual self-concept and on social and emotional functioning is important, especially given that most of the available treatments substantially compromise sexual functioning.

One of the factors that has health psychology such a vital discipline is that important psychological and social issues are raised by the changing patterns of illness that favor chronic over acute disease; paradoxically, the face of health psychology may change yet again as patterns of infectious disease have altered. Although the past century has brought substantial control over infectious diseases, they remain a public health problem globally and are still responsible for 13 million deaths each year. Moreover,

changes in society, technology, and microorganisms themselves are leading to the emergence of new diseases, the reemergence of diseases that were once successfully controlled, and problems with drug-resistant strains of once successfully controlled disorders. Just one example of such an issue is getting people to use antibiotics correctly and not to overuse them (M.L. Cohen, 2000). A bigger role in health psychology may emerge from this important and frightening trend.

As technology has improved, organ donation has become an important issue for health psychologists (Radecki & Jaccard, 1997; Shih et al., 2001). 88,166 people are currently waiting for donor organ, a statistic that does not include all of those who could profit from such intervention (United Network for Organs Sharing Bulletin, 1996). The shortage of donor organs suggests that this may be an area in which health psychologists could be increasingly helpful. There appears to be a gap in that many people who are willing in theory to donate their organs have not yet made a commitment to do so. Psychologists may be able to facilitate this process (Amir & Haskell, 1997).

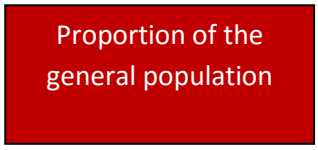
Another example of a medical trend that has fueled research and debate within health psychology is the increasing availability of risk factor testing for a variety of relatively common disorders. Tests are now available, or will shortly be available, on a widespread basis for identify genes implicated is such disease as Huntington's disease, breast cancer, and colon cancer (V.A. McKusick, 1993), yet our understanding of how people deal psychologically with the knowledge that they possess a gene or another risk factor that may ultimately lead to fatal disease is scanty.

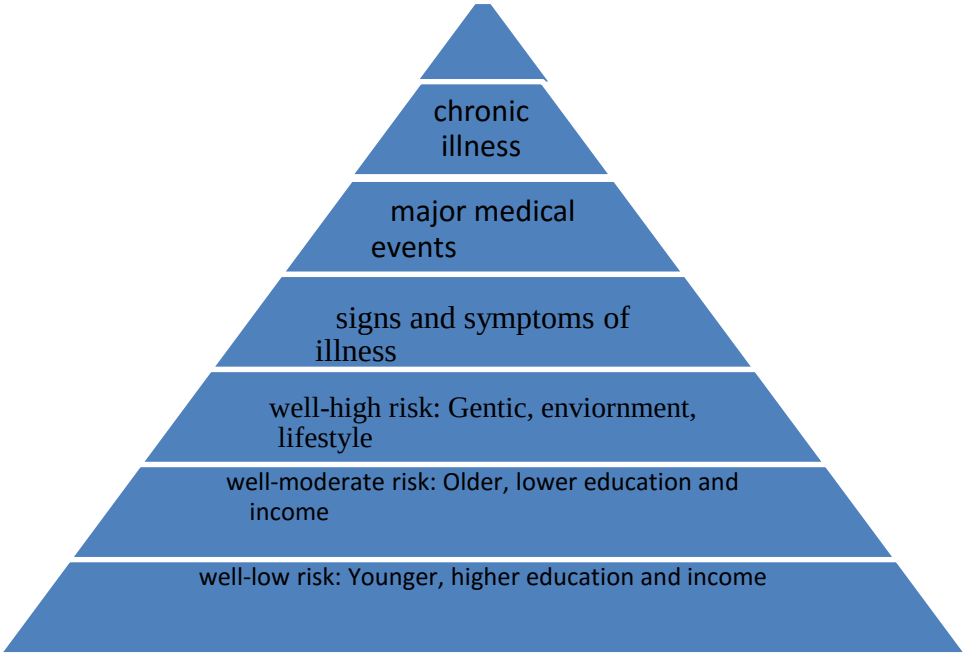
Impact of Technology

Technology advances in medicine have contributed greatly to the enormous costs of contemporary medicine (Reinhardt, 2004). These complex aspects of medicine itself also often daunting for many patients. Explaining the purpose of these technologies and using control-enhancing interventions to enable people to feel like active participants in their treatment can help reduce fear. The growth of medical technology also raises complex questions about how it should be used. Consider transplantable. At present, insufficient numbers of transplantable organs are available; organs and how to increase the supply of transplantable organs and how to develop priorities as to who should receive transplants have been highly controversial issues (Singh, Katz, Beauchamp, & Hannon, 2002). David Crosby of the musical group Crosby, Stills, Nash, and Young received a badly needed liver transplant in short order, despite the fact that some observers believed he had contributed to his liver disease through excessive alcohol consumption.

The ethics of transplantation seem tame, compared with the host of ethical questions raised by recent technological developments in human reproduction. What are the implications of cloning for human? It is now technically possible to transplant eggs and even ovaries from aborted female fetuses into women who have been unable to conceive a child on their own (Kolata, 1994). What are the ethics of using one life to create another? What psychosocial issues are raised by these dilemmas? As a science, health psychology must begin to anticipate many of these controversial issues to help provide a blueprint for considering the psychosocial and ethical issues that will arise.

As medical care has grown more technologically complex, it has also, paradoxically, begun to incorporate psychological and spiritual sides of healing, especially those approaches that draw on Eastern healing traditions. Relaxation and other nontraditional treatment methods are a boon to HMOs because those methods are typically low cost and yet can be remarkably effective for treating stress-related disorders, including such severely problematic conditions as hypertension.





Comprehensive Intervention

Another trend within medicine that affects health psychology is the movement toward **comprehensive intervention models.** There are several models that concentrate and coordinate medical and psychological expertise in well-defined areas of medical practice. One is the pain management program, in which all available treatments for pain have been brought together so that individual regimens can be developed for each patient. A second model is the hospice, in which palliative management technologies and psychotherapeutic technologies are available to the dying patient. Coordinated residential and outpatient rehabilitation programs for coronary heart disease patients, in which multiple health habits are dealt with simultaneously. Similar interventions for other chronic disease, such as cancer and AIDS, may be developed in the coming years.

Most comprehensive intervention models thus far have been geared to specific disease or disorders, but, increasingly, researchers are urging that this model be employed for concerted attacks on risk factors as well (D.B.Abrams et al., 1996). Making use of the mass media, youth prevention projects, educational interventions, social engineering solutions, and tax solutions to such problems as smoking, excessive alcohol consumption, and drug abuse, for example, may represent badly needed supplements to programs that currently focus primarily on health risks that are already in place. The coordination of public health management at the institutional and community levels, with individual health management and illness management for those already ill.

Although comprehensive interventions for particular health problem may provide the best quality of care, they can also be expensive. Some hospitals have already dismantled their pain management centers, for example lack of funds. For comprehensive intervention models to continue to define the highest quality of care, attention must be paid to cost effectiveness as well as to treatment effectiveness.

Systematic Documentation of treatment effectiveness

An important professional goal of health psychology for the future is the continued documentation of the treatment effectiveness of health psychology's technologies. We know that our behavior, cognitive, and

psychotherapeutic technologies work, but we must increasingly find ways to communicate this success to others. This issue has taken on considerable significance in recent years as debate rages over whether and to what degree behavioral and psychological interventions should be covered in managed health care systems.

Cost containment pressures have prompted the development of interventions that are limited, symptom focused, and offered as outpatient services (Sanchez & Turner, 2003), a format that is not always conducive to change through behavioral intervention. Moreover, this has been accompanied by a shift in treatment decision-making power from behavioral health care providers to policy makers. These changes affect health psychology in several ways. A lack of empirical data regarding treatment outcomes and efficacy represents a striking gap in how behavioral scientists and practitioners present their interventions to policy makers (Sanchez & Turner, 2003). This gap occurs, in part, because behavioral scientists may fail to recognize or document the treatment implications of their work and because practitioners may lack the interest or expertise to conduct the formal scientific investigations that would make the scientific case for their interventions. Continued training of health psychologists in both the science and practice aspects of health psychology is clearly needed (Center for the Advancement of Health, 2001). Developing convincing methods of measuring successful psychosocial interventions is of paramount importance.

The potential for health psychology to make major contributions to medicine and medical practice has never been greater. Evidence-based medicine is now the criterion for adopting medical standards. Evidence-based medicine refers to the conscientious, explicit, judicious use of the best scientific evidence for making decisions about the care of individual patients (Timmermans & Angell, 2001). This trend means that, with documentation of the success of health psychology interventions, the potential for empirical contribution to contribute to practice is enhanced.

Psychophysiology

"The body is the medium of experience and the instrument of action. Through its actions we shape and organize our experiences and distinguish our perceptions of the outside world from sensations that arise within the body itself" (Miler, 1978). Anatomy, physiology, and psychophysiology are all branches of science organized around bodily systems whose collective aim is to elucidate the structure and function of the parts of and interrelated system in the human body in transaction with the environment. Anatomy is the science of body structure and the relationships among structures.

Physiology concerns the study if bodily function or how the parts of the body work. For both of these disciplines, what constitutes a "body part" varies with the level of bodily organization, ranging from the molecular to cellular to tissue to organ to body system to the organism. Thus, the anatomy and physiology of the body are intricately interrelated.

Psychophysiology is intimately related to anatomy and physiology but is also concerned with psychology phenomena – the experience and behavior of organisms in the physical and social environment. The complexity added when moving from physiology to psychophysiology includes both the capacity by symbolic systems of representation (e.g., language and mathematics) to communicate and to reflect upon history and experience as well as the social and cultural influences on physiological response and behavior. These factors contribute to plasticity, adaptability, and variability in behavior. Psychology and psychophysiology share the goal of explaining human experience and behavior, and physiological constructs and processes are an explicit and integral component of theoretical thinking in psychophysiology.

The organization of psychophysiology in terms of underlying physiology system or what can be called systematic psychophysiology remains important today for theoretical and pedagogical reasons. Physiological systems provide the foundation for human processes and behavior and are often the target

of systematic observation. An understanding of the physiological system (s) under study and the bioelectrical principles underlying the responses being measured contributes to plausible hypotheses, appropriate operationalizations, laboratory safety, discrimination of signal from artifact, acquisition and analysis of physiological events, legitimate inferences based on the data, and theoretical advancement.

Psychophysiological – like anatomy, physiology, and psychology – is a broad science organized in terms of a thematic as well as a systemic focus. The organization of psychopathology in terms of topical areas of research can be called thematic psychophysiology. For instance, cognitive psychophysiology concerns the relationship between elements of human information processing and physiological events.

Psychoneuroimmunology

Psychoneuroimmunology (PNI) is a field of study that examines the interrelationships among psychosocial processes and nervous, endocrine, and immune system functioning (Ader, 1981; Ader et al., 1991; Ader & Cohen, 1985; Cohen & Herbert, 1996). The immune system is the body's means of guarding against foreign invaders such as bacteria, viruses, and carcinogenic substances. Immune system protection is of two types: (1) nonspecific immunity and (2) specific immunity. The first involves several mechanisms: actual barriers (as the skin, which can keep out invaders), phagocytosis (the process whereby special white blood cells engulf and destroy pathogens), inflammation (swelling and increased blood flow that facilitates the movement and function of white blood cells), and the secretion of toxic chemicals (to kill microorganisms such as bacteria and viruses). The second type, specific immunity, involves the body's ability to protect itself from specific invaders. An example would be the immune response to an encounter with the measles virus after antibodies have been develop from immunization. There are two kinds of immunological reactions: humoral immunity and cell-mediated immunity. Humoral immunity occurs when an antigen (a threatening agent) stimulates Blymphocytes to differentiate into cells that secrete antibodies to fight a foreign invader. Cell-mediate immunity, on the other hand, involves T-lymphocytes from the thymus gland and is slower acting. (A T-cell is a type of infection fighting white blood cell. T-cell levels and lymphocyte activity are key elements in immunity system functioning.) Since the early 1980s, phenomenal advances have been made in our understanding of the role that psychology plays in the stability of the human immune system. Much PNI research has focused on the impact that physical stress or emotions have on immune functioning

Psychobiology

Sociobiology

Sociobiology, the systematic study of the biological basis of social behaviour. The term *sociobiology* was popularized by the American biologist Edward O. Wilson in his book *Sociobiology: The New Synthesis* (1975). Sociobiology attempts to understand and explain animal (and human) social behaviour in the light of natural selection and other biological processes. One of its central tenets is that genes (and their transmission through successful reproduction) are the central motivators in animals' struggle for survival, and that animals will behave in ways that maximize their chances of transmitting copies of their genes to succeeding generations. Since behaviour patterns are to some extent inherited, the evolutionary process of natural selection can be said to foster those behavioural (as well as physical) traits that increase an individual's chances of reproducing.

Sociobiology is a field of scientific study which is based on the assumption that social behavior has resulted from evolution and attempts to explain and examine social behavior within that context. Often considered a branch of biology and sociology, it also draws from ethology, anthropology, evolution, zoology, archaeology, population genetics, and other disciplines. Within the study of human societies,

sociobiology is very closely allied to the fields of Darwinian anthropology, human behavioral ecology and evolutionary psychology.

Sociobiology investigates social behaviors, such as mating patterns, territorial fights, pack hunting, and the hive society of social insects. It argues that just as selection pressure led to animals evolving useful ways of interacting with the natural environment, it led to the genetic evolution of advantageous social behavior.

While the term "sociobiology" can be traced to the 1940s, the concept didn't gain major recognition until 1975 with the publication of Edward O. Wilson's book, *Sociobiology: The New Synthesis*. The new field quickly became the subject of heated controversy. Criticism, most notably made by Richard Lewontin and Stephen Jay Gould, centered on sociobiology's contention that genes play an ultimate role in human behavior and that traits such as aggressiveness can be explained by biology rather than a person's social environment. Sociobiologists generally responded to the criticism by pointing to the complex relationship between nature and nurture. In response to some of the potentially fractious implications sociobiology had for human biodiversity, anthropologist John Tooby and psychologist Leda Cosmides founded the field of evolutionary psychology.

Sociobiology is the study of the *biological bases for social behavior* in every species. It is essentially an *adaptationist approach* to social behavior, i.e., the application to social behavior of the *evolutionary theory* of Charles Darwin, expressed in terms of *modern genetics*--including the *selfish-gene* concept. Sociobiology is an area of biology that is closely related to biological psychology, and specifically to *evolutionary psychology*. It has been applied to explain/solve some problems in evolutionary theory, such as *altruism, infanticide, and sex/gender differences in behavior.*